ANALYTIC NUMBER THEORY NOTES

AARON LANDESMAN

1. INTRODUCTION

Kannan Soundararajan taught a course (Math 249A) on Analytic Number Theory at Stanford in Fall 2017.

These are my "live-TeXed" notes from the course. Conventions are as follows: Each lecture gets its own "chapter," and appears in the table of contents with the date.

Of course, these notes are not a faithful representation of the course, either in the mathematics itself or in the quotes, jokes, and philosophical musings; in particular, the errors are my fault. By the same token, any virtues in the notes are to be credited to the lecturer and not the scribe. [1] Please email suggestions to aaronlandesman@gmail.com

[1]This introduction has been adapted from Akhil Matthew's introduction to his notes, with his permission.

2. 9/26/17

2.1. **Overview.** This will be somewhat of an introductory course in analytic methods, but more like a second introduction. We'll assume familiarity with the prime number theorem, connecting contributions of primes from zeros of the zeta function. You might look at the first half of Davenport's book or so as a prerequisite. We'll assume the students know how to prove there are infinitely many primes in arithmetic progressions.

To get started, we'll do the first four or five lectures proving Vinogradov's three prime theorem:

Theorem 2.1 (Vinogradov)**.** *Every large odd number is the sum of three primes.*

When we say "large," one can actually compute the bound explicitly (i.e., it is effective).

Remark 2.2. Helfgott, a few years back, made the bound accessible so that one could compute exactly which odd numbers were not expressible as the sum of three primes. He showed something like all primes more than 7 could be written as the sum of three primes.

To start the proof, write $N = p_1 + p_2 + p_3$, and we'll count the number of ways to write N as such. In fact, we'll consider

$$\sum_{N=n_1+n_2+n_3} \Lambda(n_1)\Lambda(n_2)\Lambda(n_3)$$

where

$$\Lambda(n) := \begin{cases} \log p & \text{if } n = p^k \\ 0 & \text{else} \end{cases}$$

If we define

$$\Psi(x) := \sum_{n \leq x} \Lambda(n) = x + O(xe^{-c\sqrt{\log x}}).$$

This is equivalent to saying

$$\pi(x) = \mathrm{li}(x) + o(xe^{-c\sqrt{\log x}}).$$

Here

$$\mathrm{li}(x) = \int_2^x \frac{dt}{\log t}.$$

$\mathrm{li}(x)$ is about $\frac{x}{\log x}$.

2.2. **Heuristic of proof.** A first guess is that there are about $\pi(N)$ choices for each of p_1, p_2, p_3. Their sum must add up to a given number N. The chance that $p_1 + p_2 + p_3$ is exactly N is roughly $\frac{1}{N}$. Hence, the number of such ways is approximately

$$\left(\frac{N}{\log N}\right)^3 \frac{1}{N} = \frac{N^2}{(\log N)^3}.$$

We can also estimate

$$R_3(N) := \sum_{N=n_1+n_2+n_3} \Lambda(n_1)\Lambda(n_2)\Lambda(n_3) \sim N^3/N = N^2 + O(N^{1+1/2} + \log N)$$

where the error comes from contributions of powers of primes.

2.3. **Hardy and Littlewood's circle method.** Let

$$S(\alpha) := \sum_{n \le N} \Lambda(n)e(n\alpha)$$

where

$$e(x) = e^{2\pi i x}.$$

Then,

$$\begin{aligned}\int_0^1 S(\alpha)^3 e(-N\alpha)d\alpha &= \int_0^1 \sum_{n_1,n_2,n_3 \le N} \Lambda(n_1)\Lambda(n_2)\Lambda(n_3)e\left(n_1+n_2+n_3-N)\alpha\right) d\alpha \\ &= R_3(N).\end{aligned}$$

To bound this, note that $S(0) = \Psi(N) \sim N$. We'd like to bound it by about N^2.

Also, $S(1/2)$ is pretty big because

$$S(1/2) = (\Lambda(2) + \lambda(4) + \cdots) - \sum_{n \le N, n \text{ odd}} \Lambda(n)$$

because $e(x) = -1$ if x is odd.

Then, for all

$$|\lambda| \le \frac{10^{-6}}{N},$$

we have $\Re S(\alpha) > .99N$. Then,

$$\int_{|\alpha| \le 10^{-6}/N} S(\alpha)^3 e(-N\alpha)d\alpha \simeq 10^{-6}N^2.$$

We could similarly make an argument in a small neighborhood of $\frac{1}{2}$. This gives an analytic reason that the number of representations might be on the scale of N^2. So there are portions of the integral which give the correct answer.

Exercise 2.3 (Waring's problem)**.** We want to know whether we can write $N = x_1^k + x_2^k + \cdots + x_3^k$ (i.e., as a sum of four squares or nine cubes, etc.)

(1) First, find a probabilistic guess for the number of such representations.
(2) Use the circle method

$$\int_0^1 \left(\sum_{n \le N^{\frac{1}{k}}} e(n^k \alpha) \right)^S e(-N\alpha) d\alpha.$$

Then, find portions of the integrand that correspond to the right probabilistic answer.

Returning to our integral for three primes, let's think about when $S(\alpha)$ is big.

Example 2.4. Let's try $\frac{1}{3}$.

$$\sum_{n \le N} \Lambda(n) e(n/3).$$

We have a contribution from powers of 3 which is about $\log N$, so

$$\begin{aligned} \sum_{n \le N} \Lambda(n) e(n/3) &= O(\log N) + e(1/3)\Psi(N;3,1) + e(2/3)\Psi(N;3,2) \\ &\sim N/2. \end{aligned}$$

where

$$\Psi(N;3,1) \sim \frac{N}{\phi(3)} = \frac{N}{2}.$$

where $\Psi(N;a,b)$ counts the number of primes up to N which is $b \bmod a$.

Remark 2.5. Note that

$$S(\frac{a}{q})$$

counts approximately the distribution of primes in progressions mod q with $(a,q) = 1$. Sometimes when q is small, since we're only count primes coprime to q, we will get an answer substantially away from 0.

We'll later need to think through the uniformity of q in terms of N.

Remark 2.6 (Insight)**.** $S(\alpha)$ is big near most rational numbers with small denominators. It's not big near 1/4, so we're only saying it's big near certain ones. This might have something to do with whether the denominator of the rational number is square free: if it is not square free, you essentially get translates over roots of unity of that prime whose square divides q, and things cancel out

Exercise 2.7. Show

$$\sum_{k \bmod q}^{*} e(\frac{ak}{q}) = \mu(q),$$

where the $*$ means $(k,q) = 1$ and μ is the Möbius function.

Goal 2.8. If α is not near a rational number with small denominator then $|S(\alpha)|$ is small.

To accomplish this, Hardy and Littlewood decided to split $[0,1]$ into two parts - major arcs M and minor arcs m. The major arcs are close to $\frac{a}{q}$ for q small, and the minor arcs are the rest. The measure of the minor arcs are big while the measure of the major arcs have small measure. That is, the minor arcs have nearly full measure. So, there is a very small set on which the generating function is big. There is also a big set on which the generating function is small.

There is a trivial bound $|S(\alpha)| \leq \Psi(N) \sim N$, using the triangle inequality. One can also work out

Lemma 2.9. *We have*

$$\int_0^1 |S(\alpha)|^2 d\alpha \sim N \log N.$$

Proof.

$$\begin{aligned}
\int_0^1 |S(\alpha)|^2 d\alpha &= \sum_{n_1, n_2 \leq N} \Lambda(n_1)\Lambda(n_2) \int_0^1 e\left((n_1 - n_2)\alpha\right) d\alpha \\
&= \sum_{n \leq N} \Lambda(n)^2 \\
&= \sum_{n \leq N} \log n \Lambda(n) + O(\sqrt{N} \log N) \\
&= N \log N.
\end{aligned}$$

Exercise 2.10. Verify the above, where the $O(\sqrt{N} \log N)$ difference is coming from prime powers. The idea for the last step is that most numbers less than N are on the order of N. One might use partial summation, which is integration by parts.

□

Usually,

$$|S(\alpha)| \sim \sqrt{N \log N}.$$

If α is far from every rational number, such as the golden ratio, ϕ, we might try to compute $S(\phi)$. We might expect that $S(\phi) \ll N^{\frac{1}{2}+\varepsilon}$. We don't know whether this is true, but we do know

$$\sum_{n \leq N} \Lambda(n) e(n\phi) \ll N^{1-\delta}$$

for some $\delta > 0$ (and it will probably even be a pretty large δ). We will now develop a technique saying that once you are far away from a rational number, you can get this sort of power saving.

2.4. **Strategy for determining asymptotics for $R_3(N)$.** We have the integral

$$\int_0^1 S(\alpha)^3 e(-N\alpha) d\alpha = (\int_M + \int_m) S(\alpha)^3 e(-N\alpha) d\alpha$$

We want the second part over the minor arcs to be smaller than N^2. The idea is that we can bound

$$\begin{aligned} \left| \int_m S(\alpha)^3 e(-N\alpha) d\alpha \right| &\leq \left(\sum_{\alpha \in m} |S(\alpha)| \right) \int_0^1 |S(\alpha)|^2 d\alpha \\ &\ll (N \log N) \sum_m |S(\alpha)|. \end{aligned}$$

So, it is enough to have

$$\sum_{\alpha \in m} |S(\alpha)| \leq \frac{\varepsilon N}{\log N}.$$

This will show the contribution from the minor arcs is less than that of the major arcs, assuming we know the major arcs contribute N^2.

Exercise 2.11 (Roth)**.** For all $\delta > 0$, there is $N_0 = N(\delta)$ so that for all $N \geq N_0$, such that every subset $\mathscr{A} \subset [1, N]$ with $|\mathscr{A}| \geq \delta N$ has a (nontrivial) three term arithmetic progression.

Letting

$$\mathscr{A}(\alpha) = \sum_{a \in \mathscr{A}} e(a\alpha),$$

we obtain

$$\int_0^1 \mathscr{A}(\alpha)^2 \mathscr{A}(-2\alpha) d\alpha$$

counts the number of triples (x, y, z) with $x + z = 2y$. This includes $|\mathscr{A}|$ trivial solutions, so we want to see this integral is larger. We might expect $\delta^3 N^2$ solutions. But now, it's a bit hard to see how to actually bound this integral.

Exercise 2.12 (Vague exercise)**.** If, "away from 0,"

$$|\mathscr{A}(\alpha)| \leq \varepsilon |\mathscr{A}|$$

then the contribution of that portion of the integral

$$\int_0^1 \mathscr{A}(\alpha)^2 \mathscr{A}(-2\alpha) d\alpha$$

is bounded by $\varepsilon |\mathscr{A}|^2$. (We'd like to know something like $\varepsilon \leq \delta/10^6$.)

The idea is that either we have this bound above, or else we get some additive structure in $\mathscr{A}$ which we exploit to get a bigger density set.

There are notes on this on Sound's web-page from a course he taught on additive combinatorics.

Now, we want to focus on showing that for some definition of the minor arcs, the sum $S(\alpha)$ has a little bit of cancellation.

2.5. **Vinogradov's method.** Here is the key idea from Vinogradov's method. This comes up many times throughout analytic number theory. We'd like to understand the sequence

$$S(\alpha) = \sum_{n \leq N} \Lambda(n) e(n\alpha).$$

We could similarly study

$$\sum_{n \leq N} \Lambda(n) e(f(n)),$$

where, say, $f(n)$ is $e(\sqrt{n})$ or $e(\sqrt{n} + (\log n)^2)$. We could similarly study

$$\sum_{n \leq N} e(\sqrt{n})$$

or

$$\sum_{n \leq N} e(t \log n)$$

for looking at 0's of the zeta function. Let's start with the simplest version of these, where instead of summing over primes and prime powers, we only sum over all the integers. Say we want to consider

$$\sum_{n \leq N} e(n\alpha).$$

This is a geometric progression, so it is easy to sum:

$$\begin{aligned}\sum_{n \leq N} e(n\alpha) &= \frac{e(\alpha)\left(1 - e(N\alpha)\right)}{1 - e(\alpha)} \\ &= \frac{x}{\sin \pi\alpha}\end{aligned}$$

where x is bounded by 2, and the numerator is approximately $\sin \pi N\alpha$.

Exercise 2.13. Show

$$\begin{aligned}\left|\sum_{n \leq N} e(n\alpha)\right| &\leq \min\left(N, \frac{2}{\sin \pi\alpha}\right) \\ &\ll \min\left(N, \frac{1}{||\alpha||}\right).\end{aligned}$$

letting $||\alpha||$ denote the distance from the nearest integer.

Let Φ be a smooth function. Then,

$$\sum_n \Phi(n/N)e(n\alpha)$$

is some smooth version of what we are trying to approximate. We might try to use the Poisson summation formula. We can write

$$\sum_n \Phi(n/N)e(n\alpha) = N \sum_k \hat{\phi}(N(k+\alpha))$$

and work out the Poisson summation formula. For ϕ smooth, the Fourier transform is rapidly decreasing.

3. 9/28/17

Recall last time we had

$$R_3(N) := \sum_{n_1+n_2+n_3=N} \Lambda(n_1)\Lambda(n_2)\Lambda(n_3).$$

The goal was to show this asymptotes to N^2. We set

$$S(\alpha) = \sum_{n \leq N} \Lambda(n)e(n\alpha),$$

and found

$$R_3(N) = \int_0^1 S(\alpha)^3 e(-N\alpha)d\alpha.$$

The idea is to show that $S(\alpha)$ is large only near rational numbers with small denominators (the minor arcs). On the complement, we want to show $|S(\alpha)|$ is small, and then bound

$$|\int_m S(\alpha)^3 e(-N\alpha)d\alpha| \leq \left(\sup_{\alpha\in m}|S(\alpha)|\right)\int_0^1 |S(\alpha)|^2 d\alpha.$$

We then could use Parseval's identity to bound this by $N\log N$.

Toward the end of last time, we found

$$\sum_{n\leq N} e(n\alpha) \ll \min(N, \frac{1}{||\alpha||}).$$

Exercise 3.1. Count the number of ways of writing $N = n_1 + n_2 + n_3$ asymptotically by writing down the associated integral using the circle method. The exponential sum will only be big for α near 0. There is only one major arc in this case. The answer should be about $N^2/2$, and the point is to see where the $1/2$ comes from.

Recall from elementary number theory that $\Lambda(n) = \sum_{ab=n}\mu(a)\log b$. If we look at the Dirichlet series for

$$-\frac{\zeta'}{\zeta}(s) = \frac{1}{\zeta(s)} \cdot -\zeta'(s)$$

where the first term has Dirichlet series $\Lambda(n)$, $\frac{1}{\zeta}$ has Dirichlet series μ and $-\zeta'(s)$ has Dirichlet series log. Then, the convolution of μ and log is Λ. Then,

$$\begin{aligned}\int_{n\leq N}\log n e(n\alpha) &= \int_{1^-}^N \log t d\left(\sum_{n\leq t} e(n\alpha)\right)\\ &= \log N\sum_{n\leq N} e(n\alpha) - \int_{1^-}^N \frac{1}{t}\sum_{n\leq t} e(n\alpha)dt\\ &\ll (\log N)\min\left(N, \frac{1}{||\alpha||}\right).\end{aligned}$$

Then,

$$\sum_{n\leq N}\Lambda(n)e(n\alpha) = \sum_a \mu(a)\sum_{b\leq N/a}\log b e(ab\alpha).$$

Example 3.2. First, let's try the case $\alpha = 0$. Then,

$$\sum_{n \le N} \Lambda(n) = \sum_a \mu(a) \sum_{b \le N/a} \log b$$

If we knew $\sum_{a=1}^{\infty} \frac{\mu(a)}{a} = 0$ we could then prove the prime number theorem. This is essentially equivalent to proving the prime number theorem, so it would take some work. Things are good when a is small, but there is a problem when a is big.

Goal 3.3. Our overall aim is to bound $S(\alpha)$.

3.1. **Vinogradov's idea.** We'd like to somehow decompose $\Lambda(n)$ into pieces, where either we use a simple exponential sum, or using the following idea. The idea has to do with bilinear forms.

We notate $m \sim M$ meaning $M < m \le 2M$.

$$B(M, N) := \sum_{m \sim M} \sum_{n \sim N} a_m \overline{b}_n \cdot f(m, n),$$

with a_i, b_i arbitrary complex numbers, and $f(m, n)$ is an oscillatory term, such as $f(m, n) = e(mn\alpha)$. Intuitively $f(m, n)$ should have some "cancellation."

Goal 3.4. We'd like to bound the sum $B(M, N)$ by something like

$$\left(\sum_{m \sim M} |a_m|^2 \right)^{1/2} \left(\sum_{n \sim N} |b_n|^2 \right)^{1/2} \cdot N_f$$

where g is some sort of operator norm of the matrix $f(m, n)$.

(1) We think of $f(m, n)$ as something that cancels out. It does not always have the same sign.
(2) We typically have $|f(m, n)|$ small, e.g., ≤ 1.
(3) We might also imagine $a_m, b_n \le 1$.

We'd then like to compare the bound we obtain to the trivial bound MN. We'd like to beat this trivial bound.

This will be impossible to bound if

(1) $f(m, n) = 1$.
(2) $f(m, n) = \alpha(m)\beta(n)$.
(3) Both M and N are big (or at least the associated matrix has large rank).

In order to avoid these impossibilities, we will need to exploit that $f(m, n)$ is genuinely a 2-variable function, and does not decouple.

To obtain the bound, we will use Cauchy-Schwartz. We have

$$\left|\sum_{m\sim M}\sum_{n\sim N} a_m\bar{b}_n f(m,n)\right|^2 \le \left(\sum_{m\sim M}|a_m|^2\right)\left(\sum_{m\sim M}\left|\sum_{n\sim N} b_n f(m,n)\right|^2\right).$$

Let $*$ denote

$$\sum_{m\sim M}\left|\sum_{n\sim N} b_n f(m,n)\right|^2$$

Then,

$$\begin{aligned} * &= \sum_{m\sim M}\left|\sum_{n\sim N} b_n f(m,n)\right|^2 \\ &\le \sum_{n_1,n_2\sim N} b_{n_1}\bar{b}_{n_2}\sum_{m\sim M} f(m,n_1)\overline{f(m,n_2)}. \end{aligned}$$

From this we have gained that we have replaced the unknown a_m, b_n by inner products of our known matrix $f(m,n)$.

If we knew $f(m,n)$ were orthogonal, then the sum amounts to terms with $n_1 = n_2$ of the form

$$\sum_{n\sim N}|b_n|^2 M.$$

Things will never be quite so good that we will precisely get orthogonality. But, we might have some approximate orthogonality. For example, if $n_1 \neq n_2$, maybe we can bound the correlation by 1. Then, the off-diagonal terms are of the form

$$\sum_{n_1\neq n_2}|b_{n_1}b_{n_2}.$$

Using Cauchy's inequality, we obtain

$$|b_{n_1}b_{n_2}| \ll |b_{n_1}|^2 + |b_{n_2}|^2.$$

Hence,

$$\sum_{n_1\neq n_2}|b_{n_1}b_{n_2} \ll N\sum_{n\sim N}|b_n|^2.$$

In the above favorable circumstances, putting the above together, we get a bound

$$\left|\sum_{m\sim M}\sum_{n\sim N} a_m\bar{b}_n f(m,n)\right| \ll \left(\sum_{m\sim M}|a_m|^2\right)^{1/2}\left(\sum_{n\sim N}|b_n|^2\right)^{1/2}\left(\sqrt{M}+\sqrt{N}\right)$$

We might now try setting all $|a_n| = |b_n| = 1$, and then our bound is $M\sqrt{N} + N\sqrt{M}$ instead of MN so we save $\frac{1}{\sqrt{M}} + \frac{1}{\sqrt{N}}$. Again, this bound holds under various assumptions that the $f(m,n)$ are approximately orthogonal.

This is the key strategy. We now want to implement the above strategy in the situation we are in. The key point of the strategy is that we have transferred the problem from understanding the unknown a_n, b_m to the known problem of understanding the correlation of $f(m,n)$.

3.2. **Applying Vinogradov's idea.**

We now want

$$\sum_{m \sim M} \sum_{n \sim N} a_m \overline{b}_n e(mn\alpha).$$

Thinking of the a_m as $\mu(a)$ and the b_n as $\mu(n)$. We then want to bound

$$* = \sum_{n_1, n_2 \sim N} |b_{n_1} b_{n_2}| \left| \sum_{m \sim M} e\left(m(n_1 - n_2)\alpha\right) \right|.$$

Suppose we write $n_1 - n_2 =: k$. Then $|k| \leq N$. We then have

$$\begin{aligned} * &\ll \sum_{n_1, n_2 \sim N} \left(|b_{n_1}|^2 + |b_{n_2}|^2 \right) \min\left(M, \frac{1}{||(n_1 - n_2)\alpha||} \right) \\ &\ll \left(\sum_{n_1 \sim N} |b_{n_1}|^2 \right) \sum_{|k| \leq N} \min\left(M, \frac{1}{||k\alpha||} \right). \end{aligned}$$

We conclude

$$\left| \sum_{m,n} a_m \overline{b}_n e(mn\alpha) \right| \leq \left(\sum |a_m|^2 \right)^{1/2} \left(\sum_n |b_n|^2 \right)^{1/2} \left(\sum_{|k| \leq N} \min\left(M, \frac{1}{||k\alpha||} \right) \right)^{1/2}.$$

We'd like to show we get something from this if α is not close to a rational number with small denominator. So we keep in mind that α might be irrational.

We start with Dirichlet's theorem:

Theorem 3.5 (Dirichlet). *For all $Q > 1$ and all $\alpha \in \mathbb{R}$, there exists a rational number a/q with $(a,q) = 1$ and $q \leq Q$ so that*

$$\left| \alpha - \frac{a}{q} \right| < \frac{1}{qQ}.$$

So, we can get pretty good approximations to irrational numbers with small denominators. A crude version of this is

$$\left|\alpha - \frac{a}{q}\right| \le \frac{1}{q^2}.$$

We can get approximations of this type by continued fraction expansions.

Let $(*)$ denote

$$(*) := \sum_{|k| \le N} \min\left(M, \frac{1}{||k\alpha||}\right)$$

We should expect that if q is small, then we might revert to the trivial bound MN. Perhaps there is some inverse relationship with q. So maybe we get something like MN/q or $MN/\sqrt{q}$. So, the larger q gets, the more saving we should get over the trivial bound. So, very small values of q are not good, but we'd like to show that if q is in some intermediate range, we might hope to be in a good situation. So, the bound we will write down will depend on the Diophantine properties of α and the scale on which we are operating.

So, assume α has the rational approximation given by Dirichlet's theorem satisfying

$$\left|\alpha - \frac{a}{q}\right| \le \frac{1}{q^2}.$$

Split the interval from m to n of length k into several intervals of length q. How do $|k\alpha|$ vary on this interval - there is at most one value which is very close to an integer. Then, we have

$$\sum_{0 \le a \le q} \min\left(M, \frac{q}{a}\right) \ll M + q \log q.$$

The $\log q$ is unimportant and we can remove it if we'd like, using Poisson summation if we had a smooth function. It would then be $\min(M, q/a^2)$ and we could remove the log.

We now want to bound the following by dividing N into $N/q + 1$ intervals of length q.

$$\begin{aligned}
(*) &= \sum_{|k| \le N} \min\left(M, \frac{1}{||k\alpha||}\right) \\
&= (N/q + 1)(M + q \log q) \\
&\ll \left(\frac{\log q}{q}\right)(M + q)(N + q).
\end{aligned}$$

We have proven:

Proposition 3.6. *If* $|\alpha - \frac{a}{q} < \frac{1}{q^2}$ *and* $(a,q) = 1$, *then*

$$\sum_{m \sim M} \sum_{n \sim N} a_m \overline{b}_n e(mn\alpha)$$
$$\ll \left(\sum |a_n|^2\right)^{1/2} \left(\sum |b_n|^2\right)^{1/2} \left(\frac{\log q}{q}\right)^{1/2} \left(\sqrt{MN} + \sqrt{Mq} + \sqrt{Nq} + q\right).$$

Question 3.7. Why is the above bound useful?

Just to summarize, this might be helpful to think of the case that the a_i, b_j are bounded in norm by 1. We then get approximately a bound by

$$\frac{\sqrt{MN}}{\sqrt{q}}\left(\sqrt{MN} + \sqrt{q}\left(\sqrt{M} + \sqrt{N}\right) + q\right) = \frac{MN}{\sqrt{q}} + MN\left(\frac{1}{\sqrt{M}} + \frac{1}{\sqrt{N}}\right) + \sqrt{q}\sqrt{MN}.$$

The middle term is what we would get from the orthogonality relation. If q is small or large, we don't beat the trivial bound, as expected. This is the crucial bound for our particular bilinear form.

Next time, we'll try and rewrite the coefficients as a bilinear form as a function of multiple summands. The final ingredient is to write a combinatorial identity to express $\Lambda(n)$ in terms of things we understand. That is, we want to write it as something like

$$\sum (\log b)^{e(b\alpha)} + \sum_{m,n} a_m b_n e(mn\alpha).$$

where both m and n are large in the second sum.

4. 10/3/17

4.1. **Review.** Last time, we were trying to bound sums like

$$\begin{aligned}
|\sum_m \sum_n f(m,n)|^2 &\leq \left(\sum_m |a_m|^2\right) \left(\sum_m \left|\sum_n b_n f(m,n)\right|^2\right) \\
&= \sum_{n_1, n_2} |b_{n_1} b_{n_2}| \left|\sum_m f(m,n_1)\overline{f(m,n_2)}\right| \\
&= \sum_{n_1, n_2} \left(|b_{n_1}|^2 + |b_{n_2}|^2\right) \left|\sum_m f(m,n_1)\overline{f(m,n_2)}\right|,
\end{aligned}$$

and we hoped that the correlations, i.e., the terms $\left|\sum_m f(m,n_1)\overline{f(m,n_2)}\right|$ were bounded by M if $n_1 = n_2$ and $O(1)$ if $n_1 \neq n_2$. We then could bound the above by $(M+N)\sum_i |b_n|^2$.

Recall last time, we were trying to bound sums like

$$\sum_{m\sim M}\sum_{n\sim N} a_m b_n e(mn\alpha)$$

where $m \sim M$ means $M \le m \le 2M$. We had

$$\left|\alpha - \frac{a}{q}\right| \le \frac{1}{q^2},$$

with $(a,q) = 1$. We proved last time

$$\sum_{m\sim M}\sum_{n\sim N} a_m b_n e(mn\alpha)$$
$$\ll \frac{\log q}{\sqrt{q}}\left(\sum_m |a_m|^2\right)^{1/2}\left(\sum_n |b_n|^2\right)^{1/2}\left(\sqrt{MN} + \sqrt{q}\left(\sqrt{M}+\sqrt{N}\right) + q\right).$$

We had

$$\Lambda(n) = \sum \mu(a)\log b$$
$$S(\alpha) = \sum_{n\le N} \Lambda(n)e(n\alpha).$$

Our tools are

(1)
$$\sum_{n\le x} \log ne(n\alpha)$$
is some sort of geometric progression which after factoring out a log, which we understand

(2)
$$\sum_{m\sim M}\sum_{n\sim N} a_m b_n e(mn\alpha)$$
which is well bounded using Vinogradov's method discussed last time (and above this lecture) assuming the covariances are small for off-diagonal terms and on the order of the number of elements for the diagonal terms.

Our goal is now the following combinatorial one:

Goal 4.1. Write $\Lambda(n)$ in a form where we can use the above two tools.

Theorem 4.2 (Vaughan's identity)**.** *We have*

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'}{\zeta}(s)$$
$$\frac{1}{\zeta(s)} = \sum_a \frac{\mu(a)}{a^s}$$
$$-\zeta'(s) = \sum_b \frac{\log b}{b^s}.$$

Proof. This follows from straightforward manipulations of Dirichlet series, the first one comes from the derivative of $\log \zeta(s)$. □

4.2. **Mollifying $\zeta(s)$.** We now want to Mollify $\zeta(s)$. For this, we will use Selberg's sieve.

One way to study the zeta function could be an appropriate truncation. We may consider

$$M(s) = \sum_{n \leq U} \frac{\mu(n)}{n^s}.$$

which is a sort of approximation to the inverse of the Riemann ζ function, using the above identity that

$$\frac{1}{\zeta(s)} = \sum_a \frac{\mu(a)}{a^s}$$

One can compute,

$$\zeta(s)M(s) = \sum_{n=1}^{\infty} \frac{a(n)}{n^s}.$$

where $a(n)$ is defined by

$$\begin{aligned} a(n) &= \sum_{d \mid n, d \leq U} \mu(d) \\ &= \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{if } 1 < n \leq U \\ \text{the norm is bounded by } d(n) & \text{if } n > U \end{cases} \end{aligned}$$

where $d(n)$ is the number of divisors of n.

We have

$$\begin{aligned}-\frac{\zeta'}{\zeta}(s) &= -\frac{\zeta'}{\zeta}(s)\left(1-\zeta M+\zeta M\right)\\ &= -\zeta(s)M(s)-\frac{\zeta'}{\zeta}(s)\left(1-\zeta M(s)\right).\end{aligned}$$

First, we should be fairly happy with the term $-\zeta(s)M(s)$ which has Dirichlet series given by

$$\left(\sum \frac{\log b}{b^s}\right)\left(\sum_{n\le U}\frac{M(n)}{n^s}\right),$$

and we'll have a long sum in the b's, where we can hope to get some cancellation. Next, to understand

$$\frac{\zeta'}{\zeta}(s)\left(1-\zeta M(s)\right).$$

We try to think of this product as a sort of bilinear form. The terms from $1-\zeta M(s)$ only matter for n larger than U. Thinking of this term as a bilinear term, we're happy because $1-\zeta M(s)$ is large. But, we have to ensure that ζ'/ζ is not too "skinny." To deal with this, we can subtract out the small primes, and then later add them back.

To accomplish this, we define

$$P(s) := \sum_{n\le V}\frac{\Lambda(n)}{n^s}.$$

Then,

$$\begin{aligned}-\frac{\zeta'}{\zeta}(s) &= -\frac{\zeta'}{\zeta}(s)-P(s)+P(s)\\ &= \sum_{n>V}\frac{\Lambda(n)}{n^s}+\sum_{n\le V}\frac{\Lambda(n)}{n^s}\end{aligned}$$

Then,

$$\begin{aligned}-\frac{\zeta'}{\zeta}(s) &= -\zeta'(s)M(s)-\frac{\zeta'}{\zeta}(s)\left(1-\zeta M(s)\right)\\ &= -\zeta'(s)M(s)+\left(\frac{-\zeta'}{\zeta}(s)-P(s)\right)\left(1-\zeta M(s)\right)+P(s)\left(1-\zeta M(s)\right)\\ &= -\zeta'(s)M(s)+\left(\frac{-\zeta'}{\zeta}(s)-P(s)\right)\left(1-\zeta M(s)\right)+P(s)-\zeta(s)M(s)P(s)\end{aligned}$$

The point of this breakdown is that we now have three terms we can handle using our two tools we have.

The middle term decomposes into two parts, both of which are big, which gives a bilinear form.

The last term has a long sum from the $\zeta(s)$ term in simple coefficients. For $P(s)$, we can just ignore it because V is small. The first term is similarly a long some from the ζ' term.

Remark 4.3. The first term which we handle via our first summation technique is called a "type 1 sum" and the second handled via our second bilinear form summation technique is called a "type 2 sum."

Example 4.4. Let's say we want to write

$$\begin{aligned}\frac{-\zeta'}{\zeta}(s) &= \frac{-\zeta'}{\zeta}(s)\left((1-\zeta M)^2 + 2\zeta M - \zeta^2 M^2\right) \\ &= \left(-\frac{\zeta'}{\zeta} + \zeta' M\right)(1-\zeta M) - 2\zeta' M + \zeta\zeta' M^2.\end{aligned}$$

The first is a type 2 sum, the second is a type 1. The ζ and ζ' are both somewhat a simple divisor function because if the product of ζ, ζ' goes in a long range then at least one of them must be summed in a long range. So, the third term is also a type 1 sum.

Remark 4.5 (Heath Brown identity, aka binomial theorem)**.** Given

$$\frac{-\zeta'}{\zeta}(s)\left(1-\zeta M\right)^k$$

one can try expanding this in k via the binomial theorem, and try to bound various terms.

4.3. **Proving Vinogradov's theorem.** Recall we have

$$\left|\alpha - \frac{a}{q}\right| \leq \frac{1}{q^2}.$$

with $(\alpha, q) = 1$. Our goal is to bound $S(\alpha)$ in terms of q. Trivially we know $S(\alpha)$ is bounded by N, and we want to save a bit more than one log on the minor arcs, and then we'll have to concentrate on the major arcs.

Using Vaughan's identity, (where we have not yet specified U and V). There are three type 1 sums and one type 2 sum (from the bilinear form. Recall we have

$$-\frac{\zeta'}{\zeta}(s) = -\zeta'(s)M(s) + \left(\frac{-\zeta'}{\zeta}(s) - P(s)\right)(1-\zeta M(s)) + P(s) - \zeta(s)M(s)P(s)$$

and we are trying to bound the four sums. First we deal with the $P(s)$ term, which is

$$\left|\sum_{n\leq V}\Lambda(n)e(n\alpha)\right| \ll V.$$

Next, we try to bound the first term, which is the contribution from primes coming from $\zeta' M$. This term is

$$\sum_{n\leq U}\mu(n)\sum_{r\leq N/n}\log r e(nr\alpha) \ll \sum_{n\leq U}\min\left(\frac{N}{n}, \frac{1}{||n\alpha||}\right).$$

It is convenient to split over dyadic blocks $2^k < n \leq 2^{k+1}$.

Exercise 4.6. Carry out the argument from week 1 for dealing with sums like

$$\sum_{|n|\leq N}\min\left(N, \frac{1}{||n\alpha||}\right).$$

There is a small lie in what we will next do, and your job is to fix it by Thursday. You should check what happens for smaller n as well.

Pretending that only the large range matters, we can bound

$$\begin{aligned}\sum_{n\leq U}\mu(n)\sum_{r\leq N/n}\log r e(nr\alpha) &\ll \sum_{n\leq U}\min\left(\frac{N}{n}, \frac{1}{||n\alpha||}\right)\\ &\ll (\log N)\left(\frac{U}{q}+1\right)\left(\frac{N}{U}+q\log q\right)\\ &\ll (\log N)^2\left\{\frac{N}{q}+U+\frac{N}{U}+q\right\}\end{aligned}$$

Now, we'll aim to attack the last type 1 sum, which is the term corresponding to $\zeta(s)M(s)P(s)$ which is

$$\sum_{n\leq U}\sum_{\ell\leq V}\mu(n)\Lambda(\ell)\sum_{r\leq N/n\ell}e(n\ell r\alpha)$$

if we let $n\ell = a$ then $a \leq UV$. Then, the terms in a are bounded by something like $\sum_{n\ell=a}\Lambda(\ell) = \log a$ (using that the left hand side is the convolution of ζ with ζ'/ζ which is ζ' which has coefficients given by log. Therefore,

$$\begin{aligned}\sum_{n\le U}\sum_{\ell\le V}\mu(n)\Lambda(\ell)\sum_{r\le N/n\ell}e(n\ell r\alpha) &\ll \sum_{a\le UV}\log a\left|\sum_{r\le N/a}e\left(ar\alpha\right)\right|\\ &\ll (\log N)\sum_{a\le UV}\min\left(\frac{N}{a},\frac{1}{||\alpha a||}\right)\\ &\ll (\log N)^2\left\{\frac{Nq}{q}+q+UV+\frac{N}{UV}\right\}.\end{aligned}$$

Exercise 4.7. Verify the above bounds using a method similar to the type 2 bound of the first $\zeta' M$ term.

Adding up our three type one sums, and removing terms trivially bounded by others, we get

$$(\log N)^2\left\{\frac{N}{q}+q+UV+\frac{N}{U}\right\}.$$

This handles three of the four terms. The last term remaining to be handled is the type 2 sum corresponding to

$$\left(\frac{-\zeta'}{\zeta}(s)-P(s)\right)(1-\zeta M(s))$$

The first sum only contains terms larger than V and the second only contains terms larger than U. Using

$$1-\zeta M(s)=\sum_n\frac{1}{n^s}\left(\sum_{d|n,d>U}\mu(d)\right)$$

we obtain the sum

$$\sum_{n>V}\Lambda(n)\sum_{m>U}\sum_{d|m,d>U}\mu(s)e(mn\alpha)$$

with $mn\le N$.

Remark 4.8. We now have two terms with variables in our bilinear form, both with large values. Both will range over dyadic intervals. It starts to look like a bilinear form, though there is the caveat that the two variables are connected by the condition that $mn\le N$. Hence, we need some technical device to separate the variables.

Essentially, this is saying these are like points lying below a hyperbola and we would like to approximate the hyperbola by some rectangle.

Morally,

$$\sum_{n>V} \Lambda(n) \sum_{m>U} \sum_{d|m,d>U} \mu(s) e(mn\alpha)$$

with $mn \le N$. Ignoring the condition $mn \le N$, (which we will fix next time) the above sum is approximated by

$$\sum_{m\sim A} \sum_{n\sim B} a(m) b(n) e(mn\alpha).$$

for $A > U, B > V, AB \le N$. This is the kind of bilinear form we want for our type 2 sum.

We can now use the bilinear form estimate from type 2, we get the estimate

$$\left(\sum_{m\sim A} a(m)^2\right)^{1/2} \left(\sum_{n\sim B} b(n)^2\right)^{1/2} \left(\frac{\log q}{\sqrt{q}}\right) \left(\sqrt{AB} + \left(\sqrt{A} + \sqrt{B}\right)\sqrt{q} + q\right).$$

Note that the correlation is as needed because we have checked it for the particular bilinear form $a_m b_n e(mn\alpha)$. Here, $b_n = \Lambda(n)$. Then,

$$\sum_{n\sim B} \Lambda(n)^2 \ll B \log B.$$

Next,

$$\sum_{m\sim A} a(m)^2 \ll \sum_{m\sim A} d(m)^2$$

Exercise 4.9. Show

$$\sum_{n\le x} d(n) \sim x \log x.$$

(write this as $\sum_{n\le x} \sum_{d|n} 1$ and interchange the two sums).

It turns out $\sum_{n\le x} d(n)^2 \sim Cx (\log x)^3$. We end up getting a bound from

$$\sum_{m\sim A} a(m)^2 \ll \sum_{m\sim A} d(m)^2 \ll Cx (\log x)^3 .$$

So, we have some loose ends which we shall address next time including

(1) thinking about these sum of divisor functions up to x,
(2) thinking through the type one bounds for the first and fourth terms,
(3) and putting these things all together.

5. 10/5/17

5.1. **Recap of last time.**

Recall that we have defined

$$S(\alpha) := \sum_{n \leq N} \Lambda(n) e(n).$$

Our goal is to bound these exponential sums. We assume

$$\left| \alpha - \frac{a}{q} \right| \leq \frac{1}{q^2}$$

for $(a, q) = 1$. We had a way of approaching this bound with exponential sums and bilinear forms. We used the combinatorial identity

$$\frac{-\zeta'}{\zeta}(s) = P(s) + \zeta'(s)M(s) - \zeta(s)M(s)P(s) + \left(\frac{-\zeta'}{\zeta}(s) - P(s) \right)(1 - \zeta(s)M(s))$$

where

$$P(s) = \sum_{n \leq V} \frac{\Lambda(n)}{n^s}.$$

and

$$M(s) = \sum_{n \leq U} \frac{\mu(n)}{n^s}.$$

The first three terms are type 1 sums, and the last term is a type 2 sum. Last time, we discussed the bound for the type 1 sums. We saw they were bounded by

$$\ll (\log N)^2 \left\{ \frac{N}{q} + q + UV + \frac{N}{U} \right\}.$$

For example, to bound the term $\zeta'(s)M(s)$, we had to bound

$$\sum_{n \leq U} \min \left(\frac{N}{n}, \frac{1}{||n\alpha||} \right).$$

We could split this into dyadic blocks and carry out the usual sum. When $1 \leq n \leq q$, we cannot split it into intervals of length q.

Exercise 5.1. For the blocks, we should take a dyadic sum over intervals $2^k q \leq n \leq 2^{k+1} q$, and then we should pay attention to the case $1 \leq n \leq q$, and we should get a bound around $q \log q$ or something like that for the sum of the first q terms.

At the end of last class, we were discussing the type 2 sums. There were many small things we needed to keep track of. We wrote the sum as

$$\sum_{n>V} \Lambda(n) \sum_{m>U, mn\leq N} \left(\sum_{d|m, d>U} \mu(d) \right) e(mn\alpha)$$

Last time, we divided this sum into dyadic intervals with $m \sim A$ and $n \sim B$.

Remark 5.2. We have to justify why the sum can be split the sum into dyadic blocks subject to the condition that $mn \leq N$.

We then bounded the above by

$$\left(\sum_{m\sim A} d(m)^2 \right)^{1/2} \left(\sum_{n\sim B} \Lambda(n)^2 \right)^{1/2} \left(\frac{\log q}{\sqrt{q}} \right) \left(\sqrt{AB} + \left(\sqrt{A} + \sqrt{B} \right) \sqrt{q} + q \right)$$
$$\ll \left(\sum_{m\sim A} d(m)^2 \right)^{1/2} (B \log B)^{1/2} \left(\frac{\log q}{\sqrt{q}} \right) \left(\sqrt{AB} + \left(\sqrt{A} + \sqrt{B} \right) \sqrt{q} + q \right)$$

5.2. **Bounding the sum of the divisor function.** We can see

$$\begin{aligned} \sum_{n\leq x} d(n) &= \sum_{a\leq x} \sum_{b\leq x/a} 1 \\ &= \sum_{a\leq x} (\frac{x}{a} + O(1)) \\ &= x \log x + O(x). \end{aligned}$$

This is a wasteful $O(1)$ when a is small. Dirichlet's idea was to deal with the hyperbola $ab = x$ and count $b \leq B$ and $a \leq A$. One could count points a certain portion of the hyperbola based on whether A or B is smaller on the outside or inside. When one carries this out, one gets an error term on the size of $A + B$, instead of x (with $A + B = x$. One can take $A = B = \sqrt{x}$. One ends up getting an error of $x \log x + (2\gamma - 1)\, x + O(\sqrt{x})$.

Exercise 5.3. Carry out Dirichlet's idea and check this error term.

Then, we can compute

$$\begin{aligned} d_k(n) &= \sum_{a_1\cdots a_k = n} 1 \\ &= \sum_{ab=n} d_{k-1}(b), \end{aligned}$$

and use induction. If we knew $\sum_{b\le y} d_{k-1}(b)$, we could then use the hyperbola method for $a \le A, ab \le B$ and choose the parameters A, B with $AB = x$.

5.3. **A second method for bounding the sum of the divisor function.** We now want a second method of calculating this. We are trying to bound

$$\zeta(s)^2 = \sum \frac{d(n)}{n^s}.$$

We have

$$\sum_{n\le x} d(n) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \zeta(s)^2 \frac{x^s}{s} ds$$

for $c > 1$.

Exercise 5.4. Show

$$\frac{1}{2\pi i}\int_{(c)} y^s \frac{ds}{s} = \begin{cases} 1 & \text{if } y > 1 \\ 0 & \text{if } y < 1 \end{cases}$$

(see davenport's book) where the path (c) means that from $c - i\infty$ to $c + i\infty$. Essentially, one can prove this by noting the integral is 0 for very small y, and there is only one pole at $y = 1$ which has residue 1.

When we expand

$$x^s = x\left(1 + (s-1)\log x + \cdots\right)$$

and

$$\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1).$$

The residue of the pole at $s = 1$ is

$$x \log x + (2\gamma - 1)\, x.$$

It would be useful, and can be done easily, to have some bounds for

$$|\zeta(s)| \ll (1 + |t|)^{1/2}$$

where $s = \sigma + i\tau, 0 \le \sigma \le 1$.

We are trying to bound

$$\begin{aligned}&\frac{1}{2\pi i}\int_{(c)} \zeta(s)^2 \frac{x^s}{s} ds \\ &= \frac{1}{2\pi i}\int_{c-iT}^{c+iT} \zeta(s)^2 \frac{x^s}{s} ds + O(\frac{x^c}{T}).\end{aligned}$$

We can then try to bound this integral by something like $x \log x + (2\gamma - 1)\, x$, similarly to the way done in Davenport's book.

In potential hope of formalizing this method, we are trying to we want to bound

$$\sum_{n \le x} d_k(n),$$

and we consider

$$\zeta(s)^k = \sum_{n=1}^{\infty} \frac{d_k(n)}{n^s}.$$

We then can compute these by examining

$$\frac{1}{2\pi i} \int_{(c)} \frac{x^s}{s} \zeta(s)^k ds.$$

So, we want to know vaguely what the residue of this integral at $s = 1$. The residue is something like

$$\frac{x\,(\log x)^k}{(k-1)!}$$

with some lower order terms we can work out. The main term will be a polynomial of $\log x$ of degree $k - 1$.

Exercise 5.5. Show we end up getting a residue of the form

$$xP_k(\log x) + O(x^{1-\delta_k})$$

for P_k a degree $k - 1$ polynomial.

Remark 5.6. Gauss should that the number of lattice points in a circle of radius R is

$$N(R) = \pi R^2 + O(R^{1/2+\varepsilon}).$$

(the best currently known is only error $R^{2/3-\delta}$ Dirichlet's divisor problem is to show

$$\sum_{n \le x} d(n) = x \log x + (2\gamma - 1)\, x + O(x^{1/4+\varepsilon}).$$

In both cases, the main term is the area of the region you are considering. The best error known is only about $O(x^{1/3-\delta})$.

5.4. **Calculating the sum of squares of the divisor function.** Now, we'd like to calculate

$$\sum_{n \leq x} d(n)^2.$$

We will instead calculate

$$\sum_{n=1}^{\infty} \frac{d(n)^2}{n^s} = \prod_p \left(1 + \frac{4}{p^s} + \frac{9}{p^{2s}} + \cdots\right).$$

Note that this will converge absolutely whenever we are to the right of 1. We have

$$\zeta(s) = \prod_p \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots\right).$$

We can approximate

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{d(n)^2}{n^s} &= \prod_p \left(1 + \frac{4}{p^s} + \frac{9}{p^{2s}} + \cdots\right) \\ &= \zeta(s)^4 F(s)\end{aligned}$$

where

$$F(s) = \prod_p \left(1 + \frac{\alpha}{p^{2s}} + \cdots\right)$$

for some α, which converges absolutely if $\mathrm{Re}(s) > 1/2$. One then obtains that

$$\sum_{n \leq x} d(n)^2 \sim Cx \left(\log x\right)^3,$$

using the bound for ζ^4 as $xP_k(\log x)$ with P_k of degree $k-1$. with an asymptotic power saving error term. We could similarly use the hyperbola method to approximate

$$d(n)^2 = \left(d_4 * f\right)(n)$$

for a multiplicative function f with $f(p) = 0$.

Remark 5.7. When one actually calculates what

$$\sum_n \frac{d(n)^2}{n^s}$$

one might find something like

$$\frac{\zeta(s)^4}{\zeta(2s)}$$

although this identity is not relevant to finding the correct asymptotic formula.

Exercise 5.8 (Fun exercise!)**.** Let $a(n)$ be the number of abelian groups of order n. First make a guess for

$$\sum_{n \leq x} a(n)$$

asymptotically. Then compute the asymptotics. *Hint:* Use the identity for the partition function

$$\sum_{n=0}^{\infty} p(n) x^n = \prod (1 - x^n)^{-1}$$

to get the constant in the asymptotics, which ends up being something like $\zeta(2) \cdot \zeta(3) \cdots$.

Remark 5.9. One might also try computing

$$\sum_{n \leq x} d_\pi(n)$$

$$\sum_{n \leq x} d_i(n)$$

where $d_\pi(n)$ are the coefficients of the Dirichlet series of $\zeta(s)^\pi$. Relatedly, one might try to count

$$\sum_{n \leq x, n = a^2 + b^2} 1,$$

and one can work out

$$\sum_{n = a^2 + b^2} 1 = \zeta(s)^{1/2} L(s, \chi_{-4})^{1/4} F(s),$$

where $f(s)$ is regular to the left of 1.

It's not completely obvious how these functions continue analytically. We could make sense of

$$\zeta(s) = \exp\left(\pi \log \zeta(s)\right),$$

which makes sense to the right of 1. But, it also can be extended to regions where there are no zeros or poles of the ζ function. If we understand the zero-free region of the zeta function, then we can make sense of this function in this zero-free region.

In the region

$$\gamma > 1 - \frac{c}{\log T},$$

$\zeta(s) \neq 0$ as shown in davenport.

It turns out this function has a singularity which is not a pole (nor essential nor removable) and it turns out to be something like

$$\frac{x(\log x)^{\pi-1}}{\Gamma(\pi)}$$

for the function $d_\pi(n)$.

This idea is called the Selberg, Delange method (or in a paper today on arXiv, the LSD method).

Remark 5.10. We only wanted an upper bound for

$$\sum_{n \leq x} d(n)^2.$$

We didn't need an asymptotic. In analytic number theory, this is called Rankin's method. We can bound

$$\begin{aligned}
\sum_{n \leq x} d(n)^2 &\leq x^\alpha \sum_{n \leq x} \frac{d(n)^2}{n^\alpha} \\
&\leq x^\alpha \sum_{n=1}^{\infty} \frac{d(n)^2}{n^\alpha} \\
&= x^\alpha \prod_p \left(1 + \frac{4}{p^\alpha} + \cdots\right)
\end{aligned}$$

Then, we want to optimize to choose the best α. Making α close to 1, the product blows up and x^α gets small. From calculus, there will be some choice of α which minimizes this product.

For example, if you guess $\alpha = 1 + \frac{1}{\log x}$, you find x^α is about x and

$$\prod_p \left(1 + \frac{4}{p^\alpha} + \cdots\right) \sim \zeta\left(1 + \frac{1}{\log x}\right)^4.$$

This yields $x(\log x)^4$ as a bound, and you are only off by one log.

Exercise 5.11. Verify this.

Exercise 5.12. Let $p(n)$ denote the number of partitions of n. Prove that

$$p(n) \leq \exp\left(\pi\sqrt{2/3n}\right).$$

Moreover, find the optimal constant so that $p(n) \leq e^{\alpha n}$. (Sound thinks the constant above is optimal).

Hardy and Ramanujan found

$$p(n) \sim \frac{\exp\left(\pi\sqrt{2/3n}\right)}{4n\sqrt{3}}.$$

Hint: Show

$$\sum_{n=1}^{\infty} \prod (1-x^n)^{-1}$$

Then,

$$p(N) \le \prod (1-x^n)^{-1} x^{-N}.$$

Exercise 5.13 (Fun mathoverflow problem)**.** Let N be a parameter. How many subsets

$$S \subset [1, N]$$

are there so that

$$\sum_{s \in S} \frac{1}{s} < 1.$$

Obviously the answer is $\le 2^N$, and the exercise is to find a better bound. *Hint:* This is not an application of what we've discussed, but it is an application of the *ideas* we've discussed.

5.5. **Returning to our type 2 sum.** Recall we had $A > U, B > V$. We can now bound

$$\begin{aligned}
&\left(\sum_{m \sim A} d(m)^2\right)^{1/2} \left(\sum_{n \sim B} \Lambda(n)^2\right)^{1/2} \left(\frac{\log q}{\sqrt{q}}\right) \left(\sqrt{AB} + \left(\sqrt{A} + \sqrt{B}\right)\sqrt{q} + q\right) \\
&\ll \left(A(\log A)^3\right)^{1/2} (B \log B)^{1/2} \left(\frac{\log q}{\sqrt{q}}\right) \left(\sqrt{AB} + \left(\sqrt{A} + \sqrt{B}\right)\sqrt{q} + q\right) \\
&\ll (\log N)^3 \left\{\frac{AB}{\sqrt{q}} + A\sqrt{B} + B\sqrt{A} + \sqrt{qAB}\right\} \\
&\ll (\log N)^3 \left\{\frac{N}{\sqrt{q}} + \frac{N}{\sqrt{V}} + \frac{N}{\sqrt{U}} + \sqrt{qN}\right\}.
\end{aligned}$$

using for the last step that $AB \le N$. We are doing well here because both variable A and B vary only in long ranges (i.e., U and V are reasonably large). We then have to add the error from the type 1 sum which was

$$\ll (\log N)^2 \left\{\frac{N}{q} + q + UV + \frac{N}{U}\right\}.$$

Adding these together, we get

$$S(\alpha) \ll (\log N)^3 \left\{ \frac{N}{\sqrt{q}} + \frac{N}{\sqrt{V}} + \frac{N}{\sqrt{U}} + \sqrt{qN} + UV \right\}.$$

By symmetry, we may as well choose $U = V$, and so we should optimize by choosing $N/\sqrt{U} = U^2$. Hence, $U = N^{2/5}$. Then, one obtains

$$S(\alpha) \ll (\log N)^3 \left\{ \frac{N}{\sqrt{q}} + \sqrt{qN} + N^{4/5} \right\}.$$

There is one small caveat, where we must ensure how to separate the variables m and n subject to the condition $mn \leq N$. We'll have to finish this next time. Believing this for the moment, we've proven. So, if

$$\frac{N}{(\log N)^{10}} > q > (\log N)^{10}.$$

So, as long as we can approximate α by some rational q in this range, we get a good bound. These will be called the "minor arcs."

Theorem 5.14. *Let ϕ be the golden ration. Then,*

$$\sum_{n \leq N} \Lambda(n) e(n\phi) \ll N^{4/5} (\log N)^3.$$

Proof. We can plug in q to be around $\sqrt{N}$ using Fibonacci number approximations plugged into the above formula, and then the bound is $(\log N)^3 N^{4/5}$. □

Remark 5.15. The bound also works well for bounding things like

$$\sum_{n \leq N} \Lambda(n) e^{in}$$

using that

$$\left| \frac{1}{\pi} - \frac{a}{q} \right| \geq \frac{C}{q^{20}},$$

so one can always find a pretty good approximation to π. So, we get a bound of about

$$\sum_{n \leq N} \Lambda(n) e^{in} \ll N^{.99}$$

6. 10/10/17

6.1. **Exercises and questions.**

Last time, we let q be a number with

$$\left|\alpha - \frac{a}{q}\right| < \frac{1}{q^2}$$

$$\sum_{n \leq N} \Lambda(n) e(n\alpha) \ll (\log N)^3 \left(\frac{N}{\sqrt{q}} + \sqrt{qN} + N^{4/5} \right).$$

Exercise 6.1. Let ϕ be the Euler totient function, let

$$\sum_{n \leq N} \left(\frac{\phi(n)^k}{n} \right)$$

Find asymptotics for this. Why might these asymptotics be interesting.

6.2. **Recapping what we have seen in the Proof of Vinogradov's theorem.**

Last time, we had some sum in terms of $m \sim A, n \sim B$ with a condition $mn \leq N$. We want to separate m and n. We have

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \left(\frac{N}{mn} \right)^s \frac{ds}{s} = \begin{cases} 1 & \text{if } mn \leq N \\ 0 & \text{if } mn > N \end{cases}$$

When we plug this into our bilinear form

$$\sum_{m \sim A} \sum_{n \sim B} f_1 f_2 e(mn\alpha),$$

(for appropriate f_1, f_2) we get

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \sum_{m \sim A} \sum_{n \sim B} \frac{f_1 f_2 e(mn\alpha)}{m^s n^s} N^s \frac{ds}{s}.$$

This separates the variables at the cost of $\log N$ when we integrate $\frac{ds}{s}$.

Question 6.2 (Possibly open question)**.** We have

$$\sum_{n \leq N} \Lambda(n) e(n\phi) \ll N^{4/5+\varepsilon},$$

for ϕ the golden ratio. Can one say something better? Presumably the right answer is $N^{1/2}$, though that may be hard. Maybe one could show something like $N^{3/4}$. The key is that we have rational approximations at every scale.

Recapping what we have done so far, we were trying to bound

$$\int_0^1 S(\alpha)^3 e(-N\alpha) d\alpha.$$

We split this up into major and minor arcs. On the minor arcs, we bounded this by

$$\left| \int_m S(\alpha)^3 e(-N\alpha) d\alpha \right| < \left(\max_{\mathfrak{m}} |S(\alpha)| \right) N \log N.$$

We expect a main term on the order of N^2. We have a good bound on $S(\alpha)$ so long as

$$\frac{N}{(\log N)^{10}} \geq q \geq (\log N)^{10}$$

The minor arcs will be all points which satisfy an approximation of this type, and the major arcs will be all points which do not satisfy an approximation of this type.

Let $Q = \frac{N}{(\log N)^{10}}$. By Dirichlet's theorem, for all $\alpha \in (0,1)$, there exists $(a,q) = 1, q \leq Q$ and

$$\left| \alpha - \frac{a}{q} \right| \leq \frac{1}{qQ} \leq \frac{1}{q^2}.$$

Definition 6.3. We say $\alpha \in \mathfrak{m}$ (in a **minor arc**) if there exists such an approximation with

$$q \geq (\log N)^{10}.$$

Otherwise, there exists $\alpha \in \mathfrak{M}$ (in a **major arc**).

That is,

$$\left| \alpha - \frac{a}{q} \right| \leq \frac{1}{qQ}$$

with $q < (\log N)^{10}$. The major arcs $\mathfrak{M}$ are disjoint. The total measure of the major arcs is

$$|\mathfrak{M}| = \sum_{q \leq (\log N)^{10}} \frac{\phi(q)2}{qQ} \sim \frac{C (\log N)^{10}}{Q},$$

which is roughly $(\log N)^{20}/N$.

We now wish to understand $S(\alpha)$ for α on a major arc. Let $\alpha = \frac{a}{q} + \beta$ for q small, $|\beta| \leq \frac{1}{qQ}$. The idea is to understand $S(\frac{a}{q})$. Let's instead try to understand

$$\sum_{n \leq x} \Lambda(n) \exp\left(\frac{an}{q}\right).$$

6.3. **Riemann hypothesis and counting primes.** To start, let us recall what the Riemann hypothesis says about counting the number of primes up to x. Let $\Psi(x)$ be the number of primes up to x. It implies

$$\Psi(x) = x + O\left(x^{1/2+\varepsilon}\right).$$

If one further assumes the generalized Riemann hypothesis, one finds

$$\Psi\,(x; q, a) = \frac{x}{\phi(q)} + O(x^{1/2+\varepsilon}).$$

Further, the constant in $O(x^{1/2+\varepsilon})$ is independent of q. In particular, this means $\phi(q) \sim q$, Thus, we have a nice asymptotic for $q \leq x^{1/2-\varepsilon}$.

Conjecture 6.4 (Montgomery)**.** We have

$$\Psi(x; q, a) = \frac{x}{\phi(q)} + O\left(\frac{x^{1/2+\varepsilon}}{\sqrt{q}}\right).$$

Plugging in the generalized Riemann hypothesis, we get

$$\begin{aligned}
\sum_{n \leq x} \Lambda(n) \exp\left(\frac{na}{q}\right) &= \sum_{k \bmod q}^{*} \sum_{n \equiv k \bmod q, n \leq x} \Lambda(n) \exp\left(\frac{ak}{q}\right) \\
&= \sum_{k \bmod q}^{*} \exp\left(\frac{ak}{q}\right) \left(\frac{x}{\phi(q)} + O(x^{1/2+\varepsilon})\right) \\
&= \frac{\mu(q)}{\phi(q)} x + O(q x^{1/2+\varepsilon})
\end{aligned}$$

where the superscript $*$ again means k is coprime to q and we are using

Exercise 6.5. Show $\sum_{k \bmod q}^{*} \exp\left(\frac{k}{q}\right) = \mu(q)$.

Now, suppose $(n, q) = 1$ and we want to express $\exp(n/q)$ in terms of characters $\chi \bmod q$.

Letting χ_0 denote the identity on $(\mathbb{Z}/q\mathbb{Z})^\times$ We can consider

$$\begin{aligned}\sum_{k \bmod q}^{*} \exp\left(\frac{k}{q}\right)\chi_0 &= \sum_{k \bmod q}^{*} \exp\left(\frac{k}{q}\right)\frac{1}{\phi(q)}\sum_{\chi \bmod q}\chi(k)\overline{\chi(n)} \\ &= \frac{1}{\phi(q)}\sum_{\chi \bmod q}\overline{\chi(n)}\tau(\chi),\end{aligned}$$

where

$$\tau(\chi) = \sum_{k \bmod q}\chi(k)\exp\left(\frac{k}{q}\right).$$

Then,

$$\sum_{n \leq x}\Lambda(n)\exp\left(\frac{an}{q}\right) = \frac{1}{\phi(q)}\sum_{\chi \bmod q}\tau(\chi)\sum_{n\leq x}\Lambda(n)\overline{\chi}(an).$$

Define

$$\psi(x;\chi) := \sum_{n\leq x}\Lambda(n)\chi(n).$$

Then,

$$\begin{aligned}\Psi\,(x;q,a) &= \sum_{n\leq x, n\equiv a \bmod q}\Lambda(x) \\ &= \frac{1}{\phi(q)}\sum_{\chi \bmod q}\overline{\chi}(a)\Psi(x,\chi).\end{aligned}$$

The generalized Riemann hypothesis (GRH) is essentially the statement that for $\chi = \chi_0$, we have $\Psi(x,\chi) = \Psi(x)$ up to a small error (which is just the usual Riemann hypothesis) and for $\chi \neq \chi_0$, we have

$$|\Psi(x,\chi)| = O(x^{1/2+\varepsilon}).$$

In the case $\chi = \chi_0$, we get the main term with

$$\tau(\chi_0) = \sum_{k \bmod q}^{*}\exp\left(\frac{k}{q}\right) = \mu(q).$$

So, the main term is

$$\frac{\mu(q)}{\phi(q)}\Psi(x).$$

Exercise 6.6 (A bit tricky, perhaps)**.** Using orthogonality of characters show that if χ is primitive mod$q$ (meaning not having a period dividing q) then $|\tau(\chi)| = \sqrt{q}$. *Hint:* See Davenport's section on Gauss sums.

Plugging this in the above formulas and the GRH bounds, we see a refined GRH bound

$$\sum_{n \le x} \Lambda(n) \exp\left(\frac{an}{q}\right) = \frac{\mu(q)}{\phi(q)} \left(x + O\left(x^{1/2+\varepsilon}\right)\right) + O\left(\sqrt{q} x^{1/2+\varepsilon}\right).$$

So, compared to our previous error bound with $O(qx^{1/2+\varepsilon})$ error, we only get $O(\sqrt{q}x^{1/2+\varepsilon})$.

We are not assuming GRH, rather we want an unconditional proof, so the above discussion assuming GRH can now be ignored.

We have

$$\begin{aligned} \sum_{n \le x} \Lambda(n) \exp\left(\frac{an}{q}\right) &= \frac{1}{\phi(q)} \sum_{\chi \bmod q} \overline{\chi}(a) \tau(\chi) \psi(x, \overline{\chi}) \\ &= \frac{\mu(q)}{\phi(q)} \Psi(x, \chi_0) + O\left(\frac{\sqrt{q}}{\phi(q)} \sum_{\chi \neq \chi_0} |\Psi(x, \overline{\chi})|\right). \end{aligned}$$

We have

$$\Psi(x, \chi_0) = \Psi(x) + O\left((\log x)^2\right).$$

From the prime number theorem, we have

$$\Psi(x) = x + O\left(x \exp\left(-c\sqrt{\log x}\right)\right)$$

for some $c > 0$. The key step in the proof of this is that the region $\sigma > 1 - \frac{c}{\log 2+|t|}$ has no zeros of the zeta function $\zeta(s)$, where $s = \sigma + it$.

We therefore get a bound

$$\begin{aligned} \Psi(x, \chi_0) &= \Psi(x) + O(\log x)^2 \\ &\sim x - \sum_{|\rho| \le T} \frac{x^\rho}{\rho} + O\left(\frac{x}{T}\right). \end{aligned}$$

In the best case, we might have

$$\Psi(x; \chi) \ll x \exp\left(-c\sqrt{\log x}\right),$$

for $\chi \neq \chi_0 \bmod q$ and $q \le \exp\left(\sqrt{\log x}\right)$.

The conclusion is that if

$$q \leq \exp\left(c\sqrt{\log x}\right)$$

for c small, then

$$\sum_{n \leq x} \Lambda(n) \exp\left(\frac{an}{q}\right) = \frac{\mu(q)}{\phi(q)} x + O\left(x \exp\left(-c\sqrt{\log x}\right)\right).$$

In a similar way, one would like to show

$$\psi(x; q, a) = \frac{x}{\phi(q)} + O\left(x \exp\left(-c\sqrt{\log x}\right)\right).$$

The short version of the story is that we can basically do this, but with one important caveat, which is called a **Landau-Siegel zero**.

6.4. **Siegel Zeros.** We want to understand $\Psi(x;\chi)$. If $\chi \neq \chi_0$, we have

$$\Psi(x, \chi) = -\sum_{\rho_\chi, |\rho_\chi| \leq T} \frac{x^{\rho_\chi}}{\rho_\chi} + O\left(\frac{x\,(\log x)^2}{T}\right)$$

where ρ_χ are the zeros of

$$L(s, \chi) := \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

One can find proofs of all of these things in Davenport. If χ is primitive then $L(s, \chi)$ has a functional equation of the form

$$\left(\frac{q}{\pi}\right)^{s/2} \Gamma\left(\frac{s+\alpha}{2}\right) L(s, \chi)$$

where α is either 0 or 1 depending on whether $\chi(-1) = 1$, (so $\alpha = 0$) or $\chi(-1) = -1$ (so $\alpha = 1$). This yields the volume at $1 - s$. One can count the number of zeros of $\zeta(s)$ or $L(s, \chi)$ up to height T, which is approximately

$$\frac{T \log qT}{2\pi}.$$

It is also useful to know the Hadamard factorization, which, once you know this is an order 1 function, tells you this has a factorization in terms of its zeros. That is,

$$L(s, \chi) = \left(\prod_{\rho} \left(1 - \frac{s}{\rho} e^{s/\rho}\right)\right) e^{A+Bs}.$$

So the sum of the reciprocals of the squares of the 0's converge, but possibly not the sum of the reciprocals of the 0's.

For the zeta function we have,

$$\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

which kills the pole at $s = 1$. This satisfies the functional equation

$$\xi(s) = \xi(1-s).$$

The main difference between the L functions and the ζ function, we will need to know something about the zero free region. For $\zeta(s)$, there is a zero free region of the form

$$\sigma > 1 - \frac{c}{\log 2 + |t|},$$

with $\sigma = \operatorname{im} s$. We want

$$\sigma > 1 - \frac{c}{q(\log 2 + |t|)},$$

is free of zeros of $L(s, \chi)$. This would imply

$$\Psi(x, \chi) = O\left(x \exp\left(-c\sqrt{\log x}\right)\right)$$

for $q \leq \exp\left(c\sqrt{\log x}\right)$. This holds if χ is a complex character $\bmod q$.

But for quadratic characters $\chi \bmod q$, there is the unfortunate possibility that there could be one exceptional real simple zero β

Theorem 6.7 (Siegel). *Let β be the possible zero of $L(s, \chi)$ as above. Then,*

$$\beta < 1 - \frac{C(\varepsilon)}{q^{\varepsilon}}$$

for any $\varepsilon > 0$ for some constant $C(\varepsilon)$ which cannot be computed (i.e., the proof is ineffective).

Next time we'll say a bit more about Siegel's theorem. It might be helpful to review things about the prime number theorem in progressions which we will go over as needed on Thursday.

Sound also says he is happy to look at or discuss solutions if you do end up solving problems.

7. 10/12/17

7.1. **Review.** Let χ be some character $\mathbb{Z}/q\mathbb{Z} \to \mathbb{C}^{\times}$. We have

$$\Psi(x, \chi) = \sum_{n \leq x} \Lambda(n)\chi(n).$$

If $\chi = \chi_0$, we have

$$\Psi(x) + O\left((\log x)^2\right) = x + O\left(x \exp\left(-c\sqrt{\log x}\right)\right).$$

If $\chi \neq \chi_0$, GRH implies

$$|\Psi(x,\chi)| \ll x^{1/2+\varepsilon}.$$

We would like an unconditional bound around

$$|\Psi(x,\chi)| \ll x \exp\left(-c\sqrt{\log x}\right) \tag{7.1}$$

and we would like to say when $q \leq \exp\left(\sqrt{\log x}\right)$. We have

$$\sum_{n \leq x} \Lambda(n) \exp\left(\frac{an}{q}\right) = \frac{1}{\phi(q)} \sum_{\chi \bmod q} \tau(\chi)\overline{\chi}\,(a)\,\Psi(x,\overline{\chi}).$$

The main term comes from $\chi = \chi_0$ where $\tau(\chi_0) = \mu(q)$, using an exercise on computing Gauss sums from last time. The error term, assuming GRH is of the form $q^{1/2} + x^{1/2+\varepsilon}$. If Equation 7.1 holds, then we can bound

$$\sum_{n \leq x} \Lambda(n) \exp\left(\frac{an}{q}\right) \ll \frac{\mu(q)}{\phi(q)} + O\left(x \exp\left(-c\sqrt{\log x}\right)\right)$$

in the range $q \leq \exp\left(c\sqrt{\log x}\right))$.

We don't actually know Equation 7.1, but for our application to sums of three primes, we thought of q as only going up to $(\log x)^{10}$ and not all the way to $\exp\left(c\sqrt{\log x}\right)$.

If χ is complex, (i.e., not a real character) then

$$|\Psi(x,\chi)| \ll x \exp\left(-c\sqrt{\log x}\right)$$

for $q \leq \exp\left(c\sqrt{\log x}\right)$ then there are no zeros of $L(s,\chi)$ for

$$\sigma > 1 - \frac{c}{\log q(2+|t|)}.$$

with $\sigma = \operatorname{im} s$. If instead χ is real or quadratic, then the zero free region above holds except possibly for one real simple zero.

Theorem 7.1 (Siegel)**.** *The real zero β (if it exists) must satisfy*

$$\beta < 1 - \frac{C(\varepsilon)}{q^{\varepsilon}}$$

for any $\varepsilon > 0$ and some ineffective constant $C(\varepsilon)$.

Remark 7.2. If the zero does not exist, then we can obtain Equation 7.1. If there does exist a Siegel zero for χ mod q, then

$$\Psi(x,\chi) = \frac{-x^{\beta}}{\beta} + O\left(x \exp\left(-c\sqrt{\log x}\right)\right)$$

If $q \leq (\log x)^A$, we can choose ε small enough, we can ensure $\beta < 1 - \frac{C}{\sqrt{\log x}}$, and then we can absorb the main term for $\Psi(x,\chi) - \frac{x^\beta}{\beta}$ into the error term. In the presence of the Siegel zero, we can only get this uniform desired result for $q \leq (\log x)^A$, but not for $q \leq \exp\left(c\sqrt{\log x}\right)$. This is also ineffective.

Therefore, we obtain

$$\sum_{n \leq x} \Lambda(n) \exp\left(\frac{an}{q}\right) \ll \frac{\mu(q)}{\phi(q)} x + O\left(x \exp\left(-c\sqrt{\log x}\right)\right).$$

Remark 7.3. Suppose β is very close to 1. Pretend $\beta = 1$. Then there is one character χ mod q so that $\Psi(x,\chi)$ is approximately $-x$.

If you think of

$$\Psi(x;q,a) = \frac{1}{\phi(q)} \sum \overline{\chi}(a)\psi(x,\chi) = \frac{x}{\phi(q)} - \frac{x^\beta}{\beta}\frac{\chi(a)}{\phi(q)}.$$

and here χ is real so $\chi(a) = \overline{\chi}(a)$. Then, half of the progression get most of the primes and the other half get none of them (this happens depending on whether $\chi(a) = \pm 1$).

7.2. **Proving Vinogradov's theorem using Siegel's theorem.** For the moment, we'll assume Siegel's theorem and finish the proof of Vinogradov's theorem. We'll later come back to discuss Siegel's theorem.

We have seen that if

$$q \leq (\log x)^A$$

(A is around 10) then

$$\sum_{n \leq x} \Lambda(n) \exp\left(\frac{an}{q}\right) = \frac{\mu(q)}{\phi(q)} + O\left(x \exp\left(-c\sqrt{\log x}\right)\right).$$

The major arcs are of the form

$$\left|\alpha - \frac{a}{q}\right| \leq \frac{1}{qQ},$$

with

$$Q = \frac{N}{(\log N)^{10}}$$

for $q \leq (\log N)^{10}$. Recall we have already bounded the minor arcs, and we are now trying to bound the major arcs. Set $\alpha = \frac{a}{q} + \beta$. We would like to understand

$$S(\alpha) := \sum_{n \le N} \Lambda(n) \exp\left(\frac{an}{q}\right) \exp(n\beta).$$

We can think of the the product of $\Lambda(n)\exp\left(\frac{an}{q}\right)$ whose partial sums we understand and $\exp(n\beta)$ which doesn't vary very much. So,

$$\begin{aligned}
S(\alpha) &= \sum_{n \le N} \Lambda(n) \exp\left(\frac{an}{q}\right) \exp(n\beta) \\
&= \int_1^N \exp(x\beta) d\left(\sum_{n \le x} \Lambda(n) \exp\left(\frac{an}{q}\right)\right) \\
&= \frac{\mu(q)}{\phi(q)} \int_1^N \exp(x\beta) dx + O\left(N \exp\left(-c\sqrt{\log N}\right)\right) + O\left(\int_1^N \beta x \exp\left(-c\sqrt{\log x}\right) dx\right) \\
&= O\left(\left(1 + N|\beta| N \exp\left(-c\sqrt{\log N}\right)\right)\right) \\
&= O\left(N \exp\left(-c\sqrt{\log x}\right)\right).
\end{aligned}$$

Where we used integration by parts to get the above bounds on the error terms, and then we used that $\beta \le \frac{1}{qQ}$, and we might have to adjust the constant c to absorb some factors of $\log N$.

Remark 7.4. The above bound makes sense: If β is very close to $\frac{a}{q}$, we pick up the same error term we had before. But if β is very far, then the error term should group approximately proportionally to $N|\beta|$, which indeed it does.

We now want to evaluate the major arc contribution

$$\int_{\mathfrak{M}} S(\alpha)^3 e(-N\alpha)\, d\alpha.$$

We are hoping this is of size $N^2 \cdot C$ with C some constant we can evaluate.

Indeed,

$$\int_{\mathfrak{M}} S(\alpha)^3 e(-N\alpha)\, d\alpha = \sum_{q \le (\log N)^{10}} \sum_{a \bmod q}^{*} \int_{-1/qQ}^{1/qQ} S\left(\frac{a}{q} + \beta\right)^3 \exp\left(-N\left(\frac{a}{q} + \beta\right)\right) d\beta.$$

We know

$$S\left(\frac{a}{q} + \beta\right)^3 = \frac{\mu(q)}{\phi(q)^3} \left(\int_0^N \exp(x\beta)\, dx\right) + O\left(N^3 \exp -c\sqrt{\log N}\right)$$

The error term in the integral over the major arcs is then

$$O\left(\sum_{q\le(\log N)^{10}} N^3\exp\left(-c\sqrt{\log N}\right)\frac{1}{Q}\right) = O\left(N^2\exp\left(-c\sqrt{\log N}\right)\right).$$

So the error terms are under control. We now want to understand the main term. The main term is almost independent of a except for the factor $\exp\left(-N\left(\frac{a}{q}+\beta\right)\right)$. We want to understand the main term of

$$\int_{\mathfrak{M}} S(\alpha)^3 e\left(-N\alpha\right) d\alpha.$$

which is

$$\sum_{q\le(\log N)^{10}} \sum_{a \bmod q}^{*} \int_{-1/qQ}^{1/qQ} \frac{\mu(q)}{\phi(q)^3}\left(\int_0^N \exp\left(x\beta\right)dx\right)\exp\left(-N\left(\frac{a}{q}+\beta\right)\right)d\beta.$$

Recall the **Ramanujan sum**

$$c_q(N) := \sum_{(a,q)=1}\exp\left(\frac{aN}{q}\right).$$

The main term is then

$$\sum_{q\le(\log N)^{10}} \frac{\mu(q)}{\phi(q)^3}c_q(N)\int_{-1/qQ}^{1/qQ}\left(\int_0^N \exp\left(x\beta\right)dx\right)^3 e\left(-N\beta\right)d\beta.$$

We can now replace

$$\int_0^N \exp(x\beta)dx = N\int_0^1 \exp\left(Nx\beta\right)dx.$$

This yields

$$\begin{aligned}
&\int_{-1/qQ}^{1/qQ}\left(\int_0^N \exp\left(x\beta\right)dx\right)^3 e\left(-N\beta\right)d\beta\\
&= N)\int_{-1/qQ}^{1/qQ} N^3\left(\int_0^1 \exp\left(Nx\beta\right)dx\right)^3 e\left(-N\beta\right)d\beta\\
&= N^2\int_{-N/qQ}^{N/qQ}\left(\int_0^1 \exp\left(Nx\beta\right)dx\right)^3 e\left(\beta\right)d\beta\\
&= N^2\int_{-\infty}^{\infty}\left(\int_0^1 \exp\left(Nx\beta\right)dx\right)^3 e\left(\beta\right)d\beta + O\left(\frac{q^2Q^2}{N^2}\right)
\end{aligned}$$

where the last step uses that the tail is

$$\int_{|\beta|>N/qQ} O\left(\frac{d\beta}{\beta^3}\right) = O\left(\frac{q^2Q^2}{N^2}\right).$$

This integral above is called the **singular integral**. Plugging in this remainder term, we get that the error contribution is

$$\begin{aligned} O\left(Q^2 \sum_{q \leq (\log N)^{10}} \frac{1}{\phi(q)^3}\phi(q)q^2\right) &= O\left(Q^2 (\log N)^{10}\right) \\ &= O\left(\frac{N^2}{(\log N)^{10}}\right). \end{aligned}$$

So, we can replace our integral from $-N/qQ$ to N/qQ by an integral going off to infinity. This integral is essentially computing the number of ways to write N as a sum of three numbers, which is essentially $N^2/2$. But, we can also compute it since this is essentially a Fourier transform. That is,

$$N^2 \int_{-\infty}^{\infty} \left(\int_0^1 \exp\left(Nx\beta\right) dx\right)^3 e\left(\beta\right) d\beta$$

is the convolution of $\chi_{[0,1]} * \chi_{[0,1]} * \chi_{[0,1]}$ which has Fourier transform above, and for this convolution we get

$$\int_{t_1,t_2,t_3 \in [0,1]} \delta\left(t_1 + t_2 + t_3 = 1\right).$$

Then one can use Parseval's identity to compute the Fourier transform of this. Here, Parseval is counting the number of ways of writing 1 as a sum of three real numbers. Before we were writing N as a sum of three integers.

Now, let's finish our calculation. We were trying to compute

$$\sum_{q \leq (\log N)^{10}} \frac{\mu(q)}{\phi(q)^3} c_q(N) \int_{-1/qQ}^{1/qQ} \left(\int_0^N \exp\left(x\beta\right) dx\right)^3 e\left(-N\beta\right) d\beta.$$

which is approximated, using our above discussion, by

$$\sum_{q \leq (\log N)^{10}} \frac{\mu(q)}{\phi(q)^3} c_q(N) \frac{N^2}{2}.$$

This sum is called the **singular sum** The tail of this sum is roughly

$$\sum_{q>(\log N)^{10}} O\left(\frac{1}{\phi(q)}^2\right) \ll (\log N)^{-10}.$$

Therefore, the main term is of the form

$$\frac{N^2}{2}\sum_{q=1}^{\infty}\frac{\mu(q)}{\phi(q)^3}c_q(N).$$

Let

$$\mathscr{S}(N) := \sum_{q=1}^{\infty}\frac{\mu(q)}{\phi(q)^3}c_q(N).$$

We can write, using the Chinese remainder theorem so that $c_{p_1p_2}(N) = c_{p_1}(N)c_{p_2}(N)$. So we have

$$\mathscr{S}(n) = \prod_p\left(1 - \frac{1}{(p-1)^s}c_p(n)\right).$$

Then,

$$\begin{aligned} c_p(N) &= \sum_{a=1}^{p-1}\exp\left(\frac{aN}{p}\right) \\ &= \begin{cases} -1 & \text{if } p \nmid N \\ p-1 & \text{if } p \mid N \end{cases} \end{aligned}$$

Then,

$$\begin{aligned} \mathscr{S}(n) &= \prod_p\left(1 - \frac{1}{(p-1)^s}c_p(n)\right) \\ &= \prod_{p\nmid N}\left(1 + \frac{1}{(p-1)^3}\cdot\prod_{p\mid N}\left(1 - \frac{1}{(p-1)^2}\right)\right). \end{aligned}$$

Remark 7.5. We see this cancels out when N is even. When N is even, the major arc at 0 is cancelled by the major arc at $1/2$. And in general, the major arc at a/q is canceled by a similar one at $a/2q$.

Finally, we have

$$\begin{aligned}\int_0^1 S(\alpha)^3 e\left(-N\alpha\right) &= \sum_{n_1+n_2+n_3=N} \\ &= \frac{N^2}{2}\mathscr{S}(N) + O\left(\frac{N}{\log N}\right).\end{aligned}$$

Because the contribution of prime squares and cubes is negligible, we get that every sufficiently large odd number is the sum of three primes (and in fact it is the sum of three primes in many ways), where here we are using that

$$\mathscr{S}(N) \geq c$$

for all N where c is some universal constant bounded below by

$$2 \cdot \prod_{p \geq 3} \left(1 - \frac{1}{(p-1)^2}\right).$$

This finishes the proof.

Remark 7.6 (Philosophy)**.** Under suitable situation, we'd like to say we can get an answer by counting contributions at each place and then multiplying them together.

For example, say we'd like to count the number of ways to write $2N = p_1 + p_2$. We can try to do the same computation mod p (i.e., the counting the number of (a, b) so that $N = a + b$ mod p for a, b relatively prime to p, and similarly over the infinite place). We could then approximate

$$\sum_{n_1+n_2=2N} \Lambda(n_1)\Lambda(n_2) \sim \mathscr{S}(N)2N,$$

we can then try to use the circle method to approximate $\int_0^1 S(\alpha)^2 e(-2N\alpha)d\alpha$. But we can no longer use Parseval's identity to bound the minor arcs because

$$\int_0^1 |S(\alpha)|^2 = \sum_{n \leq 2N} \Lambda(n)^2 \sim 2N \log N.$$

Remark 7.7. At the beginning of this course, we mentioned we could try to count the number of ways to write N as a sum

$$N = x_1^k + \cdots + x_s^k.$$

Letting $P = N^{1/k}$, this is approximated by the integral

$$\int_0^1 \left(\sum_{x \leq P} \exp\left(n^k \alpha\right) \right)^s e\left(-N\alpha\right) d\alpha.$$

Then, we might expect

$$P^s / N = P^{s-k}.$$

There might then be local obstructions (e.g., squares are always 1 mod 8 or 0 mod 8). Then, if S is large enough in terms of $4k$, one might try to show this can be done. Instead of trying to understand exponential sums over primes, we would want to understand exponential sums over powers. But, once $S \geq k+1$ and there are no congruence constructions, this sort of result should hold. For example, every large number should be a sum of four squares. But for three squares, there is a congruence obstruction - 7 mod 8 can never be written as a sum of three squares.

It turns out you can write numbers as sums of 7 cubes. But for fifth powers, the problem turns out to be much harder.

Next time we'll talk about effectivity and Siegel's theorem.

8. 10/17/17

8.1. **Exercises to solidify the ideas thus far.** Here are two exercises, which are a bit longer and harder than usual.

Exercise 8.1 (Difficult exercise)**.** Assume GRH. Give a bound for

$$\sum_{n \leq x} \Lambda(n) \exp\left(n\alpha\right)$$

for $\left|\alpha - \frac{a}{q}\right| \leq \frac{1}{q^2}$ without using bilinear forms, but instead using GRH and thinking about the prime number theorem and arithmetic progressions. *Hint:* We discussed how to write

$$\sum_{n \leq x} \Lambda(n) \exp\left(\frac{na}{q}\right) \sim \frac{1}{\phi(q)} \sum \overline{\chi}(a)\tau(\chi)\Psi(x, \chi),$$

and one can input information about Ψ using GRH.

One would then try to write $\exp\left(\frac{na}{q}\right)$ in terms of $\exp\left(n\beta\right)$ with

$$|\beta| \leq \frac{1}{qQ}$$

and then one should try to obtain good minor arc estimates for this summation using a "quasi-Riemann hypothesis" (i.e., assuming there are no zeros with $\sigma \geq 2/3$. Unconditionally, we know information about primes in progressions up to some modulus. Assuming GRH, we know estimates for primes up to $\sqrt{x}$. We can then find approximations for numbers with the denominator up to $\sqrt{x}$. We can then use the prime number theorem for everything, and then we won't even have to worry about major and minor arcs, we can hit the whole problem in both cases using GRH.

If one is more careful (via a result due to Hardy and Littlewood) it is enough to assume there are no zeros with $\sigma \geq 3/4$.

Remark 8.2. On GRH, one should be able to prove

$$\sum_{n \leq x} \Lambda(n) \exp(n\phi) \ll x^{3/4+\varepsilon},$$

whereas Vinogradov's method only gave an $x^{4/5}$. To get the $3/4$ estimate, we would need Hardy and Littlewood's refinement. This refinement due to Hardy and Littlewood is a refinement of the Gauss sum idea. One might decompose $\exp\left(\frac{an}{q}\right)$ as a sum of multiplicative characters. This incurs a loss of $\sqrt{q}$. When one writes it as

$$\frac{1}{\phi(q)} \sum_{\chi \bmod q} \tau(\chi)\overline{\chi}(ax),$$

one rewrites a number on the order of 1 as a number on the order of $\sqrt{q}$. For $n \leq x$, one can write $\exp(n\beta)$ in terms of integral of the form

$$\int_{|y| \leq x} f(y) n^{iy} dy$$

We try to replace this additive character in x in terms of a multiplicative character in y. A gauss sum is then of the form

$$\tau(\chi) = \sum \exp\left(\frac{n}{q}\right) \chi(n).$$

There will then be an integral which is an analog of a Gauss sum. One then saves a factor of $\sqrt{q}$, instead of just writing it naively by breaking it up into progressions.

Exercise 8.3 (Difficult exercise)**.** This exercise is to prove a theorem of Davenport. Let $\mu(n)$ be the Möbius function. Show

$$\sup_{\alpha \in \mathbb{R}} \left| \sum_{n \leq x} \mu(n) \exp(n\alpha) \right| \ll_A \frac{x}{(\log x)^A}$$

for any $A > 0$. You will have to figure out what happens when α is on a major or minor arc.

When α is on a minor arc, you will want to use Vinogradov's method and use a bilinear estimate, you will have $x/\sqrt{q} + \sqrt{q} \cdot x$. We use Vaughn's identity for obtaining bilinear forms for $-\frac{\zeta'}{\zeta}(s)$, and we would need an analog for identifying $\frac{1}{\zeta}(s)$. One would take $\zeta \cdot M$ for M a modifier, and play around with powers of that.

To deal with the case when α is on a major arc. This has to do with understanding

$$\sum_{n \le x} \mu(n) \exp\left(\frac{an}{q}\right),$$

for $q \le (\log x)^A$. One might rewrite this in terms of χ mod q. The goal would then be to understand

$$\sum_{n \le x} \mu(n)\chi(n).$$

Many results holding for prime numbers also hold for the Möbius function. For studying primes we look at something like

$$\frac{1}{2\pi i} \int_{(c)} \frac{-\zeta'}{\zeta}(s) x^s \frac{ds}{s},$$

and in this case we would be looking at

$$\frac{1}{2\pi i} \int_{(c)} \frac{1}{L(s,\chi)} x^s \frac{ds}{s}.$$

For primes there is a pole at $s = 1$, but the pole at $s = 1$ becomes a 0 for $1/L$. So there is some cancellation.

On the major arcs, there are savings with powers of log, and on the minor arcs there is another method using bilinear forms which gives savings of powers of log.

Exercise 8.4. Assuming GRH, show

$$\sup_{\alpha \in \mathbb{R}} \left| \sum_{n \le x} \mu(n) \exp(n\alpha) \right| \ll x^{3/4+\varepsilon}.$$

Maybe $5/6$ instead of $3/4$ would be easier to prove. Presumably the correct answer is $x^{1/2+\varepsilon}$. The supremum does obtain $x^{1/2}$ because Parseval tells us

$$\int_0^1 \left| \sum_{n \le x} \mu(n) \exp(n\alpha) \right|^2 d\alpha = \sum_{n \le x} \mu(n)^2.$$

Remark 8.5. The minor arc technology gave use an estimate of the form

$$\frac{N}{\sqrt{q}} + \sqrt{qN} + E$$

where E is some error term endemic to the method. The first two terms are optimized when q is on the order of $\sqrt{N}$, in which case the first two terms give $N^{3/4}$, so you cannot really do better than $N^{3/4}$ with this minor arcs method. We can write $\exp(n\alpha)$ in terms of

$$\sum_{\chi} \int_t \chi(n) n^{it} f,$$

for some function f. Here $n \leq N, \alpha \in (0,1)$. One than writes $\alpha = \frac{a}{q} + \beta$. One does not want to use too many q's and too many t's. Roughly one uses q characters χ and integrates over t which is roughly $1 + |\beta| x$. One needs to balance what weight to put on the sum and what weight to put on the integral. You can always choose $q \leq Q, |\beta| \leq \frac{1}{qQ}$. Then, $1 + |\beta| N \leq 1 + \frac{N}{Q}$. One can choose $Q \sim \sqrt{N}$ so the integral goes up to $\sqrt{N}$ and the sum adds over $\sqrt{N}$ terms. One looses $N^{1/2}$ complexity when doing the above procedure. So it is very hard to beat $N^{3/4}$ in these major and minor arc estimates.

8.2. **Zeros of ζ and L-functions.** It's good to have some intuition for where the 0's come from and how you might prove these functions have a 0-free region. We'd like to prove

$$\zeta(1+it) \neq 0, L(1,\chi) \neq 0,$$

where

$$L(1,\chi) = \prod_p \left(1 - \frac{\chi(p)}{p}\right)^{-1}.$$

We can consider the product

$$\zeta(1+it) = \prod_p \left(1 - \frac{1}{p^{1+it}}\right)^{-1}.$$

How will you find t with

$$|\zeta(1+it)|$$

being large or small. The small primes have a much bigger impact on this product above than the large primes. The maximum impact occurs when the small primes are as big or small as possible.

To make

$$|\zeta(1+it)|$$

large, we would like

$$p^{it} \sim 1$$

and to make it small we would like

$$p^{it} \sim -1$$

for many small primes p.

Exercise 8.6 (Simple exercise)**.** Show that for any N, there are quadratic characters χ with $\chi(p) = 1$ for all $p \leq N$ (and similarly $\chi(p) = -1$ for all $p \leq N$).

Remark 8.7. Then, χ will occur to some modulus q, and q might be very large in terms of N. If one tries to compute one of these via the Chinese remainder theorem, one might then have the modulus exponentially large in N.

Conjecture 8.8 (Vinogradov)**.** If $\chi(p) = 1$ for all $p \leq N$, then $q > N^A$ for arbitrarily large A. Conversely, $\chi(p) = -1$ for all $p \leq N$ then $q > N^A$ for all large A.

Example 8.9. The least quadratic non-residue must be smaller than q^ε, if Conjecture 8.8 were to hold true.

Remark 8.10. The chance that all the first N primes land heads, one should expect the chance is around $1/2^N$. So, one would expect the number of primes would have to be exponentially large.

Every once in a while, there can be a surprise. For example, if $D = -163$. Then,

$$\left(\frac{-163}{p}\right) = -1$$

for $p < 41$. There are 12 such primes. If you think of things as coin tosses, there would only be a $1/2^{12}$ (since there are 12 primes up to and including 37) but 163 is substantially smaller than $2^{12} = 4096$.

$L(1, \chi_{-163})$ should be very small. Indeed, the Class number formula gives

$$L(1, \chi_{-163}) \frac{\pi h\left(\mathbb{Q}\left(\sqrt{-163}\right)\right)}{\sqrt{163}} \sim \frac{\pi}{13}.$$

and the class number is 1 here (and h denotes class number). Recall that the class number formula implies that for $\mathbb{Q}\left(\sqrt{-D}\right)$,

$$L\left(1, \chi_D\right) = \frac{\pi h(\mathbb{Q}\left(\sqrt{-D}\right)}{\sqrt{D}}.$$

Goldfeld and Gross-Zagier's result implies

$$L\left(1, \chi_{-D} \leq \frac{C \log D}{\sqrt{D}}\right).$$

Siegel's theorem implies that if χ is a quadratic character $\bmod q$, then

$$L(1, \chi) \geq C\left(\varepsilon\right) q^{-\varepsilon}$$

for all $\varepsilon > 0$.

Remark 8.11. The zero free region is determined as follows. If $L\left(\beta, \chi\right) = 0$, then

$$L\left(1, \chi\right) = \left(1 - \beta\right) L'\left(\sigma, \chi\right)$$

for some $\beta \leq \sigma \leq 1$.

Exercise 8.12. Prove that if

$$1 \leq \sigma \leq 1 - \frac{1}{\log q},$$

then

$$\left|L'\left(\sigma, \chi\right)\right| \leq C\left(\log q\right)^2.$$

See Davenport. Essentially you can make sense of this a little left of the 1 line. Then you can differentiate it and deduce this. So, you have a bound on how close a zero can be to 1. So, if there's a bad Siegel 0 it has to be bounded away from 1.

8.3. **The zero-free region of the Zeta function.** We'd like to instead look at the completed zeta function

$$\xi(s) = s\left(s - 1\right) \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

The functional equation says

$$\xi(s) = \xi(s - 1).$$

The Hadamard product formula gives

$$e^{A + Bs} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho}.$$

The trivial zeros come because the $\Gamma(s/2)$ function has zeros at $s = 0, -2, -4, \cdots$.

Exercise 8.13. The Riemann hypothesis is equivalent to

$$|\xi\left(\sigma+it\right)|$$

is monotonically increasing in $\sigma \geq 1/2$ *Hint:* Show that 0 by 0, the Hadamard product above will be increasing.

Furthermore,

$$|\zeta\left(\sigma+it\right)|$$

is monotone increasing on $\sigma \geq 1$.

Exercise 8.14. Let χ be an even character, i.e., $\chi(-1) = 1$. Define

$$\xi\left(s,\chi\right) := \pi^{-s/2}\Gamma(s/2)L(s,\chi).$$

Then prove

$$|\xi\left(s,\chi\right)|$$

is monotone increasing in $\sigma > 1$.

If $\zeta(1+it) = 0$, then

$$p^{it} \sim -1$$

for many small primes p. This implies

$$p^{2it} \sim 1$$

for many small primes p. This implies $\zeta(1+2it)$ is very big. This relates to the classical inequality

$$\zeta(\sigma)^3\left|\zeta\left(\sigma+it\right)\right|^4\left|\zeta\left(\sigma+2it\right)\right| \geq 1.$$

Then,

$$\chi(p)p^{it} \sim -1$$

implies

$$\chi(p)^2p^{2it} \sim 1,$$

which implies

$$L\left(1+2it,\chi^2\right)$$

is big. This would yield a contradiction unless

$$\chi^2 = \chi_0$$

and $t = 0$. In this case, we are considering the ζ function at 1, which is big because it has a pole. This is the Siegel zero situation where we have a quadratic character and want a lower bound for $L(1,\chi)$.

8.4. **Siegel zero situation.** We'll now discuss a proof due to Goldfeld of Siegel's theorem. We want to show that a lower bound for $L(1, \chi)$

$$L(1, \chi) \gg C(\varepsilon) q^{-\varepsilon}$$

for χ a quadratic character $\bmod q$. We look at the region

$$\left[1 - \frac{\varepsilon}{10}, 1\right].$$

Either

(1) All quadratic Dirichlet L-functions have no zero in this region We take $\beta = 1 - \frac{\varepsilon}{10}$. We define Ψ to some character $\bmod 3$.
(2) There is some quadratic character Ψ mod r for some r with $L(\beta, \Psi) = 0$ with $1 \geq \beta \geq 1 - \frac{\varepsilon}{10}$.

Consider

$$\zeta(s) L(s, \chi) L(s, \Psi) L(s, \chi\Psi) = \sum \frac{a(n)}{n^s}.$$

Exercise 8.15. Check $a(n) \geq 0$ for all n by just expanding the definitions of Dirichlet characters for the various L functions.

This function above is the Dedekind ζ function for the biquadratic extension defined by χ and Ψ. This function is always non-negative on primes because

$$(1 + \chi(p))(1 + \Psi(p)) \geq 0$$

and both χ, Ψ takes values $\pm 1, 0$.

Then, for $c > 1$, consider

$$I := \frac{1}{2\pi i} \int_{(c)} \zeta(s + \beta) L(s + \beta, \chi) L(s + \beta, \Psi) L(s + \beta, \chi\Psi) \Gamma(s) X^s ds$$

where X is some large parameter that we haven't yet defined, which is roughly $(qr)^{10}$.

Exercise 8.16. Show that

$$\frac{1}{2\pi i} \int_{(c)} X^s \Gamma(s) ds = e^{-1/x}.$$

Look at the 0's of Γ, compute the residues, and you will see the Taylor expansion for $e^{-1/x}$.

Here we have nice absolutely convergent integrals. But instead of picking up the characteristic function, we pick up a "smoothed" version of the characteristic function.

Then, we have

$$I = \sum_{n=1}^{\infty} \frac{a(n)}{n^\beta} e^{-n/x}.$$

where we are plugging in

$$\sum \frac{a(n)}{n^\beta}$$

and $(X/n)^s$ We have Then, we have

$$I = \sum_{n=1}^{\infty} \frac{a(n)}{n^\beta} e^{-n/x} \geq e^{-1/x} \geq 1/2.$$

Dirichlet L-functions are entire. The only L-function with a pole is the Riemann zeta function. For any other character, the L function terms cancel out every q-steps. For example, using integration by parts

$$\begin{aligned} L(s,\chi) &= \int_{1^-}^{\infty} \frac{1}{y^s} d\left(\sum_{n\leq y} \chi(n)\right) \\ &= s\int_{1^-}^{\infty} \frac{1}{y^{s+1}} \left(\sum_{n\leq y} \chi(n)\right) dy. \end{aligned}$$

Moving the line of integration to the left, we encounter poles at $1-\beta$ from the ζ function, there are poles from the Γ function. We take $\operatorname{Re} s = -\beta + 1/2$, so this is negative, but not as negative as -1. We encounter poles at

$$s = 1-\beta, s = 0$$

coming from $\zeta(s+\beta)$ and $\Gamma(s)$. The pole at $s = 1-\beta$ has residue

Computing the residue at $1-\beta$ we get

$$L\left(1,\chi\right) L\left(1,\Psi\right) L\left(1,\chi\Psi\right) X^{1-\beta}\Gamma(1-\beta).$$

The residue at 0 is given as follows: near 0, we have $\Gamma(0) \sim \frac{1}{s}$ using that $s\cdot\Gamma(s) = \Gamma(s+1)$ and $\Gamma(1) = 1$ and is smooth.

The residue at 0 is

$$\zeta(\beta)L(\beta,\chi)L(\beta,\Psi)L(\beta,\chi\Psi) \leq 0.$$

Indeed, if all Dirichlet functions have no 0's, $L(\beta,\chi)$ is positive and $L(\beta,\Psi), L(\beta,\chi\Psi)$ is positive, and $\zeta(\beta)$ is negative. In the second case $L(\beta,\Psi)=0$.

We then have a lower bound for the residue at $1-\beta$. This is what we want, because we want a lower bound for $L(1,\chi)$. We would be done if we had upper bounds for the latter Dirichlet L functions. We can just replace using integration by parts

$$\begin{aligned} L(s,\chi) &= \int_{1^-}^{\infty} \frac{1}{y^s} d\left(\sum_{n\leq y}\chi(n)\right) \\ &= s\int_{1^-}^{\infty}\frac{1}{y^{s+1}}\left(\sum_{n\leq y}\chi(n)\right)dy. \end{aligned}$$

as above.

Exercise 8.17. Indeed show that for χ a character $\bmod q$, show

$$|L(1,\chi)| \ll \log q.$$

for X a large power of qr (using $\mathrm{Re}(s) = -\beta + 1/2$).

Therefore, we would conclude a bound of the form

$$L(1,\chi) \gg (qr)^{-\varepsilon}.$$

We could get an effective bound, but we don't know what r is. In case 1, $r=3$, so things would be fine. But, if there is some violation to the Riemann hypothesis, then r depends on what the violation to the Riemann hypothesis is. So this r is the source of the ineffectivity in Siegel's theorem.

Next time, we'll discuss effectivity of the 3-prime theorem. Then we'll move on to discussing a theorem of Maynard:

Theorem 8.18. *There are infinitely many primes with no* 7 *in their decimal expansion.*

9. 10/24/17

9.1. **Quick recap of the proof of Siegel's theorem.** Recall that last time we proved Siegel's theorem:

Theorem 9.1 (Siegel)**.** *We have*

$$L(1,\chi) > \frac{C(\varepsilon)}{q^{\varepsilon}}$$

(with $C(\varepsilon)$ ineffective).

The idea of the proof was to construct an auxiliary character Ψ mod r. There were two cases. In the first case, all characters have no zeros $[1-\varepsilon/10, 1]$ and we took ψ mod $r = \left(\frac{1}{3}\right)$ In the second case we assume there exists some r with a zero $\beta \geq 1 - \frac{\varepsilon}{10}$. The idea was to consider

$$\frac{1}{2\pi i}\int_{(c)} \zeta(s+\beta)L(s+\beta,\chi)L(s+\beta,\Psi)L(s+\beta,\chi\Psi)X^s\Gamma(s)ds = \sum \frac{a(n)e^{-n/x}}{n^\beta} \quad \geq e^{-1/x}$$

We then move the line of integration to $\mathrm{Re}(s) = 1/2 - \beta$. This has a pole at $1-\beta$. We obtain

$$L(1,\chi)L(1,\Psi)L(1,\chi\Psi)X^{1-\beta}\Gamma(1-\beta).$$

Then, at $s = 0$, we have

$$\zeta(\beta)L(\beta,\chi)L(\beta,\Psi)L(\beta,\chi\Psi) \leq 0.$$

At the end of last time, we claimed

Lemma 9.2. *The integral on $\frac{1}{2} - \beta$ is negligible. That is,*

$$\frac{1}{2\pi i}\int_{(\frac{1}{2}-\beta)} \zeta(s+\beta)L(s+\beta,\chi)L(s+\beta,\Psi)L(s+\beta,\chi\Psi)X^s\Gamma(s)ds \ll 1$$

for appropriate values of x (we will take $(qr)^{20}$).

Proof. Indeed,

$$\frac{1}{2\pi i}\int_{(\frac{1}{2}-\beta)} \zeta(s+\beta)L(s+\beta,\chi)L(s+\beta,\Psi)L(s+\beta,\chi\Psi)X^s\Gamma(s)ds$$
$$\ll x^{1/2-\beta}\int_{-\infty}^{\infty} e^{-|t|}\left|\zeta(\frac{1}{2}+it)L(\frac{1}{2}+it,\chi)L(\frac{1}{2}+it,\Psi)L(\frac{1}{2}+it,\chi\Psi)\right|dt$$

We want some kind of polynomial bound to show this integral is negligible. We have

$$\xi(s) = s\,(s-1)\,\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

is entire of order 1 (meaning it doesn't grow more than exponentially). We want to use the maximum modulus principal in a complex strip with real part between -1 and 2. It's easy to bound ξ because ζ is a bounded function on $\mathrm{Re}(s) = 2$. Similarly, we can understand asymptotics of the other terms. By the functional equation, we then also understand the value at $\mathrm{Re}(s) = -1$. So, by this variant of the maximum modulus principal, we can bound $|\xi\,(1/2+it)|$ by, essentially, $|\xi(2+it)|$. This cannot literally be true because it would imply the Riemann hypothesis, but if we restrict to a rectangular region, bounding things from above and below, we will have good enough bounds. But, in any case, after making this precise, we can bound Γ by a sterling approximation, and then bound

$$|\zeta\left(1/2+it\right)| \ll \left(1+|t|\right).$$

Remark 9.3. If we instead carry this out between $-\varepsilon$ and $1+\varepsilon$, one can obtain the **convexity bound**

$$|\zeta(1/2+it)| \ll \left(1+|t|\right)^{1/4+\varepsilon}.$$

The Lindelöf hypothesis says we can replace $1/4+\varepsilon$ by any positive exponent.

Altogether, we can bound

$$\begin{aligned}
&\frac{1}{2\pi i}\int_{(\frac{1}{2}-\beta)} \zeta(s+\beta)L(s+\beta,\chi)L(s+\beta,\Psi)L(s+\beta,\chi\Psi)X^s\Gamma(s)ds \\
&\ll x^{1/2-\beta}\int_{-\infty}^{\infty} e^{-|t|}\left|\zeta(\frac{1}{2}+it)L(\frac{1}{2}+it,\chi)L(\frac{1}{2}+it,\Psi)L(\frac{1}{2}+it,\chi\Psi)\right| dt \\
&\ll x^{1/2-\beta}\int_{-\infty}^{\infty} e^{-|t|}\left(\left(1+|t|\right)qr\right)^4 dt \\
&\ll (qr)^4 x^{-.4}.
\end{aligned}$$

Now, choose $x = qr^{20}$ so that $(qr)^4 x^{-.4} \ll 1$.

□

Using the above bound from the lemma, together with

$$\frac{1}{2\pi i}\int_{(c)} \zeta(s+\beta)L(s+\beta,\chi)L(s+\beta,\Psi)L(s+\beta,\chi\Psi)X^s\Gamma(s)ds = \sum \frac{a(n)e^{-n/x}}{n^\beta} \quad \geq e^{-1/x}$$

(with the last term bounded by .9) we get

$$\begin{aligned}
L\left(1,\chi\right)L\left(1,\Psi\right)L\left(1,\chi\Psi\right) \geq \frac{1}{3}\frac{x^{\beta-1}}{\Gamma(1-\beta)} \\
&\geq \frac{1}{5}\left(1-\beta\right)x^{-\varepsilon/10} \\
&= \left(1-\beta\right)/5\left(qr\right)^{-2\varepsilon}.
\end{aligned}$$

Then, note

$$\begin{aligned}
L(1,\chi) &\leq c\log r, \\
L(1,\chi\Psi) &\leq c\log qr.
\end{aligned}$$

So, one obtains

$$L(1, \chi) \geq C(1-\beta)(qr)^{-3\varepsilon}.$$

The constant C is calculatable, but the reason for the ineffectivity is that we do not know what r and β are.

Remark 9.4. One can effectively prove

$$L(1, \chi) \geq \frac{C}{\sqrt{q}},$$

so β must be at least $\frac{1}{\sqrt{r}}$ away from 1, or something like that. So really the constant above only depends on r, since we can get a bound on β from that.

9.2. **Effectivity of ternary Goldbach.** Returning to ternary Goldbach, we considered

$$\sum_{n \leq N} \Lambda(n) \exp\left(\frac{an}{q}\right).$$

Using $q \leq (\log N)^{10}$, we found

$$\sum_{n \leq N} \Lambda(n)\chi(n),$$

is bounded by something like $\frac{-N^\beta}{\beta}$ for $\beta > 1 - \frac{c}{\sqrt{q}}$. Then,

$$N^\beta \leq N^{1-c/\sqrt{q}}$$

is small compared to N only when $q \leq (\log N)^{1.99}$. But, we wanted q to go up to $(\log N)^{10}$ rather than $(\log N)^2$. So, we will have to use Siegel's theorem in some range. Even though Siegel's theorem is not effective, we can use that Siegel zeros are rare to still get effectivity of ternary Goldbach.

Lemma 9.5. *There cannot exist two primitive quadratic characters*

$$\chi_1 (\operatorname{mod} q_1), \chi_2 (\operatorname{mod} q_2)$$

with $Q \leq q_1, q_2 \leq Q^{100}$ *and both* L *functions having a* 0 *at least* $1 - \frac{10^{-10}}{\log Q}$.

Proof. Suppose we have two such characters χ_1, χ_2. We'll now play these two characters against each other. Consider

$$\zeta(s) L(s, \chi_1) L(s, \chi_2) L(s, \chi_1\chi_2).$$

This is the Dedekind zeta function of a biquadratic field, so its Dirichlet coefficients are all positive.

Instead, consider

$$\xi(s)\xi(s,\chi_1)\xi(s,\chi_2)\xi(s,\chi_1\chi_2).$$

Consider its logarithmic derivative and evaluate at some real number $\sigma > 1$. We have

$$\frac{\xi'}{\xi}(\sigma) + \frac{\xi'}{\xi}(\sigma,\chi_1) + \frac{\xi'}{\xi}(\sigma,\chi_2) + \frac{\xi'}{\xi}(\sigma,\chi_1\chi_2)$$

Using the Hadamard product formula we have

$$\xi(s) = e^{A+Bs}\prod_\rho \left(1 - \frac{s}{\rho}\right) e^{s/\rho}$$

then

$$\frac{\xi'}{\xi}(s) = \sum_\rho \frac{1}{s-\rho}.$$

Then,

$$\operatorname{Re}\frac{1}{s-\rho} = \frac{\sigma-\beta}{|s-\rho|^2}.$$

with $s = \sigma + it, \rho = \beta + i\gamma$. On the one hand, the expression

$$\frac{\xi'}{\xi}(\sigma) + \frac{\xi'}{\xi}(\sigma,\chi_1) + \frac{\xi'}{\xi}(\sigma,\chi_2) + \frac{\xi'}{\xi}(\sigma,\chi_1\chi_2)$$

is always positive. On the other hand, if we have two real zeros, we would obtain

$$\frac{\xi'}{\xi}(\sigma) + \frac{\xi'}{\xi}(\sigma,\chi_1) + \frac{\xi'}{\xi}(\sigma,\chi_2) + \frac{\xi'}{\xi}(\sigma,\chi_1\chi_2) \geq \frac{1}{\sigma-\beta_1} + \frac{1}{\sigma-\beta_2}.$$

We also know

$$\xi(\sigma) = \sigma\left(\sigma - 1\right)\pi^{-\sigma/2}\Gamma(\sigma/2)\zeta(\sigma),$$

and with α equal to either 0 or 1,

$$\xi(\sigma,\chi_1) = \left(\frac{q_1}{\pi}\right)^{\sigma/2}\Gamma(\sigma+\alpha/2)L(\sigma,\chi_1).$$

We get similar expressions for the other two ξ functions. Then, we obtain

$$\begin{aligned}&\frac{\xi'}{\xi}(\sigma)+\frac{\xi'}{\xi}(\sigma,\chi_1)+\frac{\xi'}{\xi}(\sigma,\chi_2)+\frac{\xi'}{\xi}(\sigma,\chi_1\chi_2)\\&\frac{1}{1-\sigma}+\frac{1}{2}\log q_1+\frac{1}{2}\log q_2+\frac{1}{2}\log q_1q_2+O(1)\\&+\frac{\zeta'}{\zeta}(\sigma)+\frac{L'}{L}(\sigma,\chi_1)+\frac{L'}{L}(\sigma,\chi_2)+\frac{L'}{L}(\sigma,\chi_1\chi_2).\end{aligned}$$

We then obtain

$$\frac{\zeta'}{\zeta}(\sigma)+\frac{L'}{L}(\sigma,\chi_1)+\frac{L'}{L}(\sigma,\chi_2)+\frac{L'}{L}(\sigma,\chi_1\chi_2).$$

is approximated by

$$-\sum\frac{\Lambda(n)}{n^r}\left(1+\chi_1(n)+\chi_2(n)+\chi_1\chi_2(n)\right),$$

which has all Dirichlet coefficients negative. Therefore, we have

$$\begin{aligned}&\frac{\xi'}{\xi}(\sigma)+\frac{\xi'}{\xi}(\sigma,\chi_1)+\frac{\xi'}{\xi}(\sigma,\chi_2)+\frac{\xi'}{\xi}(\sigma,\chi_1\chi_2)\\&\frac{1}{1-\sigma}+\frac{1}{2}\log q_1+\frac{1}{2}\log q_2+\frac{1}{2}\log q_1q_2+O(1)\\&+\frac{\zeta'}{\zeta}(\sigma)+\frac{L'}{L}(\sigma,\chi_1)+\frac{L'}{L}(\sigma,\chi_2)+\frac{L'}{L}(\sigma,\chi_1\chi_2)\\&\le\frac{1}{\sigma-1}+\log q_1q_2+O(1).\end{aligned}$$

If β_1,β_2 were close to 1, we have a lower bound by something close to $2/1-\sigma$.

Exercise 9.6. If q_1,q_2 are comparable to each other, say $q_1\le q_2^{100}, q_2<q_1^{100}$, then we can't have both a lower bound by $\frac{1}{\sigma-\beta_1}+\frac{1}{\sigma-\beta_2}$ and an upper bound by

$$\frac{1}{\theta-1}+\log(q_1q_2)+O(1)$$

Here, we are choosing σ to be around $1+\frac{10^{-6}}{\log q_1q_2}$.

□

9.3. **Discriminants of number fields.** Let K be a number field over $\mathbb{Q}$. Then, some prime must be ramified in K because the discriminant is more than 1. In general, if K has degree n over $\mathbb{Q}$, what can we say about $d_K := \operatorname{disc} K$.

Question 9.7 (Open question)**.** Take $f \in \mathbb{Z}[x]$ of degree n irreducible. How does $\operatorname{disc}(f)$ grow?

Theorem 9.8 (Minkowski, Stark-Odlyzko)**.** *The discriminant of a number field K is bounded below by c^n with $c > 1$.*

Remark 9.9. Minkowski got this by thinking about lattices and using the geometry of numbers.

One way to think of this is the following idea going back to Stark:

Let r_1 be the number of real embeddings, r_2 be the number of complex embeddings, so that $r_1 + 2r_2 = n$. Consider the Dedekind zeta function

$$\begin{aligned}\xi_K(s) &= s(s-1)d_K^{s/2}\zeta_K(s)\left(\pi^{-s/2}\Gamma(s/2)\right)^{r_1}\left((2\pi)^{-s}\Gamma(s)\right)^{r_2}\\ &= (\cdots)\prod_{\rho_K}\left(1-\frac{s}{\rho_K}\right)(\cdots)\end{aligned}$$

where $\cdots$ indicate factors we must include to make the product converge. This will satisfy a functional equation $\xi_K(1-s) = \xi_K(s)$, and will have a Hadamard product, and so on. Then,

$$\frac{\xi_K'}{\xi_K}(\sigma) = \sum_{\rho_K}\frac{1}{\sigma-\rho_K} \geq 0.$$

Then,

$$\frac{\xi_K'}{\xi_K}(\sigma) = \frac{1}{\sigma} + \frac{1}{\sigma-1} + \frac{1}{2}\log d_K + r_1\left(\frac{-1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}(\sigma/2)\right) + r_2(\cdots) + \frac{\zeta_K'}{\zeta_K}(\sigma).$$

Using that the last term $\frac{\zeta_K'}{\zeta_K}(\sigma)$ is negative, and the whole sum is positive, choosing σ near 1 optimally and knowledge of $\frac{\Gamma'}{\Gamma}(1/2)$ and $\frac{\Gamma'}{\Gamma}(1)$ gives a lower bound for the discriminant. One must choose an appropriate value of σ, Sound suggests something like $\sigma = 1 + \frac{c}{n}$.

So, if you have a field with small discriminant, this also means there are not many primes of small norm. The zeros of such an L function are then also nicely behaved.

Exercise 9.10. Work out the details in the above remark.

There is a nice survey by Odlyzko (if one searches “discriminants Odlyzko“) and also an article by Serre on “Minorations of discriminants” (in French).

Remark 9.11. The ring of integers in a number field may not be monogenic, so the discriminant of a polynomial may be much larger than the discriminant of a number field. We don't have a good lower bound on the discriminant of a polynomial.

Suppose

$$(\log N)^{1.9}\, q \leq (\log N)^{10}\,.$$

suppose there is some χ mod q_0 with a Siegel zero at β_0. All we have to worry about are $\alpha = \frac{a}{q} + \beta$ with $q_0 \mid q$. We have an expression of the form

$$\sum_{n \leq N} \Lambda(n) \exp\left(\frac{an}{q}\right) = \frac{N\mu(q)}{\phi(q)} + (\cdots) + \frac{\tau(\chi)}{\phi(q)}\chi(a)\Psi(N,\chi).$$

The last $\Psi(N,\chi)$ is bounded by something like N^{β_0}/β_0. So, the above is approximated by

$$\frac{N\mu(q)}{\phi(q)} + \frac{\tau(\chi)}{\phi(q)}\chi(a)\frac{N^{\beta_0}}{\beta_0}$$

and then we can then approximate these things by major and minor arcs.

We then have to change whatever main term we had before with this new main term coming from

$$\frac{\tau(\chi)}{\phi(q)}\chi(a)\frac{N^{\beta_0}}{\beta_0}$$

That is, we have

$$\int_{\mathfrak{M}} S(\alpha)^3 \exp\left(-N\alpha\right) d\alpha = \sum_{q \leq (\log N)^{10}, q_0 \mid q} \int_{|\beta| \leq \frac{1}{qQ}} S\left(\frac{a}{q}+\beta\right)^3 \exp\left(-N\left(\frac{a}{q}+\beta\right)\right) d\beta.$$

Then, to find the contribution of the cube of this main term, we find

$$\sideset{}{^*}\sum_{a \bmod q} \frac{\tau(\chi)^3}{\phi(q)^3}\chi(a)\frac{N^{3\beta_0}}{\beta_0^3}\exp\left(\frac{-aN}{q}\right)$$

From $\chi(a)\exp\left(\frac{-aN}{q}\right)$ we get another Gauss sum so

$$\sideset{}{^*}\sum_{a \bmod q} \frac{\tau(\chi)^3}{\phi(q)^3}\chi(a)\frac{N^{3\beta_0}}{\beta_0^3}\exp\left(\frac{-aN}{q}\right) \sim \frac{\tau(\chi)^4}{\phi(q)^3}\frac{N^{3\beta_0-1}}{\beta_0^3}.$$

where the -1 in $3\beta_0 - 1$ is coming from the integral. Then, $\tau(\chi)$ is bounded by $\sqrt{q}$. So, the above is bounded by

$$\frac{q^2}{q^3}\frac{N^{3\beta_0-1}}{\beta_0^3}.$$

So, in conclusion, we get a bound like

$$\sum_{q\leq(\log N)^{10}, q_0|q} \frac{N^2}{q} \ll N^2 q_0 \log\log N.$$

for $q_0 > (\log N)^{1.9}$. Therefore, the proof is effective.

10. 10/26/17

Remark 10.1. Goldfeld Gross Zagier says

$$L(1,\chi) \geq \frac{c\log|D|}{\sqrt{|D|}}$$

for imaginary quadratic fields. Then, effectively, $h(-D) > c\log|D|$.

Remark 10.2 (Euler's idoneal numbers)**.** Consider $\mathbb{Q}\left(\sqrt{-D}\right)$. Can it be that all $p \leq \sqrt{|D|}$ are either ramified or inert?

Gauss' genus theorem tells us

$$h(-D) \geq \text{ 2 part of the class group } = 2^{\#\text{ primes }|D} = d(|D|).$$

The divisor function grows as $O(|D|^\varepsilon)$. The problem is to find all discriminants $-D < 0$ where

$$\mathrm{cl}\left(\mathbb{Q}\left(\sqrt{-D}\right)\right) = (\mathbb{Z}/2\mathbb{Z})^r.$$

Remark 10.3. There is work of Biro on class numbers of fields of the form $\mathbb{Q}\left(\sqrt{n^2+4}\right)$.

10.1. **Primes with missing digits.**

In general it is quite hard to answer questions of the form:

(1) If p is a prime, is $p+2$ a primes?
(2) If n is even, when is n^2+1 prime?

Here are some theorems coming out of sieve methods.

Theorem 10.4 (Piatetski-Shapiro, 1950s)**.** *If $1 < \alpha < 1.1$, then there are infinitely many primes of the form*

$$\lfloor n^\alpha \rfloor.$$

Remark 10.5. This is quite a sparse set of numbers up to x, there are only $x^{1/\alpha}$ such numbers.

Recall from Fermat that every $p \equiv 1 \bmod 4$ can be written as a sum of two squares.

Theorem 10.6 (Fouvry and Iwaniec). *Infinitely often, one can write $p \equiv 1 \bmod 4$ as $p = n^2 + m^2$ with n a prime.*

Theorem 10.7 (Friedlander and Iwaniec). *For $p \equiv 1 \bmod 4$ one can write $p = m^2 + n^4$ for infinitely many primes.*

Theorem 10.8 (Heath-Brown and Li). *There are infinitely many primes $p \equiv 1 \bmod 4$ with $p = m^2 + q^4$ for q primes.*

Theorem 10.9 (Heath-Brown). *There are infinitely many primes of the form $a^3 + 2b^3$.*

Remark 10.10. This answers an old question of Hardy and Littlewood asking if there are infinitely many primes which are sums of three cubes.

Friedlander and Iwaniec only involves pairs m, n over sets of size $x^{3/4} = x^{1/2} \cdot x^{1/4}$ and in Heath-Brown's result, this only involves a set of size $x^{2/3}$ up to some x.

Question 10.11. Are there infinitely many primes $p = a^2 + b^3$?

Remark 10.12. This is analogous to the question of whether there are infinitely many elliptic curves with prime discriminant or conductor (since the discriminant of an elliptic curve in short Weierstrass form is something like $4a^3 + 27b^2$).

The main result we'll spend the next few lectures proving is of a similar flavor.

Theorem 10.13 (Maynard). *If q is a sufficiently large base (e.g. $q = 10^7$), write $n = \sum_{j=0}^{k} n_j q^j$ with $0 \le n_j \le q - 1$ as a base q expansion. Select a forbidden digit $0 \le a_0 \le q - 1$ Let*

$$\mathscr{A} := \{n \in \mathbb{N} : n \text{ does not have the digit } a_0 \text{ base } q\}.$$

Then,

$$\begin{aligned}\#\left\{n < q^k : n \in \mathscr{A}\right\} &= (q-1)^k \\ &= \left(q^k\right)^{\log(q-1)/\log q}.\end{aligned}$$

Then,

$$\sum_{\substack{n<q^k \\ n\in\mathscr{A}}} \Lambda(n) \sim \kappa_{a_0}(q)\,(q-1)^k\,.$$

for $\kappa_{a_0}(q)$ an explicit positive constant

Remark 10.14. There is also a more involved version where one obtains a lower bound of the form $\gg (q-1)^k$ for $q = 10$.

One can also find elements of $\mathscr{A}$ that are, say, squares.

Theorem 10.15 (Mauduit and Rivat)**.** *Write primes in binary $p = \sum a_j 2^j$. Count $s(p) := \sum_j a_j$. Then, $s(p)$ is equally likely to be 0 or $1 \bmod 2$.*

More generally, this can be done with any base replacing 2, with obvious exceptions.

There is also the following cute result: One might ask if one can find Fermat primes, with two 1's in the binary expansion and all other digits 0. This might be a hard problem because the set is quite sparse, but, one can try to further ask if there are infinitely many primes with k 1's. One might try an easier problem asking if there simply exist primes with exactly k 1's in their binary expansion. The following theorem shows the answer is yes.

Theorem 10.16 (Drmota, Mauduit, and Rivat)**.** *Let K be an integer and let k be on the scale of $K/2$ (say $k - K/2 = O(\sqrt{k})$). Then,*

$$\left\{p < 2^K : \ p \text{ prime}\ , \text{ there are } k \text{ digits equal to } 1\right\}$$

has an asymptotic formula, with about $1/k$ of the numbers in this set prime.

Remark 10.17. One would expect about $\binom{K}{K/2} \sim \frac{2^K}{\sqrt{K}}$

Theorem 10.18 (Bourgain)**.** *There are primes $p \leq 2^k$ for which you can specify any αK of the binary digits, for some fixed $\alpha > 0$, where the last digit must be 1 (so that the number is not even).*

10.2. **Beginning the proof of Maynard's theorem.** We now turn to proving Maynard's theorem on primes without a specified digit. Recall we have fixed a base q, an integer k, and defined $\mathscr{A}$ as the set of primes up to q^k without a digit a_0.

We are trying to count

$$\sum_{\substack{n<q^k \\ n\in\mathscr{A}}} \Lambda(n) = \sum_{n<q^k} \Lambda(n)1_A(n).$$

$$= \int_0^1 S(\alpha)\mathscr{A}(-\alpha)d\alpha,$$

where

$$S(\alpha) = \sum_{n<q^k} \Lambda(n)\exp(n\alpha)$$

and

$$\mathscr{A}(-\alpha) = \sum_{\substack{n<q^k \\ n\in\mathscr{A}}} \exp(-n\alpha).$$

In our situation, we can understand the Fourier transform of $\mathscr{A}$ so well that we can actually understand its L_1 norm. It is binary because we are intersecting two sets here: primes and integers without a specified digit.

We could still use the circle method here, but it is a little easier to apply the circle method in a discrete setting.

10.3. **The circle method in a discrete setting.** Consider

$$\frac{1}{q^k}\sum_{a=0}^{q^k-1} S\left(\frac{a}{q^k}\right)\mathscr{A}\left(\frac{-a}{q^k}\right) = \sum_{\substack{m,n<q^k \\ m\equiv n \bmod q^k}} \Lambda(m)1_A(n)$$

$$= \sum_{n\le q^k} \Lambda(n)1_A(n).$$

The last equality holds because $m, n < q^k$. For the penultimate one, writing

$$S\left(\frac{a}{q^k}\right) = \sum_m \Lambda(m)\exp\left(\frac{am}{q^k}\right)$$

and

$$\mathscr{A}\left(\frac{-a}{q^k}\right) = \sum_n 1_A(n)\exp\left(\frac{-na}{q^k}\right),$$

and the only terms that survive are $m \equiv n \bmod q^k$. This is an excellent approximation to the integral from the circle method.

Exercise 10.19. Verify the above equalities.

We now separate terms into major and minor arcs, as in the circle method.

We now write

$$\frac{a}{q^k} = \frac{\ell}{d} + \beta$$

with $d \leq qk/2$ and $|\beta| \leq \frac{1}{dq^{k/2}}$.

We try to approximate a/q^k using rational numbers with denominator at most $q^{k/2}$. By Dirichlet's theorem, we can always write numbers in this form.

We write A as some large positive number. The major arcs are those values of a with

$$d \leq \left(\log q^k\right)^A, |\beta| \leq \frac{(\log q^k)^A}{q^k}.$$

The minor arcs are the remaining values of a. The major arcs are distinct because the denominators are small and we are taking small intervals around each rational number.

We'll first deal with the major arcs. The harder part will come later when we deal with the minor arcs.

10.4. **The major arc contribution.** There are two cases:

(1) The denominator d is a small power of q
(2) The denominator d is not a small power of q.

The main terms will come from the first case. Consider the sum $S(\alpha)$ on the major arcs of the first case, so

$$\alpha = \frac{\ell}{d} + \frac{b}{q^k}$$

with d a power of q. b is small (at most d) because β is bounded.

Consider

$$\begin{aligned} S\left(\ell/d\right) &= \sum_{n<q^k} \Lambda(n) \exp\left(\frac{\ell n}{d}\right) \\ &= \frac{\mu(d)}{\phi(d)} q^k + O\left(\frac{q^k}{\left(\log q^k\right)^{4A}}\right) \end{aligned}$$

where we have proved asymptotic formulas for this in certain ranges using Siegel's theorem.

Then,

$$S\left(\frac{\ell}{d}+\frac{b}{q^k}\right)=\frac{\mu(d)}{\phi(s)}\left(\sum_{n<q^k}\exp\left(\frac{nb}{q^k}\right)\right)+O\left(\frac{q^k}{(\log k)^{3A}}\right).$$

If $b\neq 0$, the main term vanishes and $S\left(\frac{a}{q^k}\right)=O\left(\frac{q^k}{\left(\log q^k\right)^{3A}}\right)$. So, we will only have to worry about the terms $d=1$ or $d=q$.

Let's now sum all these major arcs. The contribution from the error terms of case 1 is

$$\frac{1}{q^k}+O\left(\frac{1}{q^k}\frac{q^k}{\left(\log q^k\right)^A}(q-1)^k\right)$$

using that $|\mathscr{A}(\alpha)|\leq(q-1)^k$. We have main terms

$$\frac{1}{q^k}q^k\,(q-1)^k$$

if $d=1$ and $b=0$

$$\frac{1}{q^k}\sum_{\substack{1\leq\ell\leq q-1\\ d=q\\ b=0}}\mathscr{A}\left(\frac{-\ell}{q}\left(\frac{-1}{q-1}q^k\right)\right).$$

Now we have to think a bit about the Fourier transform. We have

$$\begin{aligned}\mathscr{A}(\alpha)&=\sum_{\substack{0\leq n_0,n_1,\ldots,n_{k-1}\leq q-1\\ n_j\neq a_0}}\exp\left(\sum_j n_jq^j\alpha\right)\\&=\prod_{j=0}^{k-1}\left(\sum_{n_j\neq a_0}\exp\left(n_jq^j\alpha\right)\right).\end{aligned}$$

Then, splitting contributions from $j=1,\ldots,k-1$ and $j=0$, we get

$$\begin{aligned}\mathscr{A}\left(\frac{-\ell}{q}\right)&=(q-1)^{k-1}\left(\sum_{n_0=0}^{q-1}e\left(\frac{-n_0\ell}{q}\right)-\exp\left(\frac{-a_0\ell}{q}\right)\right)\\&=-\exp\left(\frac{-a_0\ell}{q}\right)(q-1)^{k-1}\end{aligned}$$

Adding all these main terms together we get

$$\frac{1}{q^k}q^k(q-1)^k + \frac{1}{q^k}\sum_{1\leq \ell \leq q-1} \mathscr{A}\left(\frac{-\ell}{q}\right)\left(\frac{-1}{q-1}q^k\right) + \frac{(q-1)^{k-1}}{(q-1)}\left(\sum_{\ell=1}^{q-1}\exp\left(\frac{-a_0\ell}{q}\right)\right)$$
$$= \begin{cases} (q-1)^k \frac{q}{q-1} & \text{if } a_0 = 0 \\ (q-1)^k\left(1-\frac{1}{(q-1)^2}\right) & \text{if } a_0 \neq 0 \end{cases}$$

11. 10/31/17

11.1. **review.** Let q be a large but fixed base and let $\mathscr{A}$ be the set of numbers missing $a_0 \in [0, q-1]$.

Goal 11.1. Count
$$\sum_{\substack{n<q^k \\ n\in\mathscr{A}}} \Lambda(n)$$
and show it is asymptotically $c_{a_0}(q)(q-1)^k$ for some constant $c_{a_0}(q) > 0$.

Let
$$\mathscr{A}(\alpha)\mathscr{A}_k(\alpha) := \sum_{n<q^k} 1_{\mathscr{A}}(n)e(n\alpha),$$
with
$$S(\alpha) = \sum_{n\leq q^k} \Lambda(n)e(n\alpha).$$
Our goal was to compute the discrete Fourier transform
$$\frac{1}{q^k}\sum_{a \bmod q^k} S\left(\frac{a}{q^k}\right)\mathscr{A}\left(\frac{-a}{q^k}\right) = \int_0^1 S(\alpha)\mathscr{A}(-\alpha)d\alpha.$$
We can approximate
$$\left|\frac{a}{q^k} - \frac{\ell}{d}\right| < \frac{1}{dq^{k/2}}$$
with $d \leq q^{k/2}$. Last time, we were trying to work out the major arc contributions over intervals with $d \leq \left(\log q^k\right)^A$ and
$$\left|\frac{a}{q^k} - \frac{\ell}{d}\right| \leq \frac{\left(\log q^k\right)^A}{q^k}.$$

Last time we computed the major arc contribution in the case d was a power of q less than $\left(\log q^k\right)^A$ where we got

$$c_{a_0}(q)\,(q-1)^k = \begin{cases} \frac{q}{q-1} & \text{if } a_0 = 0 \bmod q \\ 1 - \frac{1}{(q-1)^2} & \text{if } a_0 \neq 0. \end{cases}$$

11.2. **Remaining major arcs.** We next deal with the remaining major arcs. Namely, we show those centered at $\frac{\ell}{d}$ for d not a power of q are negligible. We'll put a trivial bound on S and the cancellation will come form $\mathscr{A}(\alpha)$.

We know

$$|S(\alpha)| \leq q^k\,(1 + o(1))$$

trivially. We now look for cancellation in

$$\mathscr{A}\left(\frac{a}{q^k}\right) = \mathscr{A}\left(\frac{\ell}{d} + c\right)$$

for c at most $\frac{\left(\log q^k\right)^A}{q^k}$, as we are on a major arc.

We have

$$\mathscr{A}_k(\alpha) = \prod_{j=0}^{k-1}\left(\sum_{n_j=0, n_j\neq a_0}^{q-1} e\left(n_j q^j \alpha\right)\right).$$

A crude L^∞ bound will in fact work for us, as we now explain.

We can bound

$$\begin{aligned}\left|\sum_{n_j\neq a_0} e\left(n_j\theta\right)\right| &\leq (q-3) + |e(n\theta) + e\left((n+1)\theta\right)| \\ &\leq (q-3) + 2\left|\cos(\pi\theta)\right| \\ &= (q-1) - 2\left(1 - |\cos \pi\theta|\right) \\ &\leq (q-1)\exp\left(-c_q\,||\theta||^2\right).\end{aligned}$$

for some small $c_q > 0$, where

$$||x|| = \min_{n\in\mathbb{Z}} |x-n|.$$

Then,

$$|\mathscr{A}_k(\alpha)| \leq (q-1)^k \exp\left(-c_q \sum_{j=0}^{k-1}\left|\left|q^j\alpha\right|\right|^2\right).$$

Assuming $\alpha = \frac{\ell}{d} + O\left(\frac{\left(\log q^k\right)^A}{q^k}\right)$, we get

$$\begin{aligned}|\mathscr{A}_k(\alpha)| &\leq (q-1)^k \exp\left(-c_q \sum_{j=0}^{k-1} \left|\left|q^j \alpha\right|\right|^2\right) \\ &\ll (q-1)^k \exp\left(-c_q \sum_{j=0}^{k/2} \left|\left|q^j \frac{\ell}{d}\right|\right|^2\right).\end{aligned}$$

Remark 11.2. Note that if $||\theta|| \leq \frac{1}{2q}$ then $||q\theta|| = q\,||\theta||$.

Lemma 11.3. *For k in an interval of length $\frac{\log d}{\log q} + 1$, we can find k_0 with*

$$\left|\left|q^{k_0} \frac{\ell}{d}\right|\right| \geq \frac{1}{2q}.$$

Proof. We have

$$\left|q^k \frac{\ell}{d}\right| \geq \frac{1}{d}$$

and now using Remark 11.2, we see that powers of q increase, and eventually the value "wraps around 1" so we get some term which is not too small. $\square$

Therefore,

$$|\mathscr{A}_k(\alpha)| \ll (q-1)^k \exp\left(-c_1 \frac{k}{\log k}\right).$$

Therefore, these other major arcs contribute

$$\frac{1}{q^k} \sum_{\substack{d < \left(\log q^k\right)^A \\ (\ell, d) = 1, \\ \text{d not a power of q}}} q^k (q-1)^k \exp\left(-c_q k / \log k\right) \ll (q-1)^k \left(\log q^k\right)^{2A} \exp\left(\frac{-c_q^k}{\log k}\right).$$

which is negligible compared to the main term computed last class.

11.3. **The minor arcs.** The real crux of the matter for dealing with minor arcs is that it is possible to get good bounds for the L^1 norm of $|\mathscr{A}(\alpha)|$. We want to bound either

$$\sum_{a \bmod q^k} \left|\mathscr{A}_k\left(\frac{a}{q^k}\right)\right|$$

or

$$\int_0^1 |\mathscr{A}_k(\alpha)|\, d\alpha.$$

There will be a huge amount of cancellation here.

Let's look at

$$|\mathscr{A}_k(\alpha)| = \prod_{j=0}^{k-1} \left| \sum_{n=0}^{q-1} e(n_j q^j \alpha) - e(a_0 q^j \alpha) \right|.$$

We'll now try to bound

$$\sum_{n=0}^{q-1} e(n_j q^j \alpha) - e(a_0 q^j \alpha)$$

Let $\theta := n_j q^j \alpha$. Then, we have

$$\sum_{n=0}^{q-1} e(n_j q^j \alpha) - e(a_0 q^j \alpha) \le \min \left(q-1, 1 + \left| \frac{1-e(q\theta)}{1-e(\theta)} \right| \right)$$

We have

$$\begin{aligned} \left| \frac{1-e(q\theta)}{1-e(\theta)} \right| &\le \frac{2}{2\,|\sin \pi\theta|} \\ &= \frac{1}{2\,||\theta||}. \end{aligned}$$

Therefore,

$$\sum_{n=0}^{q-1} e(n_j q^j \alpha) - e(a_0 q^j \alpha) \le \min \left(q-1, 1 + \left| \frac{1-e(q\theta)}{1-e(\theta)} \right| \right) \le \min \left(q-1, 1 + \frac{1}{2\,||\theta||} \right).$$

We now plug this in above for each $\theta = q^j \alpha$. We have

$$|\mathscr{A}_k(\alpha)| \le \prod_{j=0}^{k-1} \min \left(q-1, 1 + \frac{1}{2\,||q^j \alpha||} \right)$$

Let's write

$$\alpha = \frac{b_1}{q} + \frac{b_2}{q^2} + \frac{b_3}{q^3} + \cdots$$

for $0 \le b_j \le q-1$. Multiplying by $q^j \alpha$ yields

$$z + \frac{b_{j+1}}{q} + \varepsilon_j$$

with z and integer and $0 < \varepsilon_j \leq \frac{1}{q}$. Therefore, the distance to the nearest integer of $q^j\alpha$ is well determined by b_{j+1}. We have good bounds whenever $b_{j+1} \neq 0, q-1$, while at $b_{j+1} = 0, q-1$, we need to use the rather poor bound of $q-1$.

Putting the above together, we have

$$\begin{aligned} |\mathscr{A}_k(\alpha)| &\leq \prod_{j=0}^{k-1} \min\left(q-1, 1 + \frac{1}{2\,||q^j\alpha||}\right) \\ &= \begin{cases} q-1 & \text{if } b_{j+1} = 0 \text{ or } q-1 \\ 1 + \frac{q}{2b_{j+1}} & \text{if } 1 \leq b_{j+1} \leq \frac{q-1}{2} \\ 1 + \frac{q}{2(q-1-b_{j+1})} & \text{if } \frac{q-1}{2} \leq b_{j+1} \leq q-2. \end{cases} \end{aligned}$$

Plugging this in a summing over all possibilities for digits, we have

$$\begin{aligned} \sum_{a \bmod q^k} \left|\mathscr{A}_k\left(\frac{a}{q^k}\right)\right| &= \prod_{j=1}^{k}\left(\alpha(q-1) + \sum_{b=1}^{q-1/2}\left(2 + \frac{q}{b_i}\right)\right) \\ &= \prod_{j=1}^{k}\left(3q + q\log q\right). \end{aligned}$$

From the above, we have deduced the following lemma.

Lemma 11.4. *We have*

$$\sum_{q \bmod q^k} \left|\mathscr{A}\left(\frac{a}{q^k}\right)\right| \leq (3q + q\log q)^k$$

and

$$\int_0^1 |\mathscr{A}_k(\alpha)|\, d\alpha \leq (3 + \log q)^k.$$

So if q is large, there is a lot of cancellation, but if q is small, we won't get very much.

Before continuing the proof, let's motivate this. Recall that in our sum of three primes problem, we had some bound of the form

$$\sum_{n \leq x} \Lambda(n)e(n\alpha) \ll \left(x^{4/5} + x/\sqrt{q} + \sqrt{qx}\right)$$

Here $|\alpha - \frac{a}{q}| = \frac{1}{q^2}$ for $x^{.9x} \geq q > x^{.1}$. On the L^1 norm in this lemma, we're only using a very small power of x. As long as the denominators are not too small or large, we are doing well on the minor arcs.

Then, we want to bound

$$\frac{1}{q^k} \sum_a \left| \mathscr{A}_k(a/q^k) S(-a/q^k) \right|$$

where the latter is bounded by $q^k/(\log q^k)^A$ and the former is bounded by the lemma. This will work out when $d \geq q^{.01k}$ in ℓ/d. So we would be happy for large q.

The second idea will be used to estimate $\sum_{j=1}^N \left|\mathscr{A}_k(\alpha_j)\right|$ for relatively few values of α_j. We'll need an additional spacing condition that $\left|\alpha_i - \alpha_j\right| \geq \delta$ if $j \neq i$. This is natural because

$$\left|\frac{\ell_1}{d_1} - \frac{\ell_2}{d_2}\right| \geq \frac{1}{d_1 d_2}.$$

Estimates like this are called **large sieve estimates**. These are usually done in L^2, but here we'll do an L^1 estimate.

Lemma 11.5. *With the spacing condition that $\left|\alpha_i - \alpha_j\right| \geq \delta$ if $j \neq i$. we have*

$$\sum_{j=1}^N \left|\mathscr{A}_k(\alpha_j)\right| \ll \left(\frac{1}{\delta} + q^k\right)(3 + \log q)^k$$

Our hope to get a bound is something like $N \int_0^1 |\mathscr{A}(\alpha)|\, d\alpha$.

Remark 11.6 (Sobolev inequality)**.** We have

$$f(t) = f(u) - \int_t^u f'(v) dv.$$

Integrating both over $u \in \left(t - \frac{\delta}{2}, t + \frac{\delta}{2}\right)$, we have

$$\delta\, |f(t)| \leq \int_{t-\delta/2}^{t+\delta/2} |f(u)| + \int_{t-\delta/2}^{t+\delta/2} \delta |f'(v)| dv.$$

Then

$$|f(t)| \ll \frac{1}{\delta} \int_{t-\delta/2}^{t+\delta/2} |f(u)| du + \int_{t-\delta/2}^{t+\delta/2} \left|f'(v)\right| dv.$$

Since all points α_j were at least δ apart, these intervals will not overlap when proving the lemma.

Proof. We have the bound

$$\sum_{j=1}^{N} \left|\mathscr{A}_k(\alpha_j)\right| \ll \frac{1}{\delta}\int_0^1 \left|\mathscr{A}_k(\alpha)\right| d\alpha + \int_0^1 \left|\mathscr{A}_k'(\alpha)\right| d\alpha$$
$$\ll \frac{1}{\delta}\left(3+\log q\right)^k + \int_0^1 \left|\mathscr{A}_k'(\alpha)\right| d\alpha$$

Using

$$\mathscr{A}(\alpha) = \sum_{n<q^k} e(n\alpha) 1_A(n),$$

we get

$$\mathscr{A}'(\alpha) 2\pi i \sum_{n<q^k} n e(n\alpha) 1_A(n)$$

Writing

$$n = \sum_{j=0}^{k-1} n_j q^j$$

we have

$$\left|\mathscr{A}_j(\alpha)\right| \ll \sum_{j=0\neq a_0}^{k-1} \sum_{n_j} n_j q^j e\left(n_j q^j \alpha\right) \prod_{i=0, i\neq j}^{k-1} \left(\sum_{n_i \neq a_0} e\left(n_i q^i \alpha\right)\right)$$
$$\ll q^k \cdot B$$

where B was the bound for $\mathscr{A}(\alpha)$. Then, integrating it out, we again get a bound $q^k\left(3+\log q\right)^k$. □

The idea to finish the proof is to use the above lemma's bound with a different k. We will continue bounding the minor arcs next time using our lemmas with other values of k.

12. 11/2/17

12.1. **Review.** Last time, we wanted to evaluate

$$\frac{1}{q^k} \sum_{a \bmod q^k} \mathscr{A}_k\left(\frac{a}{q^k}\right) S\left(\frac{-a}{q^k}\right).$$

We had the major arcs which were

$$\left\{\frac{a}{q} : \frac{a}{q} = \frac{\ell}{d} + \eta, d = \left(\log q^k\right)^A, |\eta| \leq \frac{\left(\log q^k\right)^A}{q^k}\right\}.$$

We gave an asymptotic formula for these major arcs of the form

$$c_{a_0}(q)\,(q-1)^k.$$

This did not depend on the size of q and is true for any base at least 3 or so.

It remains to deal with the minor arcs. Last time, we saw we could estimate the L^1 norm of $\mathscr{A}_k$. We showed

$$\frac{1}{q^k}\sum\left|\mathscr{A}_k\left(\frac{a}{q^k}\right)\right| \ll (3+\log q)^k$$

We also found

$$\int_0^1 |\mathscr{A}_k(\alpha)|\,d\alpha \ll (3+\log q)^k.$$

By a Sobolev type argument, we saw that if

$$\alpha_1,\ldots,\alpha_N$$

are δ spaced (i.e., $\left|\alpha_i-\alpha_j\right| \geq \delta$ if $i \neq j$.

Then,

$$\sum_{j=1}^N \left|\mathscr{A}_k(\alpha_j)\right| \leq \left(\frac{1}{\delta}+q^k\right)(3+\log q)^k.$$

12.2. **Bounding the minor arcs.** Recall that by Dirichlet's theorem,

$$\left|\frac{a}{q^k}-\frac{\ell}{d}\right| \leq \frac{1}{dq^{k/2}}$$

using Dirichlet's theorem with $Q = q^{k/2}$, with $d \leq q^{k/2}$. We can assume $d \geq \left(\log q^k\right)^A$ as we are on the minor arcs.

For now, fix a choice of B and D (where we will split d into dyadic intervals based on D and $q^k|\eta|$ into dyadic intervals based on D). We will later range over different possibilities of D and B.

We will split this into terms with $D \leq d \leq 2D$. We won't worry about over-counting because we'll ultimately estimate things by taking absolute values. Write

$$\frac{a}{q^k} = \frac{\ell}{d}+\eta$$

and so that $B \leq q^k|\eta| \leq 2B$ and

$$q^k\left(\eta+\frac{\ell}{d}\right) \in \mathbb{Z}.$$

We also need to consider the case where $q^k|\eta| \leq 1$.

The number of choices for η with $q^k|\eta|$ between B and $2B$ is roughly $2B$ (since η can be negative). We have $D \leq q^{k/2}$. Then $q^k|\eta| \ll q^{k/2}/D$. We can assume

$$B \ll q^{k/2}/D.$$

Being on a minor arc means either

(1) $D \geq \left(\log q^k\right)^A$.
(2) or if D is small then $B \geq \left(\log q^k\right)^A$.

So, being on a minor arc means BD is somewhat large.

Goal 12.1. We now want to understand the contribution of one of these dyadic blocks.

We want to understand

$$\sum_{D \leq d \leq 2D} \sum_{(\ell,d)=1} \sum_{\substack{\eta \\ q^k|\eta| \sim B \\ q^k(\eta+\ell/d) \in \mathbb{Z}}} \left| \mathscr{A}_k\left(\frac{\ell}{d}+\eta\right)\right|.$$

This is a sum containing about D^2B terms. This number of terms in the sum is then at most $q^{k/2}D \ll q^k$ since $B \ll q^{k/2}/D$.

Recall that we are trying to estimate the number of primes up to q^k not containing the digit a_0. Now, our set $\mathscr{A}_k$ is self similar meaning

$$\mathscr{A}_k\left(\frac{\ell}{d}+\eta\right) = \sum_{\substack{n_0,n_1,\ldots,n_{k-1} \\ 0 \leq n_i \leq q-1 \\ n_j \neq a_0}} e\left(\left(n_0+n_1q+\cdots+n_{k-1}q^{k-1}\right)\alpha\right)$$

where $\alpha = \frac{a}{q} = \frac{\ell}{d}+\eta$.

Proposition 12.2. *We have*

$$\sum_{D \leq d \leq 2D} \sum_{(\ell,d)=1} \sum_{\substack{\eta \\ q^k|\eta| \sim B \\ q^k(\eta+\ell/d) \in \mathbb{Z}}} \left| \mathscr{A}_k\left(\frac{\ell}{d}+\eta\right)\right|$$
$$\ll (q-1)^k \left(D^2B\right)^{\alpha_q}$$

where

$$\alpha_q = \frac{\log\left(\frac{q}{q-1}\left(B+\log q\right)\right)}{\log q}$$

Then,

$$D^{2\alpha_q} = \left(q^{k_1}\right)^{\alpha_q} = \left(\frac{q}{q-1}\left(3+\log q\right)^{k_1}\right).$$

where $q^{k_1} \sim D^2, q^{k_2} \sim B$.

Proof. We'll now split this sum into

$$\begin{cases} \text{the first } k_1 \text{ digits} \\ \text{the middle } k-k_1-k_2 \text{ digits} \\ \text{the last } k_2 \text{ digits} \end{cases}$$

with $k_1 - 4k_2 \leq k$.

The first k_1 digits is dominated by $\left|\mathscr{A}_{k_1}(\alpha)\right|$. For the middle digits, there are $(q-1)^{k-k_1-k_2}$, each bounded by 1, so we get $(q-1)^{k-k_1-k_2}$ as a bound. For the last digits, we get

$$q^{k-k_2}\left(n_{k-k_2} + n_{k-k_2+1} + \cdots\right)\alpha.$$

Therefore, multiplying the contributions from all the digits, we get

$$e\left(\left(n_0 + n_1 q + \cdots + n_{k-1}q^{k-1}\right)\alpha\right) = \left|\mathscr{A}_{k_1}(\alpha)\right|(q-1)^{k-k_1-k_2}\left|\mathscr{A}_{k_2}\left(q^{k-k_2}\alpha\right)\right|.$$

Remark 12.3. Thinking about what we are doing, there are D^2 points of the form ℓ/d and about B points η near each ℓ/d. Given a fixed ℓ/d, we are multiplying it by something corresponding to each of the B well spaced B intervals. Then, we choose q^{k_1} on the scale of D^2 and q^{k_2} on the scale of B.

We can bound

$$\left|\mathscr{A}_k\left(\frac{\ell}{d}+\eta\right)\right| \ll (q-1)^{k-k_1-k_2}\left(\sup_{|\eta|\sim B/q^k}\left|\mathscr{A}_{k_1}\left(\frac{\ell}{d}+\eta\right)\right|\right)\left|\mathscr{A}_{k_2}\left(\frac{1}{q^{k_2}}\left(q^k\left(\frac{\ell}{d}+\eta\right)\right)\right)\right|.$$

Note that the last $\mathscr{A}_{k_2}$ term corresponds to B well spaced points mod 1. That is there are $B\,\frac{1}{q^{k_2}}$ (since we chose $q^{k_2} \sim B$, and in fact we will need $q^{k_2} < B$).

Now, we sum this over ℓ, d, η. By a lemma from last time, fixing ℓ, d and summing over η, we get

$$\sum_{\eta}\left|\mathscr{A}_{k_2}\left(\frac{1}{q^{k_2}}\left(q^k\left(\frac{\ell}{d}+\eta\right)\right)\right)\right| \sim q^{k_2}\left(3+\log q\right)^{k_2}.$$

Then, we want to compute

$$\begin{aligned} &(q-1)^{k-k_1-k_2} \sum_{\ell,d,d\sim D} \left(\sup_{|\eta|\sim B/q^k} \left| \mathscr{A}_{k_1}\left(\frac{\ell}{d}+\eta\right)\right|\right) \\ &\ll (q-1)^{k-k_1-k_2} q^{k_1} (3+\log q)^{k_1} \end{aligned}$$

using the lemma from last time again and the fact that rational numbers with denominator on the order of D are $\frac{1}{D^2}$ spaces. □

The bound of the above proposition yields a useful bound when D^2B is small compared to q^k, where this bound starts to beat the L^1 bound.

We want

$$\begin{aligned} &\sum_{d\sim D}\sum_{\ell}\sum_{\eta, q^k|\eta|\sim B} |\mathscr{A}_k(\ell/d+\eta)| \left| S\left(-\left(\frac{\ell}{d}+\eta\right)\right)\right| \\ &\ll (q-1)^k \left(D^2B\right)^{\alpha_q} \max_{d\sim D, q^k|\eta|\sim B} \left| S\left(\frac{\ell}{d}+\eta\right)\right|. \end{aligned}$$

We want this to be small compared to $(q-1)^k \cdot q^k$. We already know that for these points, the size of

$$\left| S\left(\frac{\ell}{d}+\eta\right)\right|$$

Recall

$$\left| S\left(\frac{\ell}{d}+\eta\right)\right| = |\sum_{n\le q^k} \Lambda(n) e(n\alpha)|.$$

Remark 12.4. Recall that from Vinogradov's theorem, we found

$$\sum_{n\le x} \lambda(n) e(n\alpha)$$

we had an approximation of the form

$$\left|\alpha - \frac{a}{q}\right| \le \frac{1}{q^2}$$

was bounded by

$$\left(x^{4/5} + \frac{x}{\sqrt{q}} + \sqrt{xq}\right)(\log x)^3.$$

We can use the same bound here.

We have $d \le q^{k/2}$ and $|\eta| \le \frac{1}{dq^{k/2}}$. Therefore,

$$\left| S\left(\frac{\ell}{d}+\eta\right)\right| \ll \left(q^{4k/5}+\frac{q^k}{\sqrt{D}}+\sqrt{q^k D}\right)\left(\log q^k\right)^3.$$

The only worry is that if D is small then there is an issue, because $1/\sqrt{D}$ is big. In this case, we would like to look for savings in B. So, this is not quite enough when D is small.

Recall that the approximations $\frac{\ell}{d}$ to $\frac{a}{q}$ are convergents of the continued fractions. We have

$$\left|\frac{a}{q^k}-\frac{\ell}{d}\right| \sim \frac{B}{q^k}.$$

Perhaps this approximation is not too good. We could try taking later approximations of continued fractions, taking the next convergent. Choose a modulus Q and pick an approximation $\frac{u}{v}$ with $v \le Q$ and

$$\left|\frac{a}{q^k}-\frac{u}{v}\right| \le \frac{1}{vQ}.$$

Arrange this so that $\frac{1}{dQ} \ll \frac{B}{10q^k}$. That is, choose $Q = \frac{10^3 q^k}{BD}$. Then, this u/v is not the same as ℓ/d because it is a closer approximation. Further,

$$\frac{1}{dv} \le \left|\frac{\ell}{d}-\frac{u}{v}\right| \le \frac{1}{vQ}+\frac{2B}{q^k}.$$

Further, $\frac{1}{vQ}$ is small compared to $\frac{1}{dv}$, and $\frac{1}{dv} \le 3\frac{2B}{q^k}$ and we get

$$\frac{10^3 q^k}{BD} \gg v \ge \frac{1}{10}\frac{q^k}{BD}.$$

So, in this case, we can redo our previous argument with a larger denominator. Using the bound from before, we see

$$\left| s\left(\frac{\ell}{d}+\eta\right)\right| \ll \left(q^{4k/5}+\frac{q^k}{\sqrt{BD}}\right)\left(\log q^k\right)^3.$$

using that

$$\frac{q^k\sqrt{BD}}{q^{k/2}} \ll q^{3k/4}$$

and so we can absorb the third term into $q^{4k/5}$.

We are now basically done. We know BD is at least some power of $\log q^k$. We want to find

$$\begin{aligned}
&\frac{1}{q^k} \sum_{\text{minor arcs}} \mathscr{A}_k\left(\frac{a}{q^k}\right) S\left(\frac{-a}{q^k}\right) \\
&\ll \left(\log q^k\right)^5 (q-1)^k \max_{D,B,DB \geq \left(\log q^k\right)^A} \left(\frac{(D^2B)^{\alpha_q}}{q^{k/5}} + \frac{\left(D^2B\right)^{\alpha_q}}{\sqrt{DB}}\right)
\end{aligned}$$

Now, $DB \leq q^{k/2}$ and $D \leq q^{k/2}$, so we just need $\alpha_q < \frac{1}{5}$ for the first term to be sufficiently small and we need $\alpha_q < \frac{1}{4}$ for the second term to be sufficiently small. We we just need to check $\alpha_q < \frac{1}{5}$.

Let's now examine this constraint. Recall

$$\alpha_q = \frac{\log\left(\frac{q}{q-1}\left(B + \log q\right)\right)}{\log q}$$

If q is sufficiently large, this will hold. Indeed, for $q > 2 \cdot 10^6$ this will hold.

For example, if $\alpha_q = .19$, the first term is bounded by $q^{-.01k}$ and the second term is bounded by $(DB)^{-.12} \leq \left(\log q^k\right)^{-.12A}$, and we can choose A as large as we want so that this savings dominates $\left(\log q^k\right)^5$.

Exercise 12.5. Work out any changes in the case that q is composite. *Hint:* There is essentially no difference. We only used some simplifications for computing the major arcs. We divided major arcs into cases that the denominators are powers of the modulus q. One would then have to work out differences when the denominator divides q, or something like that.

Remark 12.6. For further ideas along this line, look at Piatetski-Shapiro yielding primes of the form $\lfloor n^\alpha \rfloor$ for $\alpha = 1.01$.

13. 11/7/17

Today we'll start talking about something new, the Bombieri-Vinogradov Theorem. This tells us about the distribution of primes in arithmetic progressions. Let $\Psi(x; q, a)$ denote the number of primes up to x congruent to a mod q. Let $(a, q) = 1$. We are looking to estimate

$$E\left(x; q, a\right) := \Psi\left(x; q, a\right) - \frac{x}{\phi(q)}$$

The generalized Riemann hypothesis implies

$$|E(x; q, a)| \ll x^{1/2} (\log x)^2$$

which is good for $q \leq x^{1/2} / (\log x)^2$.

In conditionally, we'll need to include Siegel zeros.

Theorem 13.1 (Bombieri-Vinogradov)**.** *For every $A > 0$ there exists a $B > 0$ so that*

$$\sum_{q \leq Q} \max_{(a,q)=1} \max_{y \leq x} |E(y; q, a)| \ll \frac{x}{(\log x)^A}$$

provided $Q \leq \frac{x^{1/2}}{(\log x)^B}$.

Remark 13.2. The generalized Riemann hypothesis yields a bound of the form $Q \cdot x^{1/2} (\log x)^2$, which is essentially the same.

Remark 13.3. There is a trivial bound of the form

$$|E(x; q, a)| \ll \frac{x}{q} \log x$$

so one trivially obtains the bound $\ll x (\log x)^2$ trivially, and Bombieri-Vinogradov lets us save arbitrary powers of $\log x$.

Remark 13.4. The key ideas in the proof are

(1) Bilinear forms and Vaughn's identity
(2) Primes in progressions and Siegel zeros
(3) Large Sieve inequalities

13.1. **Large Sieve for Additive characters.** Suppose we have $\alpha_1, \ldots, \alpha_R \in \mathbb{R}/\mathbb{Z}$ which are δ well-spaces, i.e., $|\alpha_r - \alpha_s| \geq \delta$ for $r \neq s$.

Goal 13.5. Our goal is to bound

$$\sum_{j=1}^{R} \left| \sum_{n=1} N a_n e(n\alpha_j) \right|^2.$$

for $a_n \in \mathbb{C}$.

Instead of taking the sum in the above goal from $n = 1$ to N we can re-parameterize the sum as

$$\begin{aligned}
\sum_{n=M+1}^{M+N} a_n e(n\alpha) &= \sum_{n=1}^{N} a_{M+N} e((M+n)\alpha) \\
&= \sum_{n=1}^{N} a_{M+N} e(M\alpha) \cdot e(n\alpha)
\end{aligned}$$

so this is no more general.

Remark 13.6. We can bound

$$\left(\sum_n |a_n|\right)^2 \leq N \sum_n |a_n|^2$$

by Cauchy Schwartz and we shouldn't expect anything better than this.

Suppose on the other hand, that the a_n are "wiggling around in all directions randomly" and all have norm 1. If the a_n are behaving independently for different values of n, and in this case we might expect some kind of square-root cancellation. That is, we might have

$$\left(\sum |a_n|^2\right) \cdot R$$

Maybe in place of R, we might get $\frac{1}{\delta}$ because if R points were evenly spaced, we would be using square root cancellation and averaging.

We now want to get estimates of the above form. We'll prove something stronger, but here's a first pass:

Theorem 13.7. *We have*

$$\sum_{j=1}^{R} \left| \sum_{n=1}^{N} a_n e\left(n\alpha_j\right) \right|^2 \ll \left((N + \frac{1}{\delta}) \sum_{n=1}^{N} |a_n|^2 \right)$$

We give two proofs.

First Proof. The first step to prove this is a Sobolev argument. We have

$$\begin{aligned}\left| \sum_{n=1}^{N} a_n e\left(n\alpha_j\right) \right|^2 &\ll \frac{1}{\delta} \int_{\alpha_j - \delta/2}^{\alpha_j + \delta/2} \left|\sum a_n e(n\alpha)\right|^2 d\alpha \\ &\quad + \int_{\alpha_j - \delta/2}^{\alpha_j + \delta/2} \left| \left(\sum_n a_n e(n\alpha)\right) \left(\sum_n n a_n e(n\alpha)\right) \right| d\alpha.\end{aligned}$$

Summing from 1 to R, we have

$$\begin{aligned}\sum_{j=1}^{R}\left|\sum_{n=1}^{N} a_n e\left(n\alpha_j\right)\right|^2 &\ll \frac{1}{\delta}\int_0^1\left|\sum_n a_n e(n\alpha)\right|^2 d\alpha \\ &\quad+\left(\int_0^1\left|\sum_n a_n e(n\alpha)\right|^2 d\alpha\right)^{1/2}\left(\int_0^1\left|\sum_n n a_n e(n\alpha)\right|^2 d\alpha\right)^{1/2} \\ &\ll \frac{1}{\delta}\sum_n |a_n|^2+\left(\sum_n |a_n|^2\right)^{1/2}\left(\sum_n |n a_n|^2\right)^{1/2} \\ &\ll \frac{1}{\delta}\sum_n |a_n|^2+\left(\sum_n |a_n|^2\right)^{1/2}\left(N\left(\sum_n |a_n|^2\right)^{1/2}\right).\end{aligned}$$

using Parseval's identity. □

Second proof. This argument is based on duality. Say we have $(a_{m,n})_{M\times N}$ an $M\times N$ matrix. From this we can consider three kinds of objects.

(1)

$$\sum_{m=1}^{M}\left|\sum_{n=1}^{N} a_{mn} y_n\right|^2 \leq C\sum_{n=1}^{N}|y_n|^2$$

(2)

$$\left|\sum_{m=1}^{M}\sum_{n=1}^{N} a_{mn} x_m y n\right| C\left(\sum_m |x_m|^2\right)^{1/2}\left(\sum_n |y_n|^2\right)^{1/2}$$

(3)

$$\sum_{n=1}^{N}\left|\sum_{m=1}^{M} a_{mn} x_m\right|^2 \leq C\sum_m |x_m|^2.$$

Exercise 13.8. Show one of the above three inequalities holds for all choices of x, y if and only if the other two do. I.e., show the above three statements are equivalent.

By duality, in order to give the desired bound, it suffices to bound

$$\sum_{n=1}^{N}\left|\sum_{r=1}^{R} b_r e(n\alpha_r)\right|$$

in terms of the L^2 norm of b for all choices of b. Expanding this out, we get

$$\sum_{n=1}^{N}\left|\sum_{r=1}^{R} b_r e(n\alpha_r)\right| = \sum_{r,s\leq R} b_r\overline{b}_s \sum_{n=1}^{N} e\left(n\left(\alpha_r-\alpha_s\right)\right).$$

Since α_r and α_s are all well spaced, the terms in the exponentials won't be close to integers very often.

There are two types of terms, those with $r = s$, in which case we get a contribution of $N\sum_r |b_r|^2$.

There are also the off-diagonal terms with $r \neq s$. Here,

$$\left|\sum_{n=1}^{N} e\left(n\theta\right)\right| \ll \frac{1}{||\theta||}$$

where $||\theta||$ is the integer nearest to θ. We can then estimate the sum of the off diagonal terms by

$$\begin{aligned}
\sum_{r\neq s}\left(|b_r|^2+|b_s|^2\right)\frac{1}{||\alpha_r-\alpha_s||} &\ll \sum_r |b_r|^2\left(\sum_{s\neq r}\frac{1}{||\alpha_r-\alpha_s||}\right)\\
&\ll \left(\sum_r |b_r|^2\right)\left(\sum_{j=1}^{R}\frac{1}{j\delta}\right)\\
&\ll \left(\sum_r |b_r|^2\right)\left(\frac{1}{\delta}\log R\right)\\
&\ll \left(\sum_r |b_r|^2\right)\frac{1}{\delta}\log\frac{1}{\delta}.
\end{aligned}$$

using symmetry to bound the $|b_r|^2 + |b_s|^2$ by (an implicit factor of 2 times $|b_r|^2$.

So, we have proved, in the dual form, that

$$\sum_{n=1}^{N}\left|\sum_r b_r e(n\alpha_r)\right|^2 \leq \left(N + O\left(\frac{1}{\delta}\log\frac{1}{\delta}\right)\right)\sum_r |b_r|^2.$$

We now have an extra factor of log, and we will now explain how to remove this factor of log. We will set it up, but won't really carry it out.

Recall we were trying to estimate

$$\sum_{n=1}\left|\sum_{r=1}^{R} b_r e\left(n\alpha_r\right)\right|^2$$

Say we start with the characteristic function between $1, N$ and taking a smoothing Φ of this characteristic function supported on a small interval around $(1, N)$ and always positive. We instead try to estimate

$$\sum_{n=1}\Phi(n)\left|\sum_{r=1}^{R} b_r e\left(n\alpha_r\right)\right|^2$$

One could image one might be able to smooth on an the interval

$$\left(1-\frac{1}{\delta}, N+\frac{1}{\delta}\right)$$

so that Φ is supported on this interval. We have

$$\begin{aligned}\sum_{n=1}\Phi(n)\left|\sum_{r=1}^{R} b_r e\left(n\alpha_r\right)\right|^2 &= \sum_{r,s} b_r\overline{b}_s \sum_{n}\Phi(n) e\left(n\left(\alpha_r-\alpha_s\right)\right)\\ &= \sum_{r,s} b_r\overline{b}_s \sum_{k}\hat{\Phi}\left(k+\alpha_r-\alpha_s\right).\end{aligned}$$

using Poisson summation. The Fourier transform is large at 0 (around $N+\frac{2}{\delta}$). You can get the rate of decay by integrating by parts many times. One can learn about the decay from the derivative of Φ. The Fourier transform is approximately supported on an interval of length δ , apart from some small fluctuations. Since $\hat{\Phi}(k+\alpha_r-\alpha_s)$ never gets within δ of an integer, it is always close to 0 when $r \neq s$. Therefore, including the contribution at $r = s$, we get the sum is well estimated by

$$\hat{\Phi}(0)\sum_{r=1}^{R}|b_r|^2$$

and we save the log term.

Exercise 13.9 (Involved exercise)**.** Complete the above sketch into a proof

$\square$

Remark 13.10. Here is a problem: Can on e choose $\Phi \geq 0, \Phi \geq \chi_{[1,N]}$ and $\hat{\Phi}$ supported in $(-\delta, \delta)$ minimizing $\hat{\Phi}(0)$?

There is a solution discovered by Beurling and Selberg. One obtains something like $\hat{\Phi}(0) \leq N + \frac{1}{\delta} - 1$.

We will next deduce the large sieve from the above theorem. Say we have

$$\left\{ \frac{a}{q} : q \leq Q, (a, q) = 1 \right\}$$

which is about Q^2 points each $\frac{1}{Q^2}$ spaced.

$$\sum_{q=1}^{Q} \sideset{}{^*}\sum_{a \bmod q} \left| \sum_{n=M+1}^{M+N} a_n e\left(\frac{an}{q}\right) \right|^2 \leq \left(N + O(Q^2)\right) \sum_{n=M+1}^{M+N} |a_n|^2.$$

Example 13.11 (Important example)**.** Take $a_n = 1$ if $n \in [M+1, M+N]$ is prime and 0 otherwise. To examine the left hand side,

$$\begin{aligned}
\sideset{}{^*}\sum_{a \bmod q} \sum_{n \text{ prime } \in [M+1, M+N]} e\left(\frac{an}{q}\right) &= \sum_{n \in [M+1, M+N], n \text{ prime}} \sideset{}{^*}\sum_{a \bmod q} e\left(\frac{an}{q}\right) \\
&= \sum_{n \in [M+1, M+N], n \text{ prime}} \mu(q) \\
&= \mu(q) \left(\pi(M+N) - \pi(M)\right)
\end{aligned}$$

where $\pi(k)$ is the number of primes up to k. Using Cauchy-Schwartz, we get

$$\phi(q) \sideset{}{^*}\sum_{a \bmod q} \left| \sum_{nprime} e\left(\frac{an}{q}\right) \right|^2 \geq \mu(q)^2 \left(\pi(M+N) - \pi(M)\right)^2.$$

Combining the above, the left hand side of the Large sieve is bounded by

$$\sum_{q=1}^{Q} \sideset{}{^*}\sum_{a \bmod q} \left| \sum_{n=M+1}^{M+N} a_n e\left(\frac{an}{q}\right) \right|^2 \geq \sum_{q \leq Q} \frac{\mu(q)^2}{\phi}(q) \left(\pi(M+N) - \pi(M)\right)^2$$

and the large sieve implies

$$\sum_{q \leq Q} \frac{\mu(q)^2}{\phi}(q) \left(\pi(M+N) - \pi(M)\right)^2 \leq \left(N + O(Q^2)\right) \left(\pi\left(M+N\right) - \pi(M)\right).$$

This implies that the number of primes in the interval M to $M+N$ is bounded by

$$\left(N + O\left(Q^2\right)\right)\left(\sum_{q \leq Q} \frac{\mu(q)^2}{\phi(q)}\right)^{-1}.$$

If we make the Q too big, this $O(Q^2)$ term will start to dominate. So, we might want Q^2 to be something like $Q = N^{1/2-\varepsilon}$. The bound then becomes

$$N\left(1 + o(1)\right) \frac{1}{\sum_{q \leq N^{1/2-\varepsilon}} \mu(q)^2/\phi(q)}.$$

Exercise 13.12. Show

$$\sum_{n \leq x} \frac{\mu(n)^2}{\phi(n)} \sim \log x.$$

Then, the number of primes between N and $M+N$ yields a bound of

$$N\left(1 + o(1)\right) \frac{1}{\sum_{q \leq N^{1/2-\varepsilon}} \mu(q)^2/\phi(q)} \leq \frac{2N(1+o(1))}{\log N}$$

This yields

Theorem 13.13 (Brun-Titchmarsh theorem)**.** *We have*

$$\pi(M+N) - \pi(M) \leq \frac{2\left(1 + o(1)\right)N}{\log N}.$$

Remark 13.14. The constants in this inequality can be made explicit. In fact, one can replace

$$\pi(M+N) - \pi(M) \leq \frac{2\left(1 + o(1)\right)N}{\log N}.$$

by

$$\pi(M+N) - \pi(M) \leq \frac{2N}{\log N}.$$

without any error terms. That is, the number of primes from M to $M+N$ is no more than twice the number of primes from 1 to N.

Remark 13.15. In fact, one might expect $\pi(x) + \pi(y) \geq \pi(x+y)$. This contradicts a conjecture of Hardy and Littlewood, so is expected to be false.

Exercise 13.16. Generalize the Brun-Titchmarsh theorem as follows. Use the Large Sieve appropriately to show

$$\pi\left(x;q,a\right) \leq \frac{x\left(2+o(1)\right)}{\phi(q)\log(x/q)}.$$

For example if $x = q^{1,000,000}$. Then, $\pi(x;q,a)$ is at most 2.00001 times the expected number of primes. I.e.,

$$\pi\left(x;q,a\right) \leq (2.000001)\frac{x}{\phi(q)\log x}.$$

This constant more than 2 is significant because of Siegel zeros. Then,

$$\Psi(x;q,a) = \frac{x}{\phi(q)} - \chi(a)\frac{x^{\beta}}{\phi(q)\beta}$$

for χ a quadratic character. If one could replace the 2 by 1.99 one would imply there are no Siegel zeros.

Exercise 13.17 (What is large about the large sieve)**.** For primes we used that one residue class is forbidden and so we get some imbalances. Now, more generally, suppose we have $S \subset [1, N]$ with $|S(\operatorname{mod} p)| \leq \frac{p+1}{2}$. Use the Large sieve to show

$$|S| \leq N^{1/2+\varepsilon}.$$

Here, the sieve is large because we are forbidding a large number of residue classes. Say here the primes p range up to $p \leq \sqrt{N}$.

Remark 13.18. This bound is tight because if we take S to be the set of squares, we get the claimed number of residue classes.

Remark 13.19. There is a conjecture of Helfgott and Venkatesh saying that if one is missing half the residue classes and do have half the residue classes, it should look like some quadratic polynomial.

14. 11/9/17

Last time we discussed the Large sieve in its additive form. That is, if $\alpha_1, \ldots, \alpha_R$ are δ well spaced, then

$$\sum_{r=1}^{R}\left|\sum_{M+1}^{M+N} a_n e(n\alpha_r)\right| \leq \left(N + O\left(\frac{1}{\delta}\right)\right)\sum |a_n|^2.$$

One way we'll apply this is by taking $\frac{a}{q}$ with $(a,q)=1, q \le Q, R = Q^2, \delta \sim \frac{1}{Q^2}$ and obtaining

$$\sum_{q \le Q} \sideset{}{^*}\sum_{a \bmod q} \left| \sum_{n=M+1}^{M+N} a(n) e\left(\frac{an}{q}\right) \right|^2 \le \left(N + O(Q^2)\right) \sum |a(n)|^2 .$$

14.1. **A multiplicative version of the large sieve.** We'll now formulate the large sieve in a multiplicative form in order to prove Bombieri Vinogradov. We'll average over all characters χ mod q and sum over $q \le Q$.

$$\sum_{q \le Q} \sideset{}{^*}\sum_{\chi \bmod q} \left| \sum_{n=M+1}^{M+N} a(n) \chi(n) \right|^2 \le \left(N + O(Q^2)\right) \sum |a(n)|^2$$

Remark 14.1. Here the term of size N corresponds to a particular "bad character." and the Q^2 corresponds to the sum over the remaining characters with square-root savings bounding by some L^2 norm. We'd like to think characters of different moduli are orthogonal to each other, but we don't want to recount characters, so we have the star on our sum to indicate we are summing over primitive characters χ (not induced by characters of smaller modulus).

In fact, we'll obtain something slightly more precise than the above. We want to go from multiplicative characters to something involving additive characters. We'll want to pass between $\chi(n)$ and $e\left(\frac{n}{q}\right)$. Let χ mod q be a primitive character. Let

$$\tau(\chi) = \sum_{a \bmod q} \chi(a) e\left(\frac{a}{q}\right)$$

be the Gauss sum. Suppose $(n,q)=1$. Then, consider

$$\begin{aligned} \sum_{a \bmod q} \chi(a) e\left(\frac{an}{q}\right) &= \sum_{a \bmod q} \chi(a) e\left(\frac{an}{q}\right) \chi(n)\overline{\chi}(n) \\ &= \tau(\chi)\overline{\chi}(n). \end{aligned}$$

noting that $\chi(n)\overline{\chi}(n) = 1$ if n is coprime to q. Then,

$$\chi(n) = \frac{1}{\tau(\overline{\chi})} \sum_{a \bmod q} \overline{\chi}(a) e\left(\frac{an}{q}\right).$$

This holds for all χ mod q so long as $(n,q)=1$.

Exercise 14.2. If χ is primitive, then in fact the above equality is true for all n. That is, if n has factor in common with q, then the left hand side is 0, and we have to check the right hand side is also zero so long as χ is primitive.

For example, consider q a prime. Then, every character except the principal character has right hand side evaluating to 0 when $q \mid n$.

We have

$$\sum_{n=M+1}^{M+N} a(n)\chi(n) = \frac{1}{\tau(\overline{\chi})} \sum_{a \bmod q} \overline{\chi}(a) \sum_{n=M+1}^{M+N} a(n) e\left(\frac{an}{q}\right).$$

Let

$$S\left(\frac{a}{q}\right) := \sum_{n=M+1}^{M+N} a(n) e\left(\frac{an}{q}\right).$$

We wanted to bound

$$\begin{aligned}
\sideset{}{^*}\sum_{\chi \bmod q} \left|\sum a(n)\chi(n)\right|^2 &= \frac{1}{q} \sideset{}{^*}\sum_{\chi \bmod q} \left| \sum_{a \bmod q} \overline{\chi}(a) S(\frac{a}{q}) \right|^2 \\
&\leq \frac{1}{q} \sum_{\chi \bmod q} \left| \sum_{a \bmod q} \overline{\chi}(a) S(\frac{a}{q}) \right|^2 \\
&= \frac{\phi(q)}{q} \sideset{}{^*}\sum_{a \bmod q} \left| S\left(\frac{a}{q}\right) \right|^2.
\end{aligned}$$

using that $|\tau(\overline{\chi})| = \sqrt{q}$.

So, using the above and the large sieve,

$$\begin{aligned}
\sum_{q \leq Q} \frac{q}{\phi(q)} \sideset{}{^*}\sum_{\chi \bmod q} \left| \sum_{n=M+1}^{M+N} a(n)\chi(n) \right|^2 &\leq \sum_{q \leq Q} \sideset{}{^*}\sum_{a \bmod q} \left| S\left(\frac{a}{q}\right) \right|^2 \\
&\leq \left(N + O(Q^2)\right) \sum_{n=M+1}^{M+N} |a(n)|^2.
\end{aligned}$$

Remark 14.3. The idea is that we are estimating some quantity on average, and one term is very bad and the rest of the terms have square-root cancellation.

14.2. **Proving Bombieri Vinogradov.** Let $Q = \frac{\sqrt{x}}{(\log x)^B}$. Our goal is to bound

$$\sum_{q \le Q} \max_{(a,q)=1} \left| \Psi(x; q, a) - \frac{x}{\phi(q)} \right| \ll \frac{x}{(\log x)^2}.$$

In our original statement, we also had a maximum over y up to x, which we will forget about, as it is not so important.

Recall

$$\begin{aligned} \Psi(x; q, a) &= \frac{1}{\phi(q)} \sum_{\chi} \overline{\chi}(a) \Psi(x, \chi) \\ &= \frac{1}{\phi(q)} \sum_{\chi \neq \chi_0} \overline{\chi}(a) \Psi(x, \chi) - \frac{x}{\phi(q)} + O\left(\frac{x}{\phi(q) (\log x)^{A+100}} \right) \end{aligned}$$

and this error term is bounded using

$$\sum_{q \le Q} \frac{1}{\phi(q)} \ll \log x.$$

Here we are using

$$1_{n \equiv a \bmod q} = \frac{1}{\phi(q)} \sum \overline{\chi}(a) \chi(n)$$

and so

$$\Psi(x; q, a) = \frac{1}{\phi(q)} \sum_{\chi} \overline{\chi}(a) \sum_{n \le x} \Lambda(n) \chi(n).$$

Then, we get

$$\max_{(a,q)=1} \left| \Psi(x; q, a) - \frac{x}{\phi(q)} \right| \le \frac{1}{\phi(q)} \sum_{\chi \neq \chi_0} |\Psi(x, \chi)|$$

Suppose χ mod q is induced by some primitive character $\widetilde{\chi}$ mod $\widetilde{q}$. We'll assume $\widetilde{q} > 1$ so the principal character does not show up, and then $\widetilde{q} \mid q$. Then,

$$\sum_{q \le Q} \frac{1}{\phi(q)} \sum_{\chi \bmod q, \chi \neq \chi_0} |\Psi(x, \chi)| = \sum_{1 < \widetilde{q} \le Q} \sideset{}{^*}\sum_{\widetilde{\chi} \bmod \widetilde{q}} \sum_{q \le Q, \widetilde{q} | q} \frac{1}{\phi(q)} |\Psi(x, \chi)|.$$

We have $\chi(n) = \widetilde{\chi}(n)$ if $(n, q) = 1$. If $(n, q) > 1$ bun $(n, \widetilde{q}) = 1$ then the two could be different.

We have the bound

$$|\Psi(x,\chi) - \Psi(x,\widetilde{\chi})| = \sum_{\substack{n \leq x \\ (n,q)>1 \\ (n,\widetilde{q})=1}} \Lambda(n) \ll \log x \# \{p \mid q : p \nmid \widetilde{q}\}$$
$$\ll (\log x)^2 .$$

Exercise 14.4. Show that we can bound

$$\sum_{1<\widetilde{q}\leq Q} \sideset{}{^*}\sum_{\widetilde{\chi} \bmod \widetilde{q}} \sum_{q \leq Q, \widetilde{q} \mid q} \frac{1}{\phi(q)} |\Psi(x,\chi)| .$$

by

$$\sum_{q<\widetilde{q}\leq Q} \sideset{}{^*}\sum_{\widetilde{\chi} \bmod \widetilde{q}\, \widetilde{q}} \sum_{\bmod q, q \leq Q} \frac{1}{\phi(q)} |\Psi(x,\widetilde{\chi})| + O\left(Q (\log x)^3\right) \tag{14.1}$$

using the bound on the difference between $\Psi(x,\chi)$ and $\Psi(x,\widetilde{\chi})$ above, and so it suffices to bound Equation 14.1.

The point is that the difference is bounded by $(\log x)^2$ the number of characters is $\phi(\widetilde{q})$ which cancels out and we get a factor of Q and three factors of log.

Then, bound the inner most sum by

$$\sum_{r \leq Q/\widetilde{q}} \frac{1}{\phi(\widetilde{q}r)} \ll \frac{1}{\phi(\widetilde{q})} \log x$$
$$\ll \frac{1}{\widetilde{q}} (\log x)^2 .$$

Warning 14.5. Now, we replace $\widetilde{q}$ by q to avoid writing lots of tildes.

Let's break $q \leq Q$ into dyadic blocks

$$R \leq q \leq 2R$$

with $R \leq Q$. There are on the order of $\log x$ such blocks. We can bound Equation 14.1 by

$$(\log x)^3 \max_{R \leq Q} \frac{1}{R} \sum_{R \leq q \leq 2R} \sideset{}{^*}\sum_{\chi \bmod q} |\Psi(x,\chi)| . \tag{14.2}$$

We now have two cases.

(1) R is small so $R \leq (\log x)^{10A}$.
(2) R is large (here we will use the large sieve)

Remark 14.6. We'd like to bound

$$\sum_{q \leq \sqrt{x}/(\log x)^B} \max_{\alpha} |E(x;q,a)| \ll \frac{x}{(\log x)^A}.$$

Even if we are only interested in $q \geq x^{1/3}$, we still will have to deal with small moduli because of imprimitive characters. We could avoid dealing with small moduli if we only sum over primes.

Exercise 14.7. Work out a Bombieri Vinogradov theorem in the range $x^{1/3}$ to $Q = \frac{\sqrt{x}}{(\log x)^B}$ for integers, all of whose prime factors are bigger than $x^{1/10}$.

14.3. **The case R is small.** Here we can use Siegel zeros and what we know about zero-free regions. If $q \leq (\log x)^{10A}$ then

$$|\Psi(x,\chi)| \ll \frac{x}{(\log x)^{100A+100}}$$

using Siegel's theorem.

These easily yield a bound of Equation 14.2 by

$$\frac{x}{(\log x)^{10A+10}}.$$

14.4. **The case R is large.** Now, let's deal with the range

$$(\log x)^{10A} \leq R \leq \frac{\sqrt{x}}{(\log x)\,B}.$$

We now use the trick of decomposing $\Lambda(n)$ via Vaughan's identity. We have

$$\sum \Lambda(n) e(n\alpha)$$

which by Vaughan's identity yields a good bound

$$\sum_m \sum_n a_m b_n e(mn\alpha).$$

There is an issue that if we only wanted to estimate $\sum_n \Lambda(n)\chi(n)$ for one character χ we could get something like $\sum_{m,n} a_m b_n \chi(m)\chi(n)$. But, we're only trying to average over all characters Q over ranging R. The idea is now to write down Vaughan's identity and the use the large sieve.

Recall Vaughan's identity says that for

$$P(s) = \leq_{m \leq U} \frac{\Lambda(n)}{n^s}$$
$$M(s) = \sum_{n \leq V} \frac{\mu(n)}{n^s}$$

we have

$$\frac{-\zeta'}{\zeta}(s) = \left(\frac{-\zeta'}{\zeta}(s) - P(s)\right)(1 - \zeta(s)M(s)) + P(s) - \zeta'(s)M(s) - \zeta(s)M(s)P(s)$$

with the first term on the right a type 2 sum and the latter three terms Type 1 sums.

We'd now like to try to bound what all these terms give us.

First, let's consider the type 2 sum.

14.5. **Bounding the type 2 sum.** Recall we are trying to bound some sum of terms of the form $\sum_n \Lambda(n)\chi(n)$.

Expanding $\Lambda(n)$ using Vaughan's identity, we get some bound of the form

$$\sum_{R \leq q \leq 2R} \sideset{}{^*}\sum_{\chi \bmod q} \sum_{m > U} \sum_{n > V, mn \leq x} \Lambda(m) + \left(\sum_{d \mid n, d > V} \mu(d)\right) \chi(m)\chi(n).$$

But note that the term

$$\sum_{d \mid n, d > V} \mu(d)$$

is bounded by $d(n)$, so we can essentially ignore this.

Exercise 14.8. Use a Perron type integral to separate the variables m and n. Then, group them into dyadic blocks with $M \leq m \leq 2M$, $N \leq n \leq 2N$ with the conditions $M \geq U, N \geq V, MN \ll x$ to remove the dependence $mn \leq x$.

Then, using the above, show

$$\begin{aligned}
&\sum_{R \leq q \leq 2R} \sideset{}{^*}\sum_{\chi \bmod q} \sum_{m > U} \sum_{n > V, mn \leq x} \Lambda(m) + \left(\sum_{d \mid n, d > V} \mu(d)\right) \chi(m)\chi(n) \\
&\ll (\log x)^3 \sum_{q \sim R} \sideset{}{^*}\sum_{\chi \bmod q} \left(\sum_{m \sim M} \Lambda(m)\chi(m)\right)\left(\sum_{n \sim N} a(n)\chi(n)\right).
\end{aligned}$$

We now use Cauchy-Schwartz and $\sum_{m\sim M}\Lambda(m)^2 \ll M\log M, \sum_{n\sim N} d(n)^2 \ll N\left(\log N\right)^3$ to obtain

$$\begin{aligned}
&(\log x)^3 \sum_{q\sim R}\sideset{}{^*}\sum_{\chi \bmod q}\left(\sum_{m\sim M}\Lambda(m)\chi(m)\right)\left(\sum_{n\sim N} a(n)\chi(n)\right)\\
&\ll (\log x)^3\left(\sum_q \sideset{}{^*}\sum_\chi \left|\sum_{m\sim M}\Lambda(m)\chi(m)\right|^2\right)^{1/2}\left(\sum_q \sideset{}{^*}\sum_\chi \left|\sum_{n\sim N} a(n)\chi(n)\right|^2\right)^{1/2}\\
&\ll (\log x)^3\left(\left(M+R^2\right)\sum_{m\sim M}\Lambda(m)^2\right)^{1/2}\left(\left(N+R^2\right)\sum_{n\sim N} d(n)^2\right)^{1/2}\\
&\ll (\log x)^5\left\{M^2+MR^2\right\}^{1/2}\left\{N^2+NR^2\right\}^{1/2}\\
&\ll (\log x)^5\left\{MN+\frac{MN}{\sqrt{M}}R+\frac{MN}{\sqrt{N}}R+\sqrt{MN}R^2\right\}\\
&\ll (\log x)^5\left\{x+\frac{xR}{\sqrt{U}}+\frac{xR}{\sqrt{V}}+\sqrt{x}R^2\right\}.
\end{aligned}$$

Then, taking $U = V = x^{1/10}$, we see

$$\begin{aligned}
&\max_{(\log x)^{10A}\le R\le\sqrt{x}/(\log x)^B} \frac{1}{R}(\log x)^5\left\{x+\frac{xR}{\sqrt{U}}+\frac{xR}{\sqrt{V}}+\sqrt{x}R^2\right\}\\
&\ll \frac{x}{\log x}^A + x(\log x)^5\left(\frac{1}{\sqrt{U}}+\frac{1}{\sqrt{V}}\right)+\sqrt{x}(\log x)^5 5\frac{\sqrt{x}}{(\log x)^B}
\end{aligned}$$

For $B > A + 10$ or so we can bound all the terms by $\sqrt{x}/(\log x)^A$. So, this completes the type 2 sum case.

It only remains to deal with the type 1 sum case. We'll do the trivial type 1 sum, which comes from $P(s)$. This is

$$\sum_{u\le U}\frac{\Lambda(n)}{n^s}.$$

We then have to bound

$$\begin{aligned}
\frac{1}{R}\sum_{q\sim R}\sideset{}{^*}\sum_{\chi \bmod q}\left|\sum_{n\le U}\Lambda(n)\chi(n)\right| &\ll UR\\
&\ll U\sqrt{s}.
\end{aligned}$$

Since U is small, around $x^{1/10}$, we have bounded this sum.

Next time we'll deal with the other two terms. Let's just give an idea of how to deal with one of them now. We're trying to bound the term corresponding to

$$\zeta(s)M(s)P(s).$$

We want to bound

$$\sum_{q\sim R}\sideset{}{^*}\sum_{\chi \bmod q}\sum_{m\leq U, n\leq V}\Lambda(m)\mu(n)\sum_{k\leq x/mn}\chi(k)\chi(m)\chi(n).$$

We then have the problem of evaluating

$$\sum_{k\leq x/mn}\chi(k)$$

which is certainly bounded by q, and we'd like to even improve this a bit to $\sqrt{q}$, plug it in, and take trivial estimates on everything.

15. 11/14/17

15.1. **Brun-Titchmarsh.** Recall a few classes ago, we were trying to bound $\pi(M+N)-\pi(M)$. We wanted to bound

$$\sideset{}{^*}\sum_{a \bmod q}\sum_{M+1\leq p\leq M+N} e\left(\frac{ap}{q}\right) = \mu(q)\left(\pi(M+N)-\pi(M)\right).$$

We used Cauchy Schwartz to bound

$$\mu(q)^2\left(\pi(M+N)-\pi(M)\right)^2 \leq \left(\sideset{}{^*}\sum_{a \bmod q} 1\right)\left(\sum_{a \bmod q}\left|\sum_{M+1\leq p\leq r+N} e\left(\frac{ap}{q}\right)\right|^2\right).$$

One then gets a bound to which one can now use the large sieve.

15.2. **Back to Bombieri Vinogradov.** Recall we have reduced to proof to bounding

$$(\log x)^3 \max_{R\leq\sqrt{x}/(\log x)^B}\frac{1}{R}\sum_{R\leq q\leq 2R}\sideset{}{^*}\sum_{\chi \bmod q}\left|\Psi\left(x;\chi\right)\right|.$$

We had two ranges. If R small, like $R \leq (\log x)^{10A}$ we could use our bounds for $|\Psi(x,\chi)|$. To conclude, we only needed to deal with

$$(\log x)^{10A} \leq R \leq \frac{\sqrt{x}}{(\log x)^B}$$

using Vaughan's identity and the large sieve.

Recall

$$\left(\frac{-\zeta'}{\zeta}(s) - P(s)\right)(1 - \zeta M(s)) = -\frac{\zeta'}{\zeta}(s) - P(s) + \zeta'(s)M(s) + \zeta(s)M(s)P(s).$$

with

$$P = \sum_{n \le U} \frac{\Lambda(n)}{n^s}$$
$$M = \sum_{n \le V} \frac{\mu(n)}{n^s}$$

We were able to bound the type 2 sum by

$$\ll (\log x)^5 \left(\frac{x}{R} + \frac{x}{\sqrt{U}} + \frac{x}{\sqrt{V}} + \sqrt{x}R\right) \ll \frac{x}{(\log x)^A}$$

We also bounded P by $UR \ll x^{.6}$. Next, we bound the type 1 sum ζMP given by

$$\frac{1}{R} \sum_{R \le q \le 2R} \sideset{}{^*}\sum_{\chi \bmod q} \left| \sum_{m \le U, n \le V} \Lambda(m)\mu(n) \sum_{k \le x/mn} \chi(kmn) \right|.$$

We'll also bound this crudely, forgetting about the sums over M and N, and get cancellation from the sum over k.

15.3. **Polya Vinogradov theorem.** We'll prove the Polya Vinogradov theorem:

Theorem 15.1 (Polya-Vinogradov Theorem)**.** *Suppose* $\chi \bmod q$ *is primitive. Then,*

$$\max_x \left| \sum_{n \le x} \chi(n) \right| \ll \sqrt{q} \log q$$

Remark 15.2. Assuming Polya Vinogradov, we can bound the type 1 sum ζMP by

$$\begin{aligned}
&\frac{1}{R} \sum_{R \le q \le 2R} \sideset{}{^*}\sum_{\chi \bmod q} \left| \sum_{m \le U, n \le V} \Lambda(m)\mu(n) \sum_{k \le x/mn} \chi(kmn) \right| \\
&\ll \frac{1}{R} R^2 UV \sqrt{R} \log R \\
&\ll x^{1-\varepsilon}.
\end{aligned}$$

Exercise 15.3. Show
$$\sum_{n \leq x} \chi(n) \log n \ll \sqrt{q} (\log q) (\log x)$$
using Partial summation. *Hint:* Consider
$$\int_1^x \left(\sum_{n \leq t} \chi(n) \right) \frac{dt}{t}$$
and obtain a $\log n$ from this integral.

Exercise 15.4. Bound $\zeta'(s)M(s)$ in a similar way using the previous exercise. In fact, one can bound this term by
$$\zeta'(s)M(s) \ll \frac{1}{R} R^2 V \sqrt{R} (\log R) (\log x).$$

It only remains to prove the Polya-Vinogradov Theorem.

Proof. The idea is to rewrite the character χ in terms of additive characters.
$$\sum_{a \bmod q} \overline{\chi}(a) e\left(\frac{an}{q}\right)$$
$$= \chi(n) \sum_{a \bmod q} \overline{\chi}(a)\overline{\chi}(n) e\left(\frac{an}{q}\right) \quad = \tau(\overline{\chi})\chi(n)$$
This yields,
$$\chi(n) = \frac{1}{\tau(\overline{\chi})} \sum_{a \bmod q} \overline{\chi}(a) e\left(\frac{an}{q}\right).$$
therefore, we have
$$\sum_{n \leq x} \chi(n) = \frac{1}{\tau(\overline{\chi})} \sum_{a \bmod q} \overline{\chi}(a) \sum_{n \leq x} e\left(\frac{an}{q}\right).$$
Summing
$$\sum_{n \leq x} e\left(\frac{an}{q}\right)$$
as a progression, we have
$$\sum_{n \leq x} e\left(\frac{an}{q}\right) \leq \min\left(x, \frac{1}{\left|\left|\frac{a}{q}\right|\right|}\right).$$

We also have

$$|\tau(\overline{\chi})| = O(\frac{1}{\sqrt{q}}).$$

We know $\sum_{n\leq x}\chi(n) \leq x$. Say $-q/2 \leq a \leq q/2$. Then, $\left|\left|\frac{a}{q}\right|\right| \sim \frac{|a|}{q}$ and we use the bound x if $|a| \leq \frac{q}{a}$ and the bound $q/|a|$ if $|a| > q/x$. Therefore, we obtain a bound

$$\begin{aligned}
\sum_{n\leq x}\chi(n) &= \frac{1}{\tau(\overline{\chi})}\sum_{a \bmod q}\overline{\chi}(a)\sum_{n\leq x} e\left(\frac{an}{q}\right) \\
&\ll \frac{1}{\sqrt{q}}\left(q\log q\right) \\
&= \sqrt{q}\log q.
\end{aligned}$$

□

Remark 15.5. Here is an alternate heuristic explanation of Polya-Vinogradov. We have

$$\begin{aligned}
\sum_n \chi(n)\Phi\left(\frac{n}{x}\right) &= \frac{1}{\tau(\overline{\chi})}\sum_{a \bmod q}\overline{\chi}(a)\sum_n \Phi\left(\frac{n}{x}\right)e\left(\frac{an}{q}\right) \\
&= \frac{1}{\tau(\overline{\chi})}\sum_{a \bmod q}\overline{\chi}(a)\sum_k x\widehat{\Phi}\left(x\left(k+\frac{a}{q}\right)\right) \\
&= \frac{1}{\tau(\overline{\chi})}\sum_{a \bmod q}\overline{\chi}(a)\sum_k x\widehat{\Phi}\left(x\left(\frac{kq+a}{q}\right)\right) \\
&= \frac{1}{\tau(\overline{\chi})}\sum_m \overline{\chi}(m)x\widehat{\Phi}\left(\frac{xm}{q}\right).
\end{aligned}$$

where we let $m = kq + a$. So the left hand side is a sum over χ of size x and the right hand side is a sum over $\overline{\chi}$ of size q/x. This is an involution. This explains why Polya Vinogradov holds. If $x < \sqrt{q}$ and we get a bound by $\sqrt{q}$. If not, do the flip and bound the $\overline{\chi}$ right hand side trivially which gives a bound $x/\sqrt{q}\cdot\frac{q}{x} \ll \sqrt{q}$.

Say we want to understand $\left|L\left(\frac{1}{2},\chi\right)\right|$. We have

$$\begin{aligned}L\left(\frac{1}{2},\chi\right) &= \sum_{n=1}^{\infty}\frac{\chi(n)}{\sqrt{n}}\\ &= \int_1^\infty \frac{1}{\sqrt{y}} d\left(\sum_{n\le y}\chi(n)\right)\\ &= \frac{1}{2}\int_1^\infty \frac{1}{y^{3/2}}\left(\sum_{n\le y}\chi(n)\right)dy\end{aligned}$$

where we can bound

$$\sum_{n\le y}\chi(n)\min\left(y,\sqrt{q}\log q\right)$$

using Polya Vinogradov.

Exercise 15.6. Show the above is bounded by

$$\ll q^{1/4}\log q.$$

The kind of argument we were discussing in Remark 15.5 yields

$$L\left(\frac{1}{2},\chi\right) = \sum_{n\le\sqrt{q}}\frac{\chi(n)}{\sqrt{n}} + \varepsilon(\chi)\sum_{n\le\sqrt{q}}\frac{\overline{\chi}(n)}{\sqrt{n}}$$

where $\varepsilon(x)$ is a complex number of size 1.

So, $q^{1/4}\log q$ is called the convexity bound for for $\left|L\left(\frac{1}{2},\chi\right)\right|$.

The Riemann hypothesis implies the Lindelöf hypothesis, which implies

$$\left|L\left(\frac{1}{2},\chi\right)\right| \ll q^{\varepsilon}$$

for any $\varepsilon > 0$.

In fact, we can slightly improve the above bound.

Theorem 15.7 (Burgess)**.** *We have*

$$\left|L\left(\frac{1}{2},\chi\right)\right| \ll q^{3/16+\varepsilon}$$

for q cube free.

If χ is quadratic, there is an even better result:

Theorem 15.8 (Conrey and Iwaniec)**.** *For χ quadratic,*

$$L\left(\frac{1}{2}, \chi\right) \ll q^{1/6+\varepsilon}$$

Burgess has a result saying q is cubefree if

$$\left|\sum_{n \leq x} \chi(n)\right| = o(x)$$

if $x \geq q^{1/4+\varepsilon}$.

Exercise 15.9. If χ is quadratic mod$q$ for q a prime, then then the least quadratic non-residue (lqnr) mod$q$ is at most $q^{1/2}\log q$. Gauss showed lqnr $\leq q^{1/2}$. A trick of Vinogradov allows you to save a factor of $\sqrt{e}$ and Polya Vinogradov yields

$$\text{lqnr} \leq q^{1/(2\sqrt{e})+\varepsilon}$$

Burgess implies

$$\text{lqnr} \leq q^{1/(4\sqrt{e})+\varepsilon}.$$

15.4. **Some more extended exercises.**

Exercise 15.10 (Difficult, theorem of Goldfeld)**.**

Question 15.11 (Open question, Sophie Germian)**.** Are there infinitely many primes p with $p-1 = 2q$ with q a prime.

A weakening of the above statement would be to find primes p so that $p-1$ has a large prime factor.

Let

$$P(x)$$

be the largest prime dividing x The exercise is to show that there are lots of primes $p \leq x$ so that

$$P(p-1) \geq x^{1/2+\delta}$$

for some $\delta > 0$. *Hint:* The point of this exercise is to use Bombieri Vinogradov. Here is the idea of the proof. Suppose there were lots of such primes, say

$$\sum_{p \leq x}\left(\sum_{q \mid p-1} \Lambda(d)\right) \sim x$$

Suppose $q \leq x^{1/2+\delta}$ always. Let $Q = x^{1/2+\delta}$. We can now exchange these two sums to obtain

$$\sum_{q \leq Q} \log q \left(\pi(x; q, 1)\right)$$

If $q \leq x^{1/2} (\log x)^A$ would give some bound by Bombieri Vinogradov. So, we write

$$\begin{aligned}
&\sum_{q \leq Q} \log q \left(\pi(x; q, 1)\right) \\
&= \sum_{q \leq x^{1/2}/(\log x)^B} \log q \left(\pi(x; q, 1)\right) + \sum_{x^{1/2}/(\log x)^B \leq q \leq Q} \log q \left(\pi(x; q, 1)\right)
\end{aligned}$$

and we get for the first term is asymptotic to

$$\sum_{q \leq x^{1/2}/(\log x)^B} (\log q) \frac{\pi(x)}{\phi(q)} \sim x/2$$

where we can bound the error term by Bombieri Vinogradov So, there must be a large contribution from the second term. We don't know how to control $\pi(x; q, 1)$. But, we do have an upper bound on them by Brun-Titchmarsh. Indeed, we can estimate $\pi(x; q, 1)$ from Brun Titchmarsh.

$$\pi(x; q, 1) \leq \frac{2x\left(1 + o(1)\right)}{\phi(q) \log(x/q)}.$$

Use this to show that the second term is at most $.49x$ if $\delta \leq .01$. This finishes the proof because then these two terms cannot add up to be as big as x. This would yield a contradiction.

Remark 15.12. The above theorem was published under Morris Goldfeld, but this is the same person as Dorian Goldfeld, who changed his name after he published this.

Exercise 15.13 (Difficult exercise, Titchmarsh divisor problem)**.** Pick a number at random and say its largest prime factor. "Does $p - 1$ look in some ways like a random number?"

Prove the following lemma.

Lemma 15.14. *We have*

$$\sum_{n \leq x} d(n) \sim x \log x.$$

and

$$\sum_{p \leq x} d(p-1) \sim Cx$$

for some $C > 0$.

Sketch of proof. We know

$$d(n) = 2 \sum_{d|n, d \le \sqrt{n}} 1$$

and so it suffices to bound

$$\sum_{p \le x} \sum_{d \le \sqrt{x}, d|p-1} 1 \qquad = \sum_{d \le \sqrt{x}} \pi(x; d, 1).$$

One can solve this problem by combining it with Bombieri Vinogradov in the range $d \le \sqrt{x}/\left(\log x\right)^B$. For the small region $\sqrt{x}/\left(\log x\right)^B \le d \le \sqrt{x}$, try to bound $\pi(x; d, q)$ by Brun-Titchmarsh and hope it becomes an error term. □

Question 15.15 (Open problem)**.** Let

$$d_3(n)$$

be defined by

$$\zeta(s)^3 = \sum_{n=1}^{\infty} \frac{d_3(n)}{n^s}.$$

so $d_3(n)$ is the number of ways of writing $n = abc$. Bound

$$\sum_{p \le x} d_3(p-1).$$

One can keep track of small prime factors, but occasionally it might have a very large prime factors.

Conjecture 15.16 (Montgomery's conjecture)**.** We have

$$\Psi(x; q, a) = \frac{\Psi(x)}{\phi(q)} + O\left(\frac{x^{1/2+\varepsilon}}{\sqrt{q}}\right).$$

Remark 15.17. There reasoning behind this is that the error term is approximately

$$\frac{1}{\phi(q)} \sum_{\chi \bmod q, \chi \ne \chi_0} \overline{\chi}(n) \Psi(x, \chi)$$

and we can bound $\Psi(x, \chi)$ by $x^{1/2}$ and then get some cancellation in the sums of the characters.

Montgomery's conjecture would imply the Elliott-Halberstam conjecture.

Conjecture 15.18 (Elliott-Halberstam conjecture)**.** We have

$$\sum_{q \leq x^{1-\varepsilon}} \max_{(a,q)=1} \left| \Psi(x; q, a) - \frac{\Psi(x)}{\phi(q)} \right| \ll \frac{x}{(\log x)^A}$$

for any $A > 0$.

16. 11/16/17

Today we'll begin a discussion of gaps between primes. Let p_n be the nth prime. By the prime number theory, $p_n \sim n \log n$. Hence, on average, $p_{n+1} - p_n \sim \log n$.

Question 16.1 (Open question)**.** What can we say about the distribution of

$$\frac{p_{n+1} - p_n}{\log n}$$

as n varies?

To make sense of this question, we can pick an interval $(\alpha, \beta) \subset \mathbb{R}_{>0}$ and ask about

$$\lim_{n \to \infty} \frac{1}{N} \# \left\{ n \leq N : \frac{p_{n+1} - p_n}{\log p_n} \in (\alpha, \beta) \right\}.$$

How might we guess this? There is a naive model called the Cramer model. This is clearly bogus, but also reasonably successful. Define the random variable X_n by

$$X_n := \begin{cases} 1 & \text{with probability } \frac{1}{\log n} \\ 0 & \text{with probability } 1 - \frac{1}{\log n} \end{cases}$$

for $n \geq 3$.

Now, let's count the probability

$$\operatorname{Prob}\left(X_{n+1} = X_{n+2} = \cdots = X_{n+h-1} = 0, X_{n+h} = 1, \text{ given } X_n = 1 \right).$$

This is asking for the chance of a gap of size h.

To calculate this, thinking of h as small compared to $\log n$. we see this is approximately

$$\left(1 - \frac{1}{\log n}\right)^{h-1} \frac{1}{\log n} \sim e^{-h/\log n} \frac{1}{\log n}.$$

Thinking of this a different way, looking at the interval $[n, n+h]$, we can ask for the chance there are exactly k values for which $X_m = 1$. We can also handle this quite easily. We can pick k numbers to be primes. Calculating using the binomial theorem, we see this chance is

$$\binom{h}{k}\frac{1}{(\log n)^k}\left(1-\frac{1}{\log n}\right)^{h-k} \sim \left(\frac{h}{\log n}^k \frac{1}{k!}e^{-h/\log n}\right).$$

Example 16.2. So the guess one might obtain from this is that in the interval

$$[n, n+\log n]$$

the chance of finding k primes is about

$$\frac{1}{k!}e^{-1}$$

So, we might make the following conjecture.

Conjecture 16.3. If $h = \lambda \log n$, then as $n \to \infty$ chosen randomly, then The number of primes in $[n, n+h]$ is Poisson with parameter λ. That is,

$$\frac{1}{N}\#\left\{n \leq N : [n, n+h] \text{ contains } k \text{ primes }\right\} \sim \frac{\lambda^k}{k!}e^{-\lambda}.$$

Remark 16.4. Saying this another way,

$$\frac{1}{N}\#\left\{n \leq N : \frac{p_{n+1}-p_n}{\log p_n} \in (\alpha, \beta)\right\} \sim \int_\alpha^\beta e^{-x}dx.$$

Example 16.5. One could ask the same sort of question about any subset of the integers (or discrete subset of the real numbers). For example, say we would like to know the spacing of the zeros of the zeta function. Say they are of the form $1/2 + i\gamma_n$ with

$$\gamma_n \sim \frac{2\pi n}{\log n}.$$

The spacings of

$$\gamma_{n+1} - \gamma_n \frac{\log n}{2\pi} \sim 1$$

on average. We can ask now about the distribution. These are not expected to behave like a Poisson process.

Question 16.6. Why should we believe the above conjecture?

Well, there is in fact a better conjecture going back to Hardy and Littlewood.

Definition 16.7. Let $\mathscr{H} = \{h_1, \ldots, h_k\}$ be a set of distinct integers. The **singular series** of $\mathscr{H}$ is

$$\mathfrak{S}(\mathscr{H}) := \prod_p \left(1 - \frac{\nu_{\mathscr{H}}(p)}{p}\right)\left(1 - \frac{1}{p}\right)^{-k}.$$

Remark 16.8. The singular series $\mathfrak{S}(\mathscr{H})$ is approximated by

$$\sum_{n \leq N} \Lambda(n + h_1) \cdots \Lambda(n + h_k) \sim \mathfrak{S}(\mathscr{H}) N.$$

Conjecture 16.9 (Hardy and Littlewood)**.** Let $\mathscr{H} = \{h_1, \ldots, h_k\}$ be a set of distinct integers. Then,

$$\#\{n \leq N : n + h_1, \ldots, n + h_k \text{ are all primes }\} \sim \mathfrak{S}(\mathscr{H}) \frac{N}{(\log N)^k}.$$

with $\mathfrak{S}(\mathscr{H})$ the singular series of $\mathscr{H}$

We next justify the above definition of singular series.

Remark 16.10. The Cramer model predicts

$$\#\{n \leq N : n + h_1, \ldots, n + h_k \text{ are all primes }\} \sim \frac{N}{(\log N)^k}.$$

Exercise 16.11. Let $n, n+2$ be both prime. We would like to conjecture a value for the singular series $\mathfrak{S}(\{0, 2\})$. We might expect an approximation via the circle method like

$$\int_0^1 \left(\sum_{n \leq N} \Lambda(n) e(n\alpha)\right)\left(\sum_{m \leq N} \Lambda(m) e(-m\alpha)\right) e(2\alpha) d\alpha.$$

Compute what the major arc contribution is. When one computes this, one might have a guess as to what the answer should be. There will be a major arc around 0 and a major arc around 1. *Hint:* Here, take α close to $\frac{a}{q}$ with error roughly $\frac{1}{n}$. Consider

$$\sum_n \Lambda(n) e\left(\frac{a}{q} n + n\beta\right).$$

Assume $\Lambda(n)$ behaves like $\log n$ from the prime number theorem, and similarly put in an estimate from the prime number theorem on arithmetic progressions. One should also check we get 0 as our main term if we put $e(\alpha)$ instead of $e(2\alpha)$ above.

Exercise 16.12. Suppose $n, n+2, n+6$ are all primes. Then, we might want to compute

$$\int_0^1 \left(\sum_{n \le N} \Lambda(n)\Lambda(n+2)e(n\alpha) \right) \left(\sum_{m \le N} \Lambda(m)e(-m\alpha)e(6\alpha \right).$$

Here again, compute an estimate for the major arcs.

16.1. **A probabilistic argument for the distribution of primes.** One could also think probabilistically (which, according to Hardy and Littlewood's paper, is not a notion in mathematics but rather a notion in physics or philosophy).

The idea is to add in by hand all the density information for any prime p we have. That is, given a prime p, we ask

Question 16.13. What is the probability that $n + h_1, \ldots, n + h_k$ are all coprime to p for n chosen randomly.

This is asking that n not be in the classes $-h_1, \ldots, -h_k \bmod p$. So, we need that n does not lie in the $\nu_{\mathscr{H}}(p)$ congruence classes $\bmod p$, where

$$\nu_{\mathscr{H}}(p) = \#\{h_1 \bmod p, \ldots, h_k \bmod p\}.$$

So, the probability that $n + h_1, \ldots, n + h_k$ are all coprime to p should be

$$1 - \frac{\nu_{\mathscr{H}}(p)}{p}.$$

We want to keep track of the difference between this probability and the Cramer model. The Cramer model only uses the fact that k random numbers are all coprime to p with probability $\left(1 - \frac{1}{p}\right)^k$.

Now, the guess is to take

$$\mathfrak{S}(\mathscr{H}) := \prod_p \left(1 - \frac{\nu_{\mathscr{H}}(p)}{p}\right)\left(1 - \frac{1}{p}\right)^{-k}.$$

If $p > \max(h_j)$ then $v_{\mathscr{H}}(p) = k$ so the above is $1 + O(1/p^2)$ for large p, and hence converges absolutely. This implies the above product is 0 if and only if one of the terms is equal to 0. This means there is some prime p for which

$$v_{\mathscr{H}}(p) = p$$

for some prime p.

Remark 16.14. Hardy and Littlewood did not like this probabilistic argument because it is assuming primes are independent. That is, when one considers $\pi(N) \sim N \prod_{p \leq \sqrt{N}} \left(1 - \frac{1}{p}\right) \sim \frac{2e^{-\gamma}N}{\log N}$.

Exercise 16.15. Show the above conjecture predicts the number of twin primes is roughly

$$1.33 \frac{N}{(\log N)^2}.$$

That is, because $N, N+1$ cannot both be prime, there is a little higher chance of N and $N+2$ being prime.

Exercise 16.16 (Extended exercise, due to Gallagher)**.**

$$\sum_{\substack{h_1, \ldots, h_k \leq H \\ h_i \text{ distinct}}} \mathfrak{S}\left(\{h_1, \ldots, h_k\}\right) \sim \sum_{\substack{h_1, \ldots, h_k \leq H \\ h_i \text{ distinct}}} 1.$$

Exercise 16.17 (Lead in to previous exercise)**.** Pick a prime p. Show

$$\mathbb{E}\left(\left(1 - \frac{\nu_{\mathscr{H}}(p)}{p}\right) : \mathscr{H} \subset \{0, p-1\}^k\right) = \left(1 - \frac{1}{p}\right)^k$$

where $\mathbb{E}$ denotes expected value. *Hint:* Use a sort of double counting argument

Let's now construct some sets where the singular series is nonzero.

Example 16.18.

(1) Consider
$$\{h_1, \ldots, h_k\} := \{0, k!, 2k!, \ldots, (k-1)k!\}.$$
Then,
$$v_{\mathscr{H}}(p) = \begin{cases} 1 & \text{if } p \leq k \\ > 0 & \text{if } p > k \end{cases}$$

(2) Take k distinct primes all larger than k for $\mathscr{H}$. This has a nonzero singular series.

Question 16.19. What is the largest set in $[1, 10^7]$ which is admissible (i.e., has a nonzero singular series).

Question 16.20. If we find such a large set, what is it good for?

Say the set in $[1, 10^7]$ has size k. If we do find such a large set, then there are intervals $(n, n+10^7)$ with at least k primes. Then, Hardy and Littlewood's conjecture also implies that the number of primes up to 10^7 is an upper bound for the number of primes in $(n, n+10^7)$.

Remark 16.21. Hensley and Richards (with a nice paper by Richards in the bulletin of the AMS in 1974) have a nice historical document on computing. For example, if $y = 20,000$ one can construct an admissible set of length more than 20,000 more than $\pi(y)$. They showed that Hardy and Littlewood's conjecture above contradicts Hardy and Littlewood's conjecture that $\pi(x+y) \leq \pi(x) + \pi(y)$.

Remark 16.22. The above is really a problem in sieve theory. In more detail, given an interval $[1, y]$, for each small prime p, remove one progression a_p mod p. The aim is to keep as many numbers as possible. Stop once the prime exceeds the number of remaining integers.

For example, we start at 2, remove either 0 or 1 mod2. Then, go to 3, and remove some residue mod3. Then, we stop once there are fewer integers left than the prime we have reached. This is sort of like a greedy algorithm.

Remark 16.23. Here is another variant: Consider the interval $[1, y]$. For each prime $p \leq z$, remove one progression mod p until nothing is left. How small can one make z?

We did prove something interesting about Remark 16.22 using the large sieve. Essentially, we get an upper bound, that the number is always at most $2\pi(y)$, as follows from the large sieve.

Remark 16.24. One could use the above problems to show

$$\limsup \frac{p_{n+1} - p_n}{\log n} \to \infty.$$

For a while, the best result was

Theorem 16.25 (Erdos-Rankin)**.**

$$p_{n+1} - p_n \geq \frac{c \log p_n \log_2 p_n}{(\log_3 p_n)^2} \log_4 p_n$$

where $\log_n$ *denotes the nth iterated log,* $\log n = \log \log_{n-1}$.

But, in 2014, there were some improvements:

Theorem 16.26 (Ford, Green, Konyagin, Tao, Maynard)**.** *One can bound*

$$p_{n+1} - p_n \geq \frac{c \log p_n \log_2 p_n}{\log_3 p_n} \log_4 p_n$$

The other side of the problem is to ask whether

$$\liminf \frac{p_{n+1} - p_n}{\log p_n} = 0.$$

In fact, this was shown by Goldston, Pintz, and Yildirim in 2005. However, their method did not show

$$\frac{p_{n+2} - p_n}{\log p_n} = 0.$$

This was the basis for an amazing result of Zhang in 2013 yielding bounded gaps between primes:

$$p_{n+1} - p_n < 70 * 10^7.$$

The main ingredient was Zhang's version of Bombieri Vinogradov: Let $a \neq 0$ be any integer. Then,

$$\sum_{q \leq x^{1/2+\delta}, p|q \implies p \leq x^\delta} |E(x; q, a)| \ll \frac{x}{(\log x)^A}.$$

Using this and the work of Goldston Pintz and Yildirim, he was able to get bounded gaps. However, in the same year, Maynard and Tao showed that instead of getting two primes, one could get many primes in a bounded interval. Further, one could use the original version of Bombieri Vinogradov instead of Zhang's variant.

Theorem 16.27 (Maynard, and independently, Tao)**.** *For any ℓ there exists k such that for any admissible set $\mathscr{H}$ (meaning there is no prime p with all residues* $\bmod p$ *appearing in $\mathscr{H}$ of size k, there exist many n with at least ℓ primes in $(n + h_1, n + h_2, \ldots, n + h_k)$.*

So, one can find 3 or 4 or more primes in a bounded interval. In other words:

Corollary 16.28.

$$\liminf p_{n+\ell} - p_n < \infty.$$

17. 11/28/17

Last time, we were discussing the Hardy Littlewood conjecture:

Conjecture 17.1 (Hardy-Littlewood)**.** If you have a set

$$\mathscr{H} := \{h_1, \ldots, h_k\}$$

then

$$\#\{n \leq x : n + h \text{ primes }\} \sim \mathfrak{S}(\mathscr{H}) \frac{x}{(\log x)^k}$$

where

$$\mathfrak{S}(\mathscr{H}) = \prod_p \left(1 - \frac{1}{p}\right)^{-k} \left(1 - \frac{\nu_{\mathscr{H}}(p)}{p}\right)$$

with

$$\nu_{\mathscr{H}}(p) = \#\mathscr{H} \bmod p$$

Note $\nu_{\mathscr{H}}(p) = k$ if p is large enough. At the end of last time we stated recent work of Maynard and Tao:

Theorem 17.2 (Maynard-Tao)**.** *For any s, there exists a suitably large k such that for every admissible set $\mathscr{H} = \{h_1, \ldots, h_k\}$ (meaning $\mathfrak{S} \neq 0$) there are infinitely many n with at least s primes in $n + h_1, \ldots, n + h_k$.*

Remark 17.3. Sieve methods can give upper bounds for the number of prime k-tuples asymptotic to $\frac{x}{(\log x)^k}$. We have already seen one version of this, which is the large sieve.

Exercise 17.4 (Extended exercise)**.** Use the large sieve to give such an upper bound. Recall the large sieve looked at some exponential sum. One can try to bound the number of inadmissible tuples over all possible primes.

But, now we describe a different sieve method, known as Selberg's sieve.

17.1. **Selberg's sieve.** Selberg's sieve is based on the simple idea that squares of real numbers are non-negative.

Consider

$$\sum_{\sqrt{x} \leq n \leq x} \left(\sum_{d \mid (n+h_1)\cdots(n+h_k)} \lambda_d \right)$$

For $\lambda_d \in \mathbb{R}$. We want to arrange that the square of the quantity in parentheses is always non-negative and at least 1 if $n + h_1, \ldots, n + h_k$ are prime.

We will assume $\lambda_d = 0$ for $d > R$, with $R = R(x)$ chosen as some function of x, to be decided later. We should choose $\lambda_1 = 1$.

So, with these stipulations on λ_k, we have

$$\sum_{\sqrt{x} \leq n \leq x} \left(\sum_{d \mid (n+h_1)\cdots(n+h_k)} \lambda_d \right) \geq \# \left\{ R < n \leq x : n + h_1, \ldots, n + h_k \text{ are all prime } \right\}.$$

On the other hand, expanding, we get

$$\sum_{\sqrt{x} \leq n \leq x} \left(\sum_{d \mid (n+h_1)\cdots(n+h_k)} \lambda_d \right) = \sum_{d_1, d_2} \lambda_{d_1} \lambda_{d_2} \left(\sum_{\substack{\sqrt{x} \leq n \leq x \\ [d_1, d_2] \mid (n+h_1)\cdots(n+h_2)}} 1 \right).$$

where $[d_1, d_2]$ is the least common multiple of d_1 and d_2. The parenthesized expression above is a quadratic form in λ_d's, and the problem is to minimize this quadratic form subject to the linear condition that $\lambda_1 = 1$.

We now try and minimize this quadratic form. Suppose $p \mid [d_1, d_2]$. Then, there is some i with $n \equiv -k_i \bmod p$ Then, n lies in $\nu_{\mathscr{H}}(p)$ residue classes $\bmod\, p$.

We define f to be multiplicative and $f(p) = \nu_{\mathscr{H}}(p)$. Then, $f([d_1, d_2])$

It follows that n lies in $f([d_1, d_2])$ residue classes mod $[d_1, d_2]$. So, the quadratic form

$$\sum \lambda_{d_1} \lambda_{d_2} \left(\frac{f([d_1, d_2])}{[d_1, d_2]\, x} + O\left(f\left([d_1, d_2]\right)\right) \right).$$

The function f is multiplicative by the Chinese remainder theorem, though some annoying things might happen on squares of primes.

For simplicity, we'll make the additional assumption that λ_d is 0 unless d is squarefree. Then,

$$\begin{aligned} O\left(f\left([d_1, d_2]\right)\right) &= O\left(k^{\omega([d_1, d_2])}\right) \\ &= O\left(x^{\varepsilon}\right), \end{aligned}$$

Here $\omega(n)$ is the number of distinct prime factors of n. The above is useful if $[d_1, d_2] \leq x^{1-\varepsilon}$. This is good if $R \leq x^{1/2-\varepsilon}$. We will also assume $|\lambda_d| \ll d^{\varepsilon}$. We will justify this later.

In this case, the total contribution of the error terms, is bounded by the number of terms times the bound which is $R^2 x^{\varepsilon} \leq x^{1-\varepsilon}$. Our quadratic form is then

$$\sum_{d_1, d_2} \frac{\lambda_{d_1} \lambda_{d_2}}{[d_1, d_2]} f\left([d_1, d_2]\right)$$

and we want to minimize this subject to the constraints

(1) $\lambda_1 = 1$
(2) $\lambda_d = 0$ unless $d \leq R = x^{1/2-\varepsilon}$ and d is square free
(3) $|\lambda_d| \ll d^{\varepsilon}$.

We'd now like to diagonalize this quadratic form and read of the diagonal entries by Cauchy Schwartz. Now, d_1, d_2 are tied together by the lcm function. Let (d_1, d_2) denote the gcd of d_1, d_2. Let $a = (d_1, d_2)$ be the gcd. Then, let $d_1 = ar_1, d_2 = ar_2$. We obtain

$$\sum_a \sum_{\substack{r_1, r_2, \\ (r_1, r_2)=1}} \frac{\lambda_{ar_1}\lambda_{ar_2}}{ar_1r_2} f\left(ar_1r_2\right).$$

By multiplicativity, we have

$$f\left(ar_1r_2\right) = f(a)f(r_1)f(r_2).$$

Next, use Möbius inversion to remove the coprimality condition. We have

$$\sum_{b|(r_1,r_2)} \mu(b) = \begin{cases} 1 & \text{if } (r_1, r_2) = 1 \\ 0 & \text{else} \end{cases}$$

Write $r_1 = bs_1, r_2 = bs_2$. Define

$$\xi_d = \sum_s \frac{\lambda_{ds} f(s)}{s}.$$

Then,

$$\begin{aligned}
&\sum_a \sum_{\substack{r_1, r_2, \\ (r_1, r_2)=1}} \frac{\lambda_{ar_1}\lambda_{ar_2}}{ar_1r_2} f\left(ar_1r_2\right) \\
&= \sum_a \sum_b \frac{f(a)}{a} \frac{\mu(b)f(b)^2}{b^2} \sum_{s_1, s_2} \frac{\lambda_{abs_1}\lambda_{abs_2}}{s_1s_2} f(s_1)f(s_2) \\
&= \sum_a \sum_b \frac{f(a)}{a} \frac{\mu(b)f(b)^2}{b^2} \sum_s \frac{\lambda_{abs}}{s} f(s) \\
&= \sum_d \xi_d^2 \sum_{ab=d} \frac{f(a)f(b)^2}{ab^2} \mu(b) \\
&= \sum_d \xi_d^2 \sum_{ab=d} \frac{f(d)}{d} \left(\sum_{b|d} \frac{f(b)}{b} \mu(b) \right).
\end{aligned}$$

Then, let

$$h(d) := \left(\sum_{b|d} \frac{f(b)}{b} \mu(b) \right).$$

We may observe

$$h(d) = \prod_{p \mid d} \left(1 - \frac{f(p)}{p}\right).$$

This is starting to look related to the singular series. Therefore, our quadratic form can be written as

$$\sum_d \xi_d^2 \sum_{ab=d} \frac{f(d)}{d} \left(\sum_{b \mid d} \frac{f(b)}{b} \mu(b)\right) = \sum_d \frac{f(d)}{d} h(d) \xi_d^2.$$

We have now diagonalized our quadratic form, but we now need to transform our constraint λ_1 into a constraint in terms of the ξ_d. So, we'd like to invert our linear change of variables. We want to write down λ_d in terms of things involving ξ_i. To do this, using Möbius inversion, we want to understand

$$\begin{aligned}
\lambda_d &= \sum_s \lambda_{ds} \frac{f(s)}{s} \left(\sum_{b \mid s} \mu(b)\right) \\
&= \sum_b \mu(b) \sum_t \frac{\lambda_{dbt} f(bt)}{bt} \\
&= \sum_b \frac{\mu(b) f(b)}{b} \xi_{db}.
\end{aligned}$$

Then, we want $\xi_d = 0$ unless $d \leq R$ and squarefree. We also want $\lambda_1 = 1$ if and only if

$$\sum \frac{\mu(b)}{b} f(b) \xi_b = 1.$$

We have

$$1 = \left(\sum_b \frac{\mu(b)}{b} f(b) \xi_b\right)^2 \leq \left(\sum_b \frac{\mu(b)^2}{b} \frac{f(b)}{h(b)}\right) \left(\sum_d \frac{f(d)}{d} h(d) \xi_d^2\right)$$

The equality case of Cauchy Schwartz occurs when the vectors are proportional to each other, which occurs when

$$\xi_b \sim \frac{\mu(b)}{h(b)}.$$

Therefore, the minimum of the quadratic form given by

$$\sum_d \frac{f(d)}{d} h(d) \xi_d^2$$

is bounded by

$$\left(\sum_{b \le R} \frac{\mu(b)^2}{b} \frac{f(b)}{h(b)} \right)^{-1}$$

and this is the equality case of Cauchy Schwartz, so it actually attains this bound. And further one can determine the constant of proportionality c by

$$1 = c \sum_{b \le R} \frac{\mu(b)^2 f(b)}{b h(b)}.$$

We obtain that

$$\#\{ n \le x : n + h_1, \ldots, n + h_k \text{ all prime } \} \le \frac{x}{\sum_{b \le R} \frac{\mu(b)^2 f(b)}{b h(b)}} + O(x^{1-\varepsilon})$$

where one needs to verify

Exercise 17.5. Verify $\xi_d \ll d^\varepsilon, \lambda_d \ll d^\varepsilon$

Then $R \le x^{1/2-\varepsilon}$.

Now, $f(p)$ is about k, so $f(n)$ is roughly $d_k(n)$ the k-divisor function of n. Next, $h(n)$ is roughly 1. Then,

$$\begin{aligned} \sum_{b \le R} \frac{d_k(b)}{b} &= \frac{1}{2\pi i} \int \zeta(s+1)^k R^s \frac{ds}{s} \\ &\sim \frac{(\log R)^k}{k!}. \end{aligned}$$

So, we should expect

$$\sum_{b \le R} \frac{\mu(b)^2 f(b)}{b h(b)} \sim \frac{(\log R)^k}{k!}$$

up to multiplying by some convergent Euler factor to mitigate our estimates above.

Exercise 17.6. Verify that this Euler factor is $\mathfrak{S}(\mathscr{H})^{-1}$, meaning

$$\sum_{b \le R} \frac{\mu(b)^2 f(b)}{b h(b)} = \mathfrak{S}(\mathscr{H})^{-1} \frac{(\log R)^k}{k!} + \text{ lower order terms }.$$

Combining the above, we conclude

Theorem 17.7.

$$\#\left\{n \leq x : n+h_1, \ldots, n+h_k \textit{ all prime }\right\} \leq \mathfrak{S}(\mathscr{H}) \frac{k!2^k x}{(\log x)^k}\left(1+o(1)\right).$$

This is a typical application of Selberg's sieve.

Remark 17.8. When $k = 2$ we can do a funny trick which gives a better bound. We can replace the $2!2^2$ by a factor of 4.

Remark 17.9. Recall that the optimal choice of $\xi_d = \frac{\mu(d)}{h(d)}$, up to scaling. Then,

$$\lambda_d = \sum \lambda_{ds} \frac{\mu(s) f(s)}{s}.$$

Imagine that $\xi_{ds} \sim \mu_{ds}$, Then, we get

$$\lambda_d \sim \mu(d) \frac{\left(\sum_{s \leq R/d} \frac{\mu(s)^2 f(s)}{sh(s)}\right)}{\sum_{s \leq R} \frac{\mu(s)^2 f(s)}{sh(s)}}$$

Then,

$$\lambda_d = \mu(d) \left(\frac{\log R/d}{\log R}\right)^k.$$

Exercise 17.10. Show

$$\sum_{d|n} \mu(d) \left(\log \frac{n}{d}\right)^k = 0 \text{ unless } n \text{ has at most } k \text{ prime factors.}$$

Remark 17.11. Consider the simplest case $k = 1$. This could be tricky because we might be trying to count primes in a short interval.

Exercise 17.12. Check that one gets exactly the same upper bound for $\pi(x+y) - \pi(x)$ as with the large sieve using Theorem 17.7. (Not only asymptotically the same, but rather exactly the same expression.)

Recall our quadratic form in the case of sieving out a 1-tuple is

$$\sum_{d_1, d_2 \leq R} \frac{\lambda_{d_1} \lambda_{d_2}}{[d_1, d_2]}$$

with

$$\lambda_d = \mu(d) \left(\frac{\log(R/d)}{\log R}\right).$$

The optimal answer ended up being

$$\sum_{d_1,d_2\le R} \frac{\lambda_{d_1}\lambda_{d_2}}{[d_1,d_2]} = \frac{1}{\log R}.$$

Then,

$$\sum_{d_1,d_2\le R} \frac{\lambda_{d_1}\lambda_{d_2}}{[d_1,d_2]} = \sum_{\substack{a,r,s\\ r\le R/a\\ s\le R/a}} \frac{\mu(a)^2}{a}\frac{\mu(r)}{r}\frac{\mu(s)}{s}.$$

for $R/2 \le a \le R$.

Exercise 17.13 (difficult extended exercise)**.** Then,

$$\sum_{d_1,d_2\le R} \frac{\mu(d_1)\mu(d_2)}{[d_1,d_2]} \to c \neq 0$$

as $R \to \infty$.

The sieve accounts for the above by replacing

$$\sum_{\substack{a,r,s\\ r\le R/a\\ s\le R/a}} \frac{\mu(a)^2}{a}\frac{\mu(r)}{r}\frac{\mu(s)}{s}.$$

by

$$\sum_{\substack{a,r,s\\ r\le R/a\\ s\le R/a}} \frac{\mu(a)^2}{a}\frac{\mu(r)}{r}\frac{\mu(s)}{s}\left(\frac{\log R/ar}{\log R}\right)\left(\frac{\log R/as}{\log R}\right).$$

Here is a variant useful for what we will do next. We want to find when $n+h_1,\dots,n+h_k$ are all prime.

Fix $n+h_1 = p$. Sieve $n+h_1,\dots,n+h_k$

$$\sum_{n+h_1=p\le x}\left(\sum_{d\mid(n+h_2)\cdots(n+h_k)} \lambda_d\right)^2$$

with $\lambda_1 = 1, \lambda_d = 0$ unless $d \le R$ and d is squarefree. This lets us count

$$\pi\left(x;[d_1,d_2],a\right)$$

and handle this using Bombieri-Vinogradov with $R \le x^{1/4-\varepsilon}$.

Exercise 17.14. Show that

$$\pi\left(x; [d_1, d_2], a\right) \leq \mathfrak{S}(\mathscr{H}) 4^{k-1}(k-1)! \frac{x}{\log x}^{k}$$

using that we have a $k-1$ dimensional sieve. When $k = 2$ this gives a better bound with 4 instead of 8, but it is worse for $k > 2$.

18. 11/30/17

Last time, we discussed Selberg's sieve. We proved bounds like

$$\#\left\{n \leq x : n + h_1, \ldots, n + h_k \text{ are all prime }\right\} \leq (1 + o(1))\, 2^k k! \frac{\mathfrak{S}(\mathscr{H})x}{(\log x)^k}.$$

This was shown by considering a quadratic form

$$\left(\sum_{d \mid (n+h_1)\cdots(n+h_k)} \lambda_d\right)^2$$

with

$$\lambda_d \sim \mu(\lambda) \left(\frac{\log R/d}{\log R}\right)^k.$$

with $R = x^{1/2-\varepsilon}$.

Exercise 18.1. Another way to find this is to consider

$$\sum_{n \leq x, n+h_1 \text{ prime}} \left(\sum_{d \mid (n+h_1)\cdots(n+h_k)} \lambda_d\right)^2$$

taking $R = x^{1/4-\varepsilon}$. Then, show that one gets a bound which is better when $k = 2$, but not for other k, of the form

$$\sum_{n \leq x, n+h_1 \text{ prime}} \left(\sum_{d \mid (n+h_1)\cdots(n+h_k)} \lambda_d\right)^2 \leq (1 + o(1))\, 4^{k-1}(k-1)! \frac{\mathfrak{S}(\mathscr{H})x}{(\log x)^k}.$$

Exercise 18.2. Use Selberg's sieve to bound

$$\#\left\{n \leq x : n^2 + 1 \text{ is prime}\right\}.$$

Use sieve weights summing over polynomial values. That is, bound

$$\#\left\{n \leq x : n^2 + 1 \text{ is prime}\right\} \leq \sum_{n \leq x} \left(\sum_{d \mid n^2+1} \lambda_d\right)^2$$

with $\lambda_d = 1$. We get $n^2 + 1 \equiv 0 \bmod [d_1, d_2]$. Probably take $\lambda_d = 0$ for primes 3 mod 4 since such primes won't divide this. Then diagonalize the quadratic form and see what you get. The solutions to this will be given by some multiplicative function of the form

$$x\frac{f\left([d_1, d_2]\right)}{[d_1, d_2]}.$$

which is 2 if the prime is 1 mod 4 and 0 if it is 3 mod 4.

Derive a similar bound for other polynomials.

Theorem 18.3 (Goldston-Pintz-Yildirim)**.**

$$\liminf_{n \to \infty} \frac{p_{n+1} - p_n}{\log p_n} = 0.$$

The idea of proof is relatively simple. Start with an admissible k-tuple, meaning $\mathfrak{S}(\mathscr{H}) \neq 0$. The idea is to look for a non-negative function $a(n) \geq 0$ so that we can make

$$\sum_{n \leq x} a(n) < \sum_{j=1}^{k} \sum_{n \leq x, n+h_j \text{ prime}} a(n).$$

Then, there is some n with at least two primes among $n + h_1, \ldots, n + h_k$, essentially by pigeonhole principal. Then, motivated by Selberg's sieve, we will take

$$a(n) = \left(\sum_{d \mid (n+h_1) \cdots (n+h_k)} \lambda_d \right)^2$$

with $\lambda_1 = 1$ and $\lambda_d = 0$ unless $d \leq R$ is squarefree. We'd like the desired equality above with k as small as possible.

We won't be able to solve this, (it would imply bounded gaps) but we can tweak it a bit to get Theorem 18.3.

In Selberg's sieve, we wished to minimize a quadratic form given a linear form. Here, the real problem is to maximize the ratio of quadratic forms.

Let

$$Q_1(\lambda) := \sum_{n \leq x} a(n)$$

and

$$Q_2(\lambda) := \sum_{n \leq x, n+h_j \text{ prime}} a(n).$$

Then, we can write $Q_1(\lambda)$ as

$$x \cdot \sum_{d_1, d_2} \lambda_{d_1} \lambda_{d_2} \frac{f([d_1, d_2])}{[d_1, d_2]} + O(R^2 x^{\varepsilon})$$

with

$$f(p) := \nu_{\mathscr{H}}(p)$$

(recall $\nu_{\mathscr{H}}(p)$ is usually k.) This is good if $R \leq x^{1/2-\varepsilon}$.

We can write $Q_2(\lambda)$ as

$$k \sum_{d_1, d_2} \lambda_{d_1} \lambda_{d_2} \sum_{\substack{n \leq x, \\ n+h_1 \text{ prime }, \\ (n+h_2)\cdots(n+h_k) \equiv 0 \bmod [d_1, d_2]}} 1$$

To evaluate this sum, we take all possible choice in $\mathscr{H}$ mod p, and rule out the single choice $n \equiv -h_1 \bmod p$. So, n lies in $g([d_1, d_2])$ residue classes mod $[d_1, d_2]$ with $g(p) = f(p) - 1$.

For our inner sum, we get an estimate of the form

$$\frac{x}{\log x} \frac{g([d_1, d_2])}{\phi([d_1, d_2])}.$$

Averaging over all d_1, d_2 and using Bombieri-Vinogradov, the error terms are under control so long as $R \leq x^{1/4-\varepsilon}$.

So, we can approximate $Q_2(\lambda)$ by

$$k \frac{x}{\log x} \sum_{d_1, d_2} \frac{\lambda_{d_1} \lambda_{d_2}}{\phi([d_1, d_2])} g([d_1, d_2]).$$

Let's simplify and assume that

$$\lambda_d = \mu(d) P\left(\frac{\log R/d}{\log R}\right).$$

where P is a polynomials vanishing to order k at 0.

It is now just a calculation to figure out these two quadratic forms and see if we can find a suitable polynomial P.

Again $Q_1(\lambda)$ is given by

$$\begin{aligned}
&\sum_a \frac{f(a)}{a} \sum_{s_1,s_2,(s_1,s_2)=1} \frac{\lambda_{as_1}\lambda_{as_2} f(s_1) f(s_2)}{s_1 s_2} \\
&= \sum_a \frac{f(a)}{a} \frac{\mu(b) f(b)^2}{b^2} \left(\sum_s \frac{\lambda_{abs} f(s)}{s} \right)^2 \\
&= \sum_d \frac{f(d)}{d} \prod_{p|d} \left(1 - \frac{f(p)}{p}\right) \left(\sum \frac{\lambda_{ds} f(s)}{s} \right)^2.
\end{aligned}$$

Similarly, we can write $Q_2(\lambda)$ as

$$\begin{aligned}
&\frac{kx}{\log x} \frac{\sum_{a,b} g(a)\mu(b)g(b)^2}{\phi(a)\phi(b)^2} \left(\sum_s \frac{\lambda_{abs} g(s)}{\phi(s)} \right)^2 \\
&= \frac{kx}{\log x} \sum_d \frac{g(d)}{\phi(d)} \prod_{p|d} \left(1 - \frac{g(p)}{\phi(p)}\right) \left(\sum \frac{\lambda_{ds} g(s)}{\phi(s)} \right)^2.
\end{aligned}$$

For both these cases, the first step is to understand the rightmost terms

$$\left(\sum \frac{\lambda_{ds} f(s)}{s} \right)^2 \text{ and } \left(\sum \frac{\lambda_{ds} g(s)}{\phi(s)} \right)^2.$$

So, for $d \le R$, we want to evaluate

$$\sum_{l \le R/d} \mu(dl) \frac{h(l)}{l} P\left(\frac{\log R/dl}{\log R} \right)$$

where $h(l)$ is either $f(l)$ or $\frac{g(l)l}{\phi(l)}$. In both cases, h is multiplicative. Usually $h(p) \sim w$ with $w = k-1$ or k.

Take

$$P(y) = \sum_{t \ge k} \frac{p^{(t)}(0)}{t!} y^t$$

Lemma 18.4. *For $c > 0$ and (c) the corresponding vertical line in the complex plane,*

$$\frac{1}{2\pi i} \int_{(c)} \frac{z^s}{s^{t+1}} ds = \begin{cases} \frac{(\log z)^t}{t!} & \text{if } z > 1 \\ 0 & \text{if } z < 1 \end{cases}$$

Proof. Either move the line of integration to the left picking up the pole at $t = 0$. If $z < 1$ move the line of integration to the right, and there is no pole so the integral is 0. □

We now want to understand

$$\sum_{t \geq k} \frac{P^{(t)}(0)}{2\pi i} \int_{(c)} \left(\sum_{\ell=1}^{\infty} \frac{\mu(dl)}{l^{1+s}} h(l) \right) \frac{R^s}{d} \frac{ds}{s^{t+1}} \frac{1}{(\log R)^t}.$$

Using the lemma, we can evaluate this, which only gives a nonzero result if $d < R$. We have

$$\sum_{\ell=1}^{\infty} \frac{\mu(dl)}{l^{1+s}} h(l)$$

can be approximated by something like $\frac{1}{\zeta^w(s+1)}$ with w either $k-1$ or k, using that a power of ζ gives the series for the divisor function, and the Möbius function inverts this.

Then, we get $\zeta(s+1)^{-w}$ up to some Euler product involving terms of primes squared, which can be thought of as quite tame. Note that $\frac{1}{\zeta(s+1)^w} \frac{1}{\zeta^{t+1}(s)}$ has a pole of order $t - w + 1$ at $s = 0$. The idea to evaluate our desired integral is to move contours using the zero free region for $\zeta(s)$. We will pick up a contribution of the pole at $s = 0$.

Then,

$$\sum_{t \geq k} \frac{p^{(t)}(0)}{(\log R)^t} \frac{(\log R/d)^{t-w}}{(t-w)!} \cdot T|_{s=0}$$

for T the tame Euler factor from above, and $T|_{s=0}$ denoting the evaluation of T at $s = 0$.

We can simplify the above to

$$\frac{T|_{s=0}}{(\log R)^w} P^{(w)} \left(\frac{\log R/d}{\log R} \right).$$

To finish this argument, we compute

$$\begin{aligned} T|_{s=0} &= \prod_p \left(1 - \frac{1}{p}\right)^s \prod_{p \nmid d} \left(1 - \frac{h(p)}{p}\right) \mu(d) \\ &= \left(\prod_{p \mid d} \mu(p) \left(1 - \frac{1}{p}\right)^{-w} \right) \left(\prod_{p \nmid d} \left(1 - \frac{h(p)}{p}\right) \left(1 - \frac{1}{p}\right)^{-w} \right). \end{aligned}$$

Let's plug this in to our first quadratic form and see what we get.

For $Q_1(\lambda)$, we get

$$\frac{x}{(\log R)^{2k}}\sum_d \mu(d)^2\frac{f(d)}{d}\left(\prod_{p|d}\left(1-\frac{f(p)}{p}\right)\left(1-\frac{1}{p}\right)^{-2k}\right)$$
$$\cdot\left(\prod_{p\nmid d}\left(1-\frac{1}{p}^{-2k}\right)\left(1-\frac{f(p)}{p}\right)^2\right)\left(P^{(k)}\left(\frac{\log R/d}{\log R}\right)\right)^2.$$

We want to find

$$\sum_n \frac{\mu(n)^2 f(n)}{n^{s+1}}\left(\prod_{p|d}\left(1-\frac{f(p)}{p}\right)\left(1-\frac{1}{p}\right)^{-2k}\right)\left(\prod_{p\nmid d}\left(1-\frac{1}{p}^{-2k}\right)\left(1-\frac{f(p)}{p}\right)^2\right)$$
$$=\zeta(s+1)^k T_2|_{s=0}$$

and then we want to understand the other corresponding term. We have

$$\prod_p\left(1-\frac{1}{p}\right)^k\left\{\left(1-\frac{1}{p}\right)^{-2k}\left(1-\frac{f(p)}{p}\right)^2+\frac{f(p)}{p}\left(1-\frac{f(p)}{p}\right)\left(1-\frac{1}{p}\right)^{2k}\right\}.$$

This should, hopefully, turn out to be the Hardy-Littlewood constant.

$$\prod_p\left(1-\frac{1}{p}\right)^k\left\{\left(1-\frac{1}{p}\right)^{-2k}\left(1-\frac{f(p)}{p}\right)^2+\frac{f(p)}{p}\left(1-\frac{f(p)}{p}\right)\left(1-\frac{1}{p}\right)^{2k}\right\}$$
$$=\prod_p\left(1-\frac{f(p)}{p}\right)\left(1-\frac{1}{p}\right)^{-k}$$
$$=\mathfrak{S}(\mathscr{H}).$$

Then, by partial summation,

$$\frac{x}{(\log R)^{2k}}\sum_d \mu(d)^2\frac{f(d)}{d}\left(\prod_{p|d}\left(1-\frac{f(p)}{p}\right)\left(1-\frac{1}{p}\right)^{-2k}\right)$$
$$\cdot\left(\prod_{p\nmid d}\left(1-\frac{1}{p}^{-2k}\right)\left(1-\frac{f(p)}{p}\right)^2\right)\left(P^{(k)}\left(\frac{\log R/d}{\log R}\right)\right)^2$$
$$\sim\frac{x}{(\log R)^{2k}}\int_1^R P^{(k)}\left(\frac{}{\log R/z\log R}\right)^2 d\left(\mathfrak{S}(\mathscr{H})\frac{(\log z)^k}{k!}\right)$$
$$\sim\frac{x\mathfrak{S}(\mathscr{H})}{(\log R)^k}\int_0^1\frac{y^{k-1}}{(k-1)!}P^{(k)}\left(1-y\right)^2 dy.$$

One does a similar calculation for $Q_2(\lambda)$. One can similarly compute the Euler product, and one again gets

$$Q_2(\lambda) = k \frac{x \mathfrak{S}(\mathscr{H})}{\log x \, (\log R)^{k-1}} \int_0^1 \frac{y^{k-2}}{(k-2)!} P^{(k-1)} (1-y)^2 dy.$$

All we need is to find a polynomial P to make the ratio $Q_2(\lambda)/Q_1(\lambda) > 1$ subject to the condition that $R < x^{1/4}$. It's advantageous to make R as large as possible in terms of x but $R < x^{1/4}$. These quadratic forms we have only depend on the polynomial P. Next time, we'll finish this.

It will end up happening that when $R = x^{1/4-\varepsilon}$ you get this ratio to be just under 1.

But, one can even get this ratio to tend to infinity thinking of this as a higher dimensional problem, using an argument of Maynard.

19. 12/5/17

Let $\mathscr{H} := \{h_1, \ldots, h_k\}$ be an admissible tuple so that $\mathfrak{S}(\mathscr{H}) \neq 0$. We want to compare

$$Q_1 := \sum_{n \leq N} \left(\sum_{d | (n+h_1) \cdots (n+h_k)} \lambda_d \right)^2$$

with

$$Q_2 := \sum_{j=1}^{k} \sum_{n \leq N, n+h_j \text{ prime}} \left(\sum_{d | (n+h_1) \cdots (n+h_k)} \lambda_d \right)^2 .$$

We took $\lambda_d = 0$ unless $d \leq R$ is square free. Then, $R \leq N^{1/4-\varepsilon}$ by Bombieri Vinogradov. We can take

$$\lambda_d = \mu(d) P \left(\frac{\log R/d}{\log R} \right)$$

for P a polynomial vanishing to order k at 0. We computed the main terms of these two sums.

We found that for $R \leq N^{1/2-\varepsilon}$

$$Q_1 \sim \frac{\mathfrak{S}(\mathscr{H}) N}{(\log R)^k} \int_0^1 \frac{y^{k-1}}{(k-1)!} P^{(k)} \left(1-y\right)^2 dy.$$

We found also that for $R \leq N^{1/4-\varepsilon}$

$$Q_1 \sim \frac{k\mathfrak{S}(\mathscr{H})N}{(\log N)(\log R)^{k-1}} \int_0^1 \frac{y^{k-2}}{(k-2)!} P^{(k-1)} (1-y)^2 \, dy$$

Let $Q(y) = P^{(k-1)}(y)$ be a polynomial vanishing to order at least 1 at 0. Then, we want to know if

$$\frac{k \log R}{\log N} \int_0^1 \frac{y^{k-2}}{(k-2)!} Q (1-y)^2 \, dy > \int_0^1 \frac{y^{k-1}}{(k-1)!} Q'(1-y)^2 dy. \tag{19.1}$$

so that we can understand the ratio Q_1/Q_2. In Selberg's sieve one typically takes $Q(y) = y$ so that $P \sim y^k$. In GPY, they took $Q(y) = y^l$ with l chosen to be large.

We now have to compute these integrals, which are examples of the β function. Recall

Lemma 19.1.

$$\int_0^1 y^a (1-y)^b \, dy = \frac{a!b!}{(a+b+1)}.$$

Proof. Take $n = a+b+1$. We put down n numbers at random

$$x_1, \ldots, x_n \in (0,1)$$

independently and uniformly. We now ask what the chance is that x_n is in position $a+1$? It is easy to see, by symmetry, there is a $\frac{1}{n}$ chance.

On the other hand, we can order the n objects, and we can integrate over the possible positions of the nth object such that it is in position $a+1$. By choosing the ordering for the other objects, we see this probability is

$$\binom{a+b}{a} \int_0^1 x_n^a (1-x_n)^b$$

Therefore,

$$\frac{1}{a+1} = \binom{a+b}{a} \int_0^1 x_n^a (1-x_n)^b.$$

□

Simplifying (19.1) we get that it suffices to check

$$(k-1) \frac{k \log R}{\log N} \frac{(k-2)!\,(2l)!}{(k-1+2l)!} < \frac{(k-1)!l^2\,(2l-2)!}{(k+2l-2)!}.$$

Simplifying further, we want to compare

$$\frac{k}{k-1+2l}\frac{\log R}{\log N}2l\left(2l-1\right) < l^2.$$

Now say k is large and $l \sim \sqrt{k}$, we get roughly that it suffices to check

$$\left(4-\varepsilon\right)\frac{\log R}{\log N} > 1.$$

But, we chose $R = N^{1/4-\varepsilon}$. If we could chose R larger than $N^{1/4}$ we could prove bounded gaps between primes. But, with Bombieri Vinogradov, this barely fails to give bounded gaps between primes.

Exercise 19.2. Assuming Elliott-Halberstam, obtain a bound for $\lim\left(p_{n+1}-p_n\right)$.

We'll next talk about Maynard's refinement. The additional idea in GPY is the following. We have

$$Q_1 = \sum_{h_1,\ldots,h_k \le H \text{ distinct}} \sum_{n \le N} \left(\sum_{d|(n+h_1)\cdots(n+h+k)} \lambda_d\right)^2$$

$$Q_2 = \sum_{h \le H}{}' \sum_{h_1,\ldots,h_k \le H \text{ distinct}} \sum_{n \le N, n+h \text{ prime}} \left(\sum_{d|(n+h_1)\cdots(n+h_k)} \lambda_d\right)^2.$$

If every interval $[n, n+H]$ contains at most one prime than $Q_2 \le Q_1$.

If $h \ne h_1, \ldots, h_k$ the second form is of size

$$\frac{N}{\left(\log N\right)\left(\log R\right)^k}.$$

One has many more possibilities coming from h not in $h_1, \ldots, h_k$. One has Q_2 is almost Q_1 when $h \in \mathscr{H}$, but it is then pushed over by allowing $h \notin \mathscr{H}$. Multiplying this by the size of H which is $\delta \log N$. Therefore, we get enough extra help from elements of $[n, n+H]$ not lying in the tuple $\mathscr{H}$.

Here, k is very large, so we are looking at a high dimensional sieve. This is a different optimization problem and has some surprises.

Maynard and Tao have a method which gives many primes in bounded gaps. GPY gives only 2 primes in bounded gaps, but not many.

Let

$$\mathscr{H} = \{h_1, \ldots, h_k\}$$

be admissible. Consider

$$\sum_{n\sim x}\left(\sum_{\substack{d_1|(n+h_1)\\ d_2|(n+h_2)\\ \vdots \\ d_k|(n+h_k)}}\lambda_{d_1,\dots,d_k}\right)^2$$

and compare it to

$$\sum_{\substack{n\sim x\\ n+h_j \text{ prime}}}\left(\sum_{d_i|(n+h_i)}\lambda_{d_1,\dots,d_k}\right)^2.$$

Before we had

$$\sum_{d|(n+h_1)\cdots(n+h_k)}\lambda_d$$

and

$$\lambda_d=\sum_{d_1\cdots d_k=d}\lambda_{d_1,\dots,d_k}$$

where we might have in mind that

$$\lambda_{d_1,\dots,d_k}=\mu(d_1)\cdots\mu(d_k)$$

equal to a function

$$F\left(\frac{\log d_1}{\log R},\dots,\frac{\log d_k}{\log R}\right)$$

whereas we are allowing $F(x_1,\dots,x_k)$ supported on $x_1+\cdots+x_k\le 1$ rather than just the function $G(x_1+\cdots x_k)$. So there is more flexibility in allowing functions of many variables rather than just of a single variable.

We now introduce the trick, previously known as using a small sieve, but after Green Tao it is known as the W-trick.

The most naive sieve can be used to count

$$\sum_{\substack{n\le x\\ p|n\implies p>z}}1$$

To count this, if
$$\prod_{p \le z} p$$
is very small, we can easily sieve this. The product above is around e^z. For example if $z \le \frac{\log x}{10^6}$ this is very easy to sieve. For example, take
$$W = \prod_{p \le \log \log \log x} p,$$
then $W = (\log \log x)^{O(1)}$. When studying these, we insist n lies in some progression $\nu \bmod W$. That is, we want to understand
$$\sum_{n \sim x, n \equiv \nu \bmod W} \left(\sum_{\substack{d_1 | (n+h_1) \\ d_2 | (n+h_2) \\ \vdots \\ d_k | (n+h_k)}} \lambda_{d_1, \ldots, d_k} \right)^2$$
and compare it to
$$\sum_{\substack{n \sim x \\ n+h_j \text{ prime} \\ n \equiv \nu \bmod W}} \left(\sum_{d_i | (n+h_i)} \lambda_{d_1, \ldots, d_k} \right)^2.$$

Then, $(n+h_i, n+h_j) = 1$ for all i and j with $n \equiv \nu \bmod W$.
So, we want to understand the quadratic form
$$\sum_{\substack{n \sim x \\ n \equiv \nu \bmod W}} \sum_{\substack{d_1, \ldots, d_k \\ e_1, \ldots, e_k \\ [d_i, e_i] | (n+h_i)}} \lambda_{d_1, \ldots, d_k} \lambda_{e_1, \ldots, e_k}.$$

We have $\lambda_{d_1, \ldots, d_k} = 0$ unless

(1) $d_1, \ldots, d_k \le R$ and are squarefree
(2) $d_1 \cdots d_k$ is coprime to W.
(3) $(d_i, d_j) = 1$.

These above conditions are automatic, but there is an additional condition: Note that if $i \neq j$ we must have $(d_i, e_j) = 1$, as otherwise the sum would be 0.

So, suppose now

$$\sum_{\substack{n\sim x \\ n\equiv\nu \bmod W}} \sum_{\substack{d_1,\dots,d_k \\ e_1,\dots,e_k \\ [d_i,e_i]|(n+h_i)}} \lambda_{d_1,\dots,d_k}\lambda_{e_1,\dots,e_k} = \sum_{\substack{d_1,\dots,d_k \\ e_1,\dots,e_k}} \lambda_{d_1,\dots,d_k}\lambda_{e_1,\dots,e_k} \frac{x}{W\prod_{i=1}^k [d_i,e_i]}.$$

The error term is ok if $R \le x^{1/2-\varepsilon}$.

If d_i, e_i have a common factor this will only appear once as in the denominator. But if d_i and d_j have a common factor, this will appear in the denominator at least to the power 2. But it turns out these form part of the tail of a convergent sum which will go to 0. Hence, we will ignore the condition that if $i \neq j$ implies $(d_i, e_j) = 1$. To justify this, we need to check that the terms with $(d_i, e_i) > 1$ will contribute a small amount compared to the main term (assuming we have removed small prime factors.

Then,

$$\begin{aligned}\sum_{\substack{n\sim x \\ n\equiv\nu \bmod W}} \sum_{\substack{d_1,\dots,d_k \\ e_1,\dots,e_k \\ [d_i,e_i]|(n+h_i)}} \lambda_{d_1,\dots,d_k}\lambda_{e_1,\dots,e_k} &= \sum_{\substack{d_1,\dots,d_k \\ e_1,\dots,e_k}} \lambda_{d_1,\dots,d_k}\lambda_{e_1,\dots,e_k} \frac{x}{W\prod_{i=1}^k [d_i,e_i]} \\ &\sim \frac{x}{W} \sum_{\substack{d_1,\dots,d_k \\ e_1,\dots,e_k}} \frac{\lambda_{d_1,\dots,d_k}\lambda_{e_1,\dots,e_k}}{d_1\cdots d_k e_1 \cdots e_k} \prod_{i=1}^k (d_i, e_i).\end{aligned}$$

Then, we can write

$$(d_i, e_i) = \sum_{f_i|(d_i,e_i)} \phi(f_i).$$

We can write our quadratic form Q_1 as approximately

$$Q_1 \sim \frac{x}{W} \sum_{f_1,\dots,f_k} \prod_{i=1}^k \phi(f_i) \left(\sum_{\substack{d_1,\dots,d_k \\ f_i|d_i}} \frac{\lambda_{d_1,\dots,d_k}}{d_1\cdots d_k} \right)^2.$$

We the set

$$y_{f_1,\dots,f_k} = \left(\prod_{i=1}^k \mu(f_i)\phi(f_i) \right) \left(\sum_{f_i|d_i} \frac{\lambda_{d_1,\dots,d_k}}{d_1\cdots d_k} \right).$$

Then,

$$\lambda_{d_1,\dots,d_k} \sim \mu(d_1)\cdots\mu(d_k).$$

The Möbius function of the f_i then cancel out and the f_i mostly cancel with the $\phi(f_i)$. Therefore, the choice of $y_{f_1,\ldots,f_k}$ will look like

$$y_{f_1,\ldots,f_k} \sim F\left(\frac{\log f_1}{\log R}, \ldots, \frac{\log f_n}{\log R}\right).$$

Then, after an invertible change of variables,

$$\lambda_{d_1,\ldots,d_k} = (\prod \mu(d_j)d_j) \sum_{d_j | a_j} \frac{y_{a_1,\ldots,a_k}}{\phi(x_1)\cdots\phi(x_k)}.$$

Then,

$$Q_1 \sim \frac{x}{W} \sum_{f_1,\ldots,f_k} \frac{y^2_{f_1,\ldots,f_k}}{\phi(f_1)\cdots\phi(f_k)}.$$

So, $y_{f_1,\ldots,f_k} = 0$ unless $f_1 \cdots f_k \leq R$ is squarefree and coprime to W. We make the choice

$$y_{f_1,\ldots,f_k} = F\left(\frac{\log f_1}{\log R}, \ldots, \frac{\log f_k}{\log R}\right).$$

Then, Q_1 is approximately

$$\frac{x}{R}\left(\frac{\phi(W)}{W}\right)^k (\log R)^k \int_{x_1,\ldots,x_k} F\left(x_1, \ldots, x_k\right)^2 dx_1 \cdots dx_k$$

where

$$F\left(x_1, \ldots, x_k\right) = 0$$

unless $x_1 + \cdots x_k \leq 1$.

Remark 19.3. The $(\log R)^k$ in the numerator (instead of the denominator) is a scaling fact relating to how we chose the $y_{f_1,\ldots,f_k}$.

Let's now start understanding Q_2, which we will finish on Thursday. This will be similar. We have

$$\begin{aligned} Q_2 &\sim k \sum_{\substack{n \sim x \\ n+h_k \text{ prime} \\ n \equiv \nu \bmod W}} \left(\sum_{d_i | n+h_i} \lambda_{d_1,\ldots,d_k}\right)^2 \\ &= k \sum_{\substack{d_1,\ldots,d_k \\ e_1,\ldots,e_k \\ d_k=1 \\ e_k=1}} \lambda_{d_1,\ldots,d_k}\lambda_{e_1,\ldots,e_k} \sum_{\substack{n \sim x \\ n \equiv \nu \bmod W \\ n+h_k \text{ prime} \\ n+h_i \equiv 0 \bmod [d_i,e_i]}} 1 \end{aligned}$$

Again, on the last line above there will be a coprimality condition on (d_i, e_j) which we can ignore as was done above. Then, we can simplify

$$\sum_{\substack{n\sim x \\ n\equiv \nu \bmod W \\ n+h_k \text{ prime} \\ n+h_i\equiv 0 \bmod [d_i,e_i]}} 1 \qquad \sim \frac{x}{(\log x)\,\phi(W)\prod_{i=1}^{k-1}\phi\left([d_i,e_i]\right)}.$$

We'll finish understanding this quadratic form on Thursday.

20. 12/7/17

Recall last time we had an admissible set

$$\mathscr{H} = \{h_1, \ldots, h_k\}$$

and we chose

$$W = \prod_{p\leq \log\log\log x} p$$

where

$$\nu \bmod W \text{ with } (\nu + h_i, W) = 1 \text{ for all } i$$

We had

$$Q_1 = \sum_{\substack{n\leq x \\ n\equiv \nu \bmod W}} \left(\sum_{d|n+h_i} \lambda_{d_1,\ldots,d_k}\right)^2.$$

When $R \leq x^{1/2-\varepsilon}$, we evaluated the above as

$$\frac{x}{W}\sum_{r_1,\ldots,r_k} \frac{y^2_{r_1,\ldots,r_k}}{\operatorname{pr}\phi(r_j)}$$

with

$$y_{r_1,\ldots,r_k} = \left(\prod \mu(r_i)\phi(r_i)\right)\sum_{r_i|d_i}\frac{\lambda_{d_1,\ldots,d_k}}{d_1\cdots d_k}$$

where

$$\lambda_{d_1,\ldots,d_k} = \left(\prod \mu(d_i)d_i\right)\sum_{d_i|r_i}\frac{y_{r_1\ldots,r_k}}{\phi(r_1)\cdots\phi(r_k)}.$$

We chose

$$y_{r_1,\ldots,r_k} = F\left(\frac{\log r_1}{\log R}, \ldots, \frac{\log r_k}{\log R}\right)$$

with

$$F(x_1, \ldots, x_k)$$

supported on $x_1 + \cdots + x_k \leq 1$.

We then get

$$Q_1 = \frac{x}{W}\left(\frac{\phi(W)}{W}\right)^k (\log R)^k \int_{x_1,\ldots,x_k} F(x_1,\ldots,x_k)^2\, dx_1 \cdots dx_k.$$

We have

$$\frac{1}{\phi([d,e])} = \frac{\phi((d,e))}{\phi(d)\phi(e)}$$

using that

$$\phi(n) = \sum_{d|n} g(d)$$

with g multiplicative (on relatively prime inputs) with $g(p) = p-2$.

We compare this Q_1 to (with the function g defined above)

$$\begin{aligned} Q_2 &= \sum_{\substack{n \leq x \\ n+h_k \text{ prime} \\ n \equiv \nu \bmod W}} \left(\sum \lambda_{d_1,\ldots,d_k}\right)^2 \\ &= \sum_{\substack{d_1,\ldots,d_k \\ e_1,\ldots,e_k \\ d_k=e_k=1}} \frac{\lambda_{d_1,\ldots,d_k}\lambda_{e_1,\ldots,e_k}}{\phi(W)\log x \prod \phi([d_i,e_i])} \\ &= \frac{x}{\phi(W)\log x} \sum_{f_1,\ldots,f_k}\left(\prod_{i=1}^k g(f_i)\right) \sum_{\substack{d_i,e_i \\ f_i|d_i, f_i|e_i \\ d_k=e_k=1}} \frac{\lambda_{d_1,\ldots,d_k}}{\prod \phi(d_j)}\frac{\lambda_{e_1,\ldots,e_k}}{\prod \phi(e_j)}. \end{aligned}$$

Note that above we had a coprimality condition $[e_i, d_i] \mid n + h_i$ and $(d_i, e_j) = 1$ for $i \neq j$.

Then, we let

$$y^{(k)}_{f_1,\ldots,f_k} = \left(\prod \mu(f_i) g(f_i)\right) \cdot \sum_{\substack{d_1,\ldots,d_k \\ d_k=1, f_i|d_i}} \frac{\lambda_{d_1,\ldots,d_k}}{\phi(d_1)\cdots\phi(d_k)}.$$

By convention we set $y^{(k)}_{f_1,\ldots,f_k} = 0$ unless $f_k = 1$.

We then have

$$Q_2 = \frac{x}{\phi(W)\log x}\sum_{f_1,\dots,f_k}\frac{\left(y^{(k)}_{f_1,\dots,f_k}\right)^2}{\prod_{j=1}^k g(f_j)}.$$

Lemma 20.1. *Letting $f_k = d_k = 1$,*

$$y^{(k)}_{f_1,\dots,f_k} \sim \sum_{r_k}\frac{y_{f_1,\dots,f_{k-1},r_k}}{\phi(r_k)}.$$

Proof. We have

$$\begin{aligned}
y^{(k)}_{f_1,\dots,f_k} &= \left(\prod \mu(f_i)g(f_i)\right)\sum_{\substack{d_1,\dots,d_k\\ f_i|d_i}}\frac{\lambda_{d_1,\dots,d_k}}{\phi(d_1)\cdots\phi(d_k)}\\
&= \left(\prod \mu(f_i)g(f_i)\right)\sum_{\substack{f_i|d_i\\ d_k=1}}\frac{1}{\phi(d_1)\cdots\phi(d_k)}\left(\prod \mu(d_j)d_j\right)\sum_{\substack{r_1,\dots,r_k\\ d_j|r_j}}\frac{y_{r_1,\dots,r_k}}{\phi(r_1)\cdots\phi(r_k)}\\
&= \left(\prod \mu(f_j)g(f_j)\right)\sum_{\substack{r_1,\dots,r_k\\ f_j|r_j}}\frac{y_{r_1,\dots,r_k}}{\prod\phi(r_j)}\sum_{\substack{d_1,\dots,d_1=1\\ f_i|d_i|r_i}}\prod_{j=1}^k\frac{\mu(d_j)d_j}{\phi(d_j)}
\end{aligned}$$

Fixing f_i, r_j we want to compute

$$\sum_{\substack{d_1,\dots,d_1=1\\ f_i|d_i|r_i}}\prod_{j=1}^k\frac{\mu(d_j)d_j}{\phi(d_j)}.$$

if $k = j$, then the term is 1. If $j < k$ the term is

$$\frac{f_j\mu(f_j)}{\phi(f_j)}\prod_{p|r_j/f_j}\left(1-\frac{p}{p-1}\right).$$

Then, $1-\frac{p}{p-1}$ the above term is relatively small, unless $r_j = f_j$, in which case the product is the empty product and goes away.

Therefore, we can simplify

$$\begin{aligned}
&\left(\prod \mu(f_j)g(f_j)\right) \sum_{\substack{r_1,\ldots,r_k \\ f_j \mid r_j}} \frac{y_{r_1,\ldots,r_k}}{\prod \phi(r_j)} \sum_{\substack{d_1,\ldots,d_1=1 \\ f_i \mid d_i \mid r_i}} \prod_{j=1}^{k} \frac{\mu(d_j)d_j}{\phi(d_j)} \\
&\sim \prod_j \left(\frac{\mu(f_j)g(f_j)f_j\mu(f_j)}{\phi(f_j)}\right) \sum_{r_k} \frac{y_{f_1,\ldots,f_{k-1},r_k}}{\phi(f_1)\cdots\phi(f_{k-1})\phi(r_k)} \\
&= \prod_j \left(\frac{\mu(f_j)g(f_j)f_j\mu(f_j)}{\phi(f_j)^2})\right) \sum_{r_k} \frac{y_{f_1,\ldots,f_{k-1},r_k}}{\phi(r_k)} \\
&\sim \sum_{r_k} \frac{y_{f_1,\ldots,f_{k-1},r_k}}{\phi(r_k)}.
\end{aligned}$$

□

Using the lemma, let's continue to evaluate Q_2. Recall we chose F so that

$$y_{r_1,\ldots,r_k} = F\left(\frac{\log r_1}{\log R}, \ldots, \frac{\log r_k}{\log R}\right).$$

By Lemma 20.1,

$$\begin{aligned}
y^{(k)}_{f_1,\ldots,f_k} &\sim \sum_{r_k} \frac{y_{f_1,\ldots,f_{k-1},r_k}}{\phi(r_k)} \\
&= \frac{\phi(W)}{W}(\log R)\left(\int F\left(\frac{\log f_1}{\log R}, \ldots, \frac{\log f_{k-1}}{\log R}, x_k\right) dx_k\right).
\end{aligned}$$

Plugging this in for Q_2, we have

$$\begin{aligned}
Q_2 &= \frac{x}{\phi(W)\log x} \sum_{f_1,\ldots,f_k} \frac{\left(y^{(k)}_{f_1,\ldots,f_k}\right)^2}{\prod_{j=1}^k g(f_j)} \\
&= \frac{x}{\phi(W)\log x}\left(\frac{\phi(W)}{W}\log R\right)^2 \left(\frac{\phi(W)}{W}\log R\right)^{k-1} \\
&\int_{x_1,\ldots,x_{k-1}} \left(\int_{x_k} F\left(x_1,\ldots,x_{k-1},x_k\right) dx_k\right)^2 dx_1 \cdots dx_{k-1}.
\end{aligned}$$

(where here we are really multiplying by k for choosing a particular h_i, and we are assuming $F(x_1, \ldots, x_k)$ is symmetric).

So, for comparison, we have

$$Q_1 = \frac{x}{W}\left(\frac{\phi(W)}{W}\log R\right)^k \int_{x_1,\ldots,x_k} F\left(x_1,\ldots,x_k\right)^2 dx_1\cdots dx_k$$
$$Q_2 = \frac{kx}{W}\left(\frac{\phi(W)}{W}\log R\right)^k \left(\frac{\log R}{\log x}\right) \int_{x_1,\ldots,x_{k-1}} \left(\int_{x_k} F\left(x_1,\ldots,x_k\right) dx_k\right)^2 dx_1\cdots dx_{k-1}.$$

We then see that this almost matches up with our first quadratic form Q_1. The only difference is that we have two different quadratic forms based on the function F we are choosing. So, we have boiled everything down to a problem of optimizing F.

Recall we want $F(x_1,\ldots,x_k)$ to be symmetric and $x_1+\cdots+x_k \leq 1$. We will choose

$$F\left(x_1,\ldots,x_k\right) = \prod_{i=1}^k g(kx_i)$$

for some fixed function g on $x_1+\cdots+x_k \leq 1$.

The numerator of the ratio Q_2/Q_1 is given by

$$\left(k\frac{\log R}{\log x}\right)\int_{x_1\cdots x_{k-1}}\left(\int_{\substack{x_k \\ x_1+\cdots+x_k\leq 1}} \prod g(kx_i)dx_k\right)^2 dx_1\cdots dx_{k-1}$$
$$= \frac{k}{k^{k+1}}\left(\frac{\log R}{\log x}\right)\int_{u_1,\ldots,u_{k-1}}\left(\int_{u_k, u_1+\cdots u_k\leq k} g(u_k)du_k\right)^2 \prod_{j=1}^{k-1} g\left(u_j\right)^2 du_j$$

with $kx_i = u_i$.

Then, the denominator is similarly given by

$$\frac{1}{k^k}\int_{\substack{u_1,\ldots,u_k \\ u_1+\cdots+u_k\leq k}} \prod g(u_j)^2 du_j.$$

Next, we will give an upper bound on the denominator and a lower bound on the numerator. For the denominator, we have the upper bound

$$\int_{\substack{u_1,\ldots,u_k \\ u_1+\cdots+u_k\leq k}} \prod g(u_j)^2 du_j \leq \left(\int_0^\infty g(u)^2 du\right)^k.$$

So, g is a function on the positive reals, but we may as well take g supported on $[0,k]$, since we are only integrating over u_i with $\sum_i u_i \leq k$. Let's assume that g is supported on $[0,B]$ with B a bit smaller than k, say $B \sim k/100$. Now, let's obtain a lower bound for the numerator.

We'll let u_k go up to B. We'll then make an upper bound for the numerator

$$\int_{\substack{u_1,\ldots,u_{k-1} \\ u_1+\cdots+u_{k-1}\leq k-B}} \left(\int_{u_k} g(u_k)\right)^2 \prod_{j=1}^{k-1} g(u_j)^2 du_j. \tag{20.1}$$

If we ignore the restriction that $\sum_i u_i \leq k-B$, then then Q_2/Q_1 is bounded below by, up to some constant,

$$\frac{\left(\int g(u)\right)^2}{\int g(u)^2}\left(\frac{\log R}{\log x}\right).$$

Remark 20.2. But now, how can we ignoring the restriction that $u_1 + \cdots + u_{k-1} \leq k-B$? This might seem like a serious issue, but we now discuss the answer.

The key additional observation is the following. If we know $\int ug(u)^2 du \leq \frac{1}{2}\int g(u)^2$, then most of the weight is concentrated on values of $u \leq 1/2$. Then, $\sum_i u_i + \cdots + u_{k-1}$ is at most $k/2$, most of the time. So we should then be able to ignore $\sum_{i=1}^{k-1} u_i \leq k-B$ condition. Let's now make this idea more precise.

We now observe that

$$\begin{aligned}
&\int_{\substack{u_1,\ldots,u_{k-1} \\ u_1+\cdots+u_{k-1}\leq k-B}} \left(\int_{u_k} g(u_k)\right)^2 \prod_{j=1}^{k-1} g(u_j)^2 du_j \\
&\geq \left(\int g(u)\right)^2 \int_{u_1,\ldots,u_{k-1}} \left(1-\left(\frac{u_1+\cdots+u_{k-1}}{k-B}\right)^2\right) \prod_{j=1}^{k-1} g(u_j)^2 du_j \\
&\geq \left(\int g(u)\right)^2 \{\left(\int g(u)^2\right)^{k-1} - \frac{k-1}{(k-B)^2}\left(\int u^2 g(u)^2 du\right)\left(\int g(u)^2\right)^{k-1} \\
&- \frac{(k-1)(k-2)}{(k-B)^2}\left(\int ug(u)^2\right)^2\left(\int g(u)^2\right)^{k-3}\}.
\end{aligned}$$

Then,

$$\int u^2 g(u)^2 du \sim \frac{B}{2}\left(\int g(u)^2\right)$$

and the whole term

$$\frac{k-1}{(k-B)^2}\left(\int u^2 g(u)^2 du\right)\left(\int g(u)^2\right)^{k-1} \leq \frac{1}{200}\left(\int g(u)^2\right)^{k-1}.$$

So, we can bound Equation 20.1 by

$$\frac{1}{2}\left(\int g(u)\right)^2\left(\int g(u)^2\right)^{k-1}.$$

So, we now want

$$\int ug(u)^2 \le \frac{1}{2}\int g(u)^2 du$$

and we want to make $\left(\int g(u)\right)^2$ large in comparison to $\int g(u)^2$.

The condition vaguely means that most of the mass should appear on small numbers. You can start to guess what function might work for g (or you can try to use calculus of variations). We can try $g(u) = \frac{1}{u}$, though this has a pole at 0 Let's try

$$g(u) = \begin{cases} \frac{1}{u} & \text{if } \frac{1}{A} \le u \le B \\ 0 & \text{else} \end{cases}$$

Then,

$$\int g(u) = \log AB$$
$$\int g(u)^2 \sim A$$
$$\int ug(u)^2 du = \log AB.$$

So, we need something like $\log AB \le A/2$. So, let's take $B = k/100$. We then, have $\log AB \sim \log k$. We can take $A = 3\log k$. It then meets the condition that $B \le k/100$ and

$$\int ug(u)^2 \le \frac{1}{2}\int g(u)^2 du.$$

To conclude, we now want to compute Q_2/Q_1. Indeed,

$$\begin{aligned}\frac{\left(\int g(u)\right)^2}{\int g(u)^2} &= \frac{(\log AB)^2}{A} \\ &\sim \frac{(\log k)^2}{3\log k} \\ &= \frac{\log k}{3}.\end{aligned}$$

And indeed, this goes to ∞ so long as $k \to \infty$.

So, in any tuple where k is sufficiently large, where you expect to find k primes, you can at least find $\log k$ number of primes.